

**ANNAMACHARYA INSTITUTE OF TECHNOLOGY AND SCIENCES, TIRUPATI
(AUTONOMOUS)**

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE MATERIAL

CLOUD COMPUTING

B. Tech III-II Sem. (CSE)



**Annamacharya Institute of Technology & Sciences, Tirupati
(Autonomous)**



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AITs – TIRUPATI, Dept Of

UNIT-I

INTRODUCTION:

The Vision of Cloud Computing:

In Simplest terms, [cloud computing](#) means storing and accessing the data and programs on remote servers that are hosted on internet instead of computer's hard drive or local server. Cloud computing is also referred as Internet based computing.

These are following **Vision of Cloud Computing** :

Cloud computing provides the facility to provision virtual hardware, runtime environment and services to a person having money.

These all things can be used as long as they are needed by the user.

3. The whole collection of computing system is transformed into collection of utilities, which can be provisioned and composed together to deploy systems in hours rather than days, with no maintenance cost.

4. The long term vision of a cloud computing is that IT services are traded as utilities in an open market without technological and legal barriers.

5. In the future, we can imagine that it will be possible to find the solution that matches with our requirements by simply entering out request in a global digital market that trades with cloud computing services.

6. The existence of such market will enable the automation of discovery process and its integration into its existing software systems.

7. Due to the existence of a global platform for trading cloud services will also help service providers to potentially increase their revenue.

8. A cloud provider can also become a consumer of a competition service in order to fulfill its promises to customers.

9. In the near future we can imagine a solution that suits our needs by simply applying our application to the global digital market for cloud computing services.

10. The presence of this market will enable the acquisition process to automatically integrate with its integration into its existing software applications. The availability of a global cloud trading platform will also help service providers to increase their revenue.

11. A cloud provider can also be a buyer of a competitive service to fulfill its promises to customers.

Defining a cloud:

Cloud computing is the delivery of different services through the Internet. These resources include tools and applications like data storage, servers, databases, networking, and software.

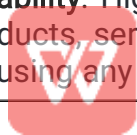
Rather than keeping files on a proprietary hard drive or local storage device, [cloud-based storage](#) makes it possible to save them to a remote database. As long as an electronic device has access to the web, it has access to the data and the software programs to run it.

Cloud computing is a popular option for people and businesses for a number of reasons including cost savings, increased productivity, speed and efficiency, performance, and security.

Characteristics of Cloud:

Agility: In a cloud computing context, agility often refers to the ability to rapidly develop, test and launch applications that drive business growth in a constantly changing IT environment. Cloud technology offers businesses a key means of promoting agility, and is a vital tool in the enterprise push toward better adaptability. It works in distributed computing environment, shares resources among users and works very fast.

High Availability and reliability: High availability is the ultimate goal of moving to the cloud. The idea is to make your products, services, and tools available to your customers and employees at any time from anywhere using any device with an Internet connection. Its servers availability is high



and more reliable in case of infrastructure failure.

High Scalable: Cloud computing scalability refers to how well your system can react and adapt to changing demands. As your company grows, you want to be able to seamlessly add resources without losing quality of service or interruptions. On demand provisioning of resources on a large scale, no need of having engineers of peak load.

Multi Sharing: Multiple users and applications can work more efficiently with less cost by sharing resources.

Device and location Independence: Cloud computing is location-independent computing, whereby shared servers provide resources, software, and data to computers and other devices on demand, as with the electricity grid.

Maintenance: Cloud computing operation and maintenance is the process of managing the cloud infrastructure, applications, and data. It involves the monitoring of the system performance, the security of the system, and the maintenance of the system.

Low Cost: Cloud computing is cost-effective because costs usually directed at purchasing hardware and software are immediately eliminated. Operational, maintenance and upgrade costs are all expenses that disappear. The IT staff that does setup and maintenance is eliminated too.

Services in the pay use mode: Here Applications are provided to the user for accessing services of cloud.

Cloud Providers:

1. Amazon Web Services

[Amazon Web Services](#) (AWS) is the world-leading cloud vendor with over 200 integrated features and services. AWS offers a free tier that enables you to test various services free of charge and without any commitments.

With 77 Availability Zones and 24 geographic regions, AWS makes up over 30% of the cloud market share.

2. Microsoft Azure

Like AWS, [Microsoft Azure](#) offers an abundance of on-demand computing services designed for efficient building in the cloud.

Azure offers four different kinds of cloud computing: Infrastructure as a service (IaaS), platform as a service (PaaS), software as a service (SaaS), and serverless.

3. Google Cloud Platform

The Google Cloud Platform (GCP) is an assemblage of services that operates on the same infrastructure used to power Google Search, Youtube, and several other tech behemoth offerings.

4. Alibaba Cloud

Alibaba is a global leader in the cloud with a focus on security and artificial intelligence. The Alibaba Cloud is the second-largest cloud service provider around the world and powers almost half of China's 4.97 million websites.

5. Salesforce

Salesforce offers cloud-based CRM, customer service, marketing automation, sales, and much more. Their customer relationship tools suite can help you grasp a data-driven approach for your business.

6. IBM



IBM blends PaaS with IaaS to provide a holistic cloud experience. Like AWS and Microsoft Azure, IBM has an abundance of services to help simplify many software development tasks.

7. Digital Ocean

Digital Ocean is designed to meet developers' needs for deploying and scaling applications that run simultaneously on multiple computers. As the third-largest hosting company globally, Digital Ocean has two primary product offerings - compute and storage.

8. Dell

Dell technology combines the capacity of VMware software and Dell Technologies infrastructure to provide an integrated multi-cloud approach. As a hybrid, multi-cloud provider, Dell focuses its core offering around storage and data protection.

9. Adobe

Adobe Creative Cloud is unlike the other vendors and focuses on the creative sector of cloud computing. With a Software-as-a-Service (SaaS) offering, Adobe provides an array of tools geared towards graphic design, video editing, photography, and more.

10. Dropbox

Dropbox is a robust cloud storage solution designed for various business sizes to store files and documents on a remote cloud server. Dropbox allows users to access any saved data or content from any device with an internet connection.

Operations Performed by Cloud:

- Developing new Application and Services.
- Storage, Backup, Recovery of Data.
- Hosting Blogs and Websites.
- Delivery of Software on Demand.
- Analysis of Data.
- Streaming Videos and Audios.

Pros & Cons of Cloud Computing:

Pros:

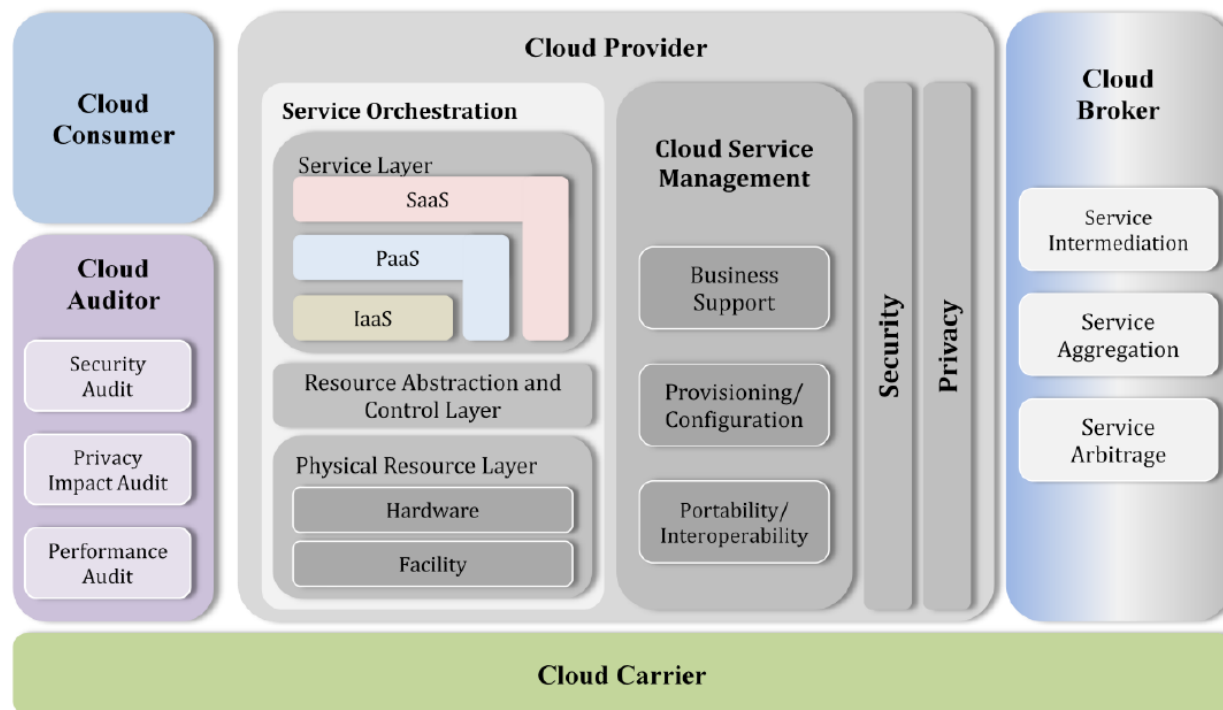
1. Lower cost computers for users.
2. Improved Performance.
3. Lower IT infrastructure costs.
4. Fewer Maintenance cost.
5. Instant software updates.
6. Increased computing power.
7. Unlimited storage capacity.
8. Increased data safety.
9. Improved compatibility between operating system.
10. Improved document format compatibility.
11. Easier group collaboration.
12. Universal access to documents.
13. Latest version availability.

Cons:



1. Requires a constant internet connection.
2. Doesn't work well with low speed connection.
3. It can be slow.
4. Stored data might not be secured.
5. Problems will arise if data loss occurs.
6. Features might be limited.

Cloud Computing Reference Model:



Cloud Consumer: is a person or organization that maintains a business relationship with, and uses service from, cloud providers. Anyone who purchases a cloud service is a consumer, and within the consumer there could be an array of roles responsible for configuring and managing the resources from the cloud provider depending on the services obtained.

Cloud Provider: is a person, organization, or entity responsible for making a [service available](#) to interested parties. A [cloud provider](#) would have a significant number of roles responsible for the management of its cloud resources including those responsible for selling, onboarding, configuring and supporting cloud services for its consumers.

Cloud Auditor: is a party that can conduct independent assessment of [cloud services](#), information system operations, performance, and security of the cloud implementation. Generally, cloud auditors are categorized based on intent. For the most part, their focus is on risk and compliance, especially around [information security](#). Other auditors can provide advisory services especially to consumers looking to cut down their bills or raise the level of efficiency in the resources consumed.

Cloud Broker: is any entity that manages the use, performance, and delivery of cloud services, and negotiates relationships between Cloud Providers and Cloud Consumers. Cloud brokers support consumers to get value for money by playing the advisory role especially for consumers who have a hybrid mix of resources from multiple providers.

Cloud Carrier: is an intermediary that provides connectivity and transport of cloud services from Cloud Providers to Cloud Consumers. Most ISPs have taken the role of cloud carriers as they provide the requisite bandwidth needed to connect consumers with providers as well as capabilities that support the connectivity.



Challenges of cloud:

1. Security and Privacy
2. Portability
3. Interoperability
4. Computing Performance
5. Reliability and Availability

Historical Developments:

1. In 1950 the main frame and time sharing are born, introducing the concept of shared computer resources.
2. During this time word cloud was not in use.
3. Cloud computing is believed to have been invented by Joseph Carl Robnett Licklider in 1960s with his work on ARPANET to connect people and data from anywhere at any time.
4. In 1969 the first working prototype of ARPANET is launched.
5. In 1970 the word "Client-Server" come in to use.
6. Client server defines the computing model where client access the data and applications from a central server.
7. In 1995, pictures of cloud are started showing in diagrams, for not technical people to understand.
8. At that time AT & T had already begun to develop an architecture and system where data would be located centrally.
9. In 1999 the salesforce.com was launched, the first company to make enterprise applications available from a website.
10. In 1999, the search engine Google launches.
11. In 1999, Netflix was launched, introducing the new revenue way.
12. In 2003, web2.0 is born, which is characterized by rich multimedia. Now user can generate content.
13. In 2004 Facebook launches giving users facility to share themselves.
14. In 2006, Amazon launched Amazon Web Services(AWS), giving users a new way.
15. In 2006, Google CEO Eric Schmidt uses the word "cloud" as an industry event.
16. In 2007, Apple launches iPhone, which could be used on any wireless network.
17. In 2007, Netflix launches streaming services, and live video watching is born.
18. In 2008, private cloud come in to existence.
19. In 2009, browser based application like google apps are introduced.
20. In 2010, hybrid cloud (private+public cloud) comes in to existence.
21. In 2012, Google launches google drive with free cloud storage.
22. Now cloud adoption is present, which makes cloud computing more stronger.
23. The IT services progressed over the decades with the adoption of technologies such as Internet Service Providers (ISP) Application Service Providers.

Applications of Cloud Computing:**Healthcare:**

Cloud computing in healthcare



- Cloud-based healthcare is the process of integrating cloud computing technology into healthcare services for cost savings, easy data sharing, personalized medicine, telehealth apps, and more benefits. Many healthcare providers also use cloud-based systems in healthcare for safe data storage, backups, and digital record retrieval.

Cloud-based healthcare solutions help with the following aspects in the healthcare industry:

1. Better analysis and monitoring of data related to diagnosis and treatment of different diseases.
 2. Massive storage resources for large datasets for EHR (Electronic Health Record) and radiology images.
 3. Ability to provide on-demand access to computing resources.
 4. Sharing of EHR only among authorized physicians, doctors, and hospitals in different locations around the world, providing timely access to information critical for saving lives and reducing the need for duplicate testing.
 5. Improved data analysis.
 6. Enhanced tracking of patients' healthcare data.
 7. Adopting cloud healthcare computing reduces the overhead of tasks like upgrading records.
- Cloud technology in the healthcare industry helps develop a proper plan for monitoring patients remotely. Additionally, cloud technology in healthcare provides new and valuable insights for different healthcare management solutions.

Applications of Cloud Computing in Healthcare

Below are some of the most useful healthcare cloud applications:

E-health and Telemedicine

Cloud computing is now widely used for E-health, which refers to providing healthcare services electronically through the internet. The cloud allows different medical professionals to collaborate and give their input on complex medical cases, such as telesurgery, teleradiology, etc.

E-health provides the right clinical treatment to patients residing anywhere in the world. Telemedicine projects use cloud computing as an **information and Communications Technology (ICT)** infrastructure for better doctor-to-doctor and doctor-to-patient communication and interaction.

Cloud solutions for healthcare and telemedicine also help with:

Sharing patients' medical data in real-time across different geographical locations

Minimizing unnecessary visits to hospitals, thereby saving time and money

Drug Discovery

Cloud computing plays a significant role in drug recovery because drug recovery requires a large number of computing resources for discovering different compounds from billions of chemical structures. Infrastructure as a Service (IaaS) services are especially useful in simplifying this process.

Healthcare Information Systems



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The healthcare industry uses cloud-based management information systems to provide improved patient care, manage human resources, better-querying services, and billing and finance. Cloud computing in the healthcare industry is also used to develop, test and deploy these systems.

Cloud computing helps with fast, collaborative development, better integration of the system with other healthcare systems, and cross-platform compatibility.

Personal Health Records

Another cloud computing application in healthcare is managing access to personal health records (PHR) and Electronic Health Records (EHR). With cloud-based programs for PHR, users can easily access and manage the PHR database and share data. These programs have advanced sharing features that give users high control over shared data. Some programs also use protocols like [SOAP](#) and can be used for mobile and desktop applications.

Clinical Decision Support System (CDSS)

CDSS is an advanced system that uses the knowledge and behavior of a medical professional to provide advice on the patient record analysis. This system is used for diagnosing conditions and prescribing medication. Cloud computing can be used to create such systems that provide better patient care.

Additionally, with the advancement of technology, smartphones and fitness trackers now monitor heart rate, diabetes, and blood pressure. Cloud-based systems can be used for efficient real-time diagnosis; patients share their data and get the right advice from their physicians at the right time.

Cloud-based Digital Libraries

Libraries are an important asset and source of knowledge for medical students, practitioners, and researchers. Cloud-based libraries have massive file storage, query languages, and indexing service.

With cloud-based libraries, medical students and practitioners can access information on demand to stay up to date on the latest research. Additionally, physicians can access useful information to understand the progress in the healthcare industry and improve their work practice.

Advantages and Disadvantages of Cloud Computing in Healthcare:

Advantages of Cloud Computing in Healthcare

Cloud computing in healthcare offers benefits for both healthcare providers and patients. Healthcare providers use the cloud to lower costs and provide personalized medicine. On the other hand, with cloud adoption in healthcare, patients are getting faster delivery of healthcare services and are able to engage more with their care and treatment.

Some of the key benefits of cloud technology in healthcare:

Cost Savings

A big impact of cloud computing on the healthcare industry is that it helps reduce costs. When healthcare institutions manage medical billing systems and [Electronic Medical Records \(EMR\)](#) on-premise, costs increase.

The medical institution must upgrade records, utilize space, and ensure data security. However, with the cloud, healthcare providers can use the resources of cloud computing healthcare companies, so they don't have to purchase and maintain expensive equipment and systems. Healthcare companies can also reduce costs by getting rid of in-house storage.

Because cloud computing is based on a subscription model, cloud providers for healthcare charge you only for the resources you use. By using cloud computing in healthcare, you also don't need large in-house IT teams, which further reduces costs.

Cloud computing in healthcare even boasts indirect cost savings, such as better staff productivity and less system downtime thanks to reliable cloud-based apps.

Customization

Before cloud computing in healthcare, electronic health records required experienced programmers and IT professionals to develop customized solutions. However, cloud-based solutions for healthcare are completely customizable and come with built-in care plans and features. A variety of easy interfaces and templates are available for customized cloud-based solutions.



High Data Storage Capacity

The digital data that hospitals and healthcare practices produce is increasing day-by-day, such as medical records, prescriptions, and lab tests. Storing such data on-premise requires purchasing expensive storage devices.

However, the cloud offers unlimited storage and can efficiently manage large volumes of data. One of the key benefits of cloud storage in healthcare is that you can add more storage, depending on your requirements.

Better Collaboration

Cloud computing in healthcare has made sharing data easy and simple. Cloud computing allows safe, secure, and real-time data sharing with relevant doctors and care providers. Cloud computing also helps with remote conferencing and instant updates on patients' conditions.

Better Patient Experience

Healthcare cloud computing companies improve patient engagement and allow them to access their medical information, doctor's prescription, and test results anytime from anywhere. This allows patients to get a better understanding of their medical conditions. Additionally, quick data sharing via the cloud eliminates unnecessary testing and prevents overprescription because doctors can easily access patients' medical histories.

Disadvantages of Cloud Computing in Healthcare

Challenges of cloud computing in healthcare include:

Security Concerns

Security is one of the major issues with the adoption of the cloud in any industry. In the healthcare industry, it is even more important to keep patients' confidential medical data secure.

This is why it is crucial to choose a secure and reliable cloud provider like [Ridge](#). In addition to being highly secure, Ridge offers the flexibility of the public cloud along with the power of local data centers for cloud computing.

Data Availability

Another concern of healthcare providers regarding cloud computing in healthcare is the lack of control over their data and apps in case the cloud service provider faces a downtime or any other issue. However, with a reliable cloud provider, this isn't usually an issue because the cloud runs on multiple servers with built-in redundancy to reduce downtime.

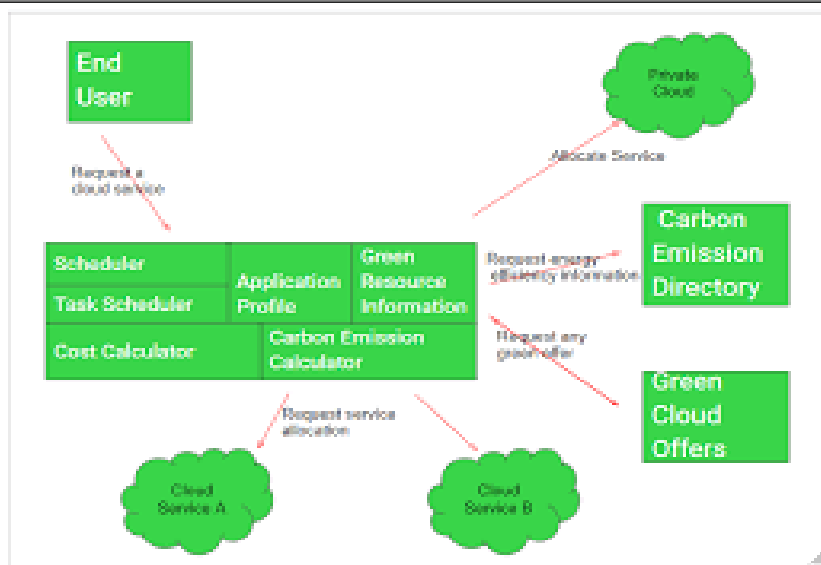
Internet Access

Since data in the cloud can only be accessed via the internet, access to the data can also be limited if there are any internet connectivity issues. This can greatly affect the performance and services of the healthcare institute.

Energy System:

Cloud computing is an internet based services to consumer. Provides computing resources through virtualization over internet. Data centre is the most prominent in cloud computing which contains collection of servers on which business information is stored and applications run.





Data centre consumes more power and release huge carbon dioxide(co2) to environment.

Green computing:

It is the eco-friendly use of computer and their resources.

Defines the study and practices of designing, engineering, manufacturing and disposing computing resources with minimal environmental damage.

Helps in reduces environmental effects and less environment effect.

Precautions/measures to be taken

Use renewable energy sources.

Make the data centre more energy efficient.

Reuse waste heat from computer servers.

All hardware is properly recycle at the end of its life.

Use hardware that has a long lifespan and contains little to no toxic materials.

Now-a-days it companies are ready to share business information.

In order to satisfy the consumer 's exceptions several data centre are established.

Each data centre are contains thousands of servers.

Small amount of workload on servers consumes 50% of power supply.

Provides information continuously to user they need to ON all the time.

Power consumptions was increase to maintain.

Idle servers and overloaded wasted huge amount of energy.

Techniques such as load balancing, vm virtualization, vm migration, resources allocation and job scheduling etc are used to solve problems.

Transporting of data between centre and home computer use more energy to store.

Transportation:

Where we use cloud computing in transportation?

1. Fleet management
2. Freight Tracking
3. E-Ticket Management
4. Passenger data
5. Route planning
6. Real Time Updates





Advantages of Cloud computing in transportation:

1. Reduces operational cost
2. Scalable to access real time data
3. Increased collaboration between employees
4. Improve security
5. Increased efficiency and productivity
6. Eliminate internal hardware
7. Quick ROI

Manufacturing:

Where we use cloud computing in Manufacturing Sector

1. To improve operational efficiency
2. Reduce costs
3. Enable better collaboration across geographically dispersed teams.
4. To aid in planning
5. Executing
6. Managing marketing campaigns

Advantages:

Cost savings: Cloud computing allows manufacturers to reduce hardware, software, and maintenance costs by leveraging cloud-based services instead of investing in on-premises infrastructure.

Improved collaboration: Cloud computing enables manufacturers to share data and collaborate with partners, suppliers, and customers in real-time, leading to faster innovation and improved customer service.

Increased efficiency: Cloud-based automation tools and analytics help manufacturers optimize production processes and supply chain management, leading to improved efficiency, reduced downtime, and better quality control.

Scalability: Cloud computing provides manufacturers with the ability to quickly scale up or down depending on demand, without the need for significant upfront investment.

Enhanced security: Cloud-based solutions offer improved security and data management, enabling manufacturers to protect their intellectual property and comply with industry regulations.

Overall, cloud computing can help manufacturers improve operational efficiency, reduce costs, and increase agility and innovation, making it a valuable tool for staying competitive in today's fast-paced business environment.

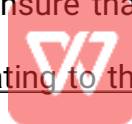
Disadvantages:

Dependence on internet connectivity : Cloud computing requires a reliable and fast internet connection, which can be a challenge in some areas or during periods of high traffic.

Data security concerns: Storing data in the cloud may raise concerns about data security and the potential for data breaches or cyber attacks. Manufacturers need to carefully evaluate the security measures of cloud service providers and take steps to ensure the safety of their data.

Regulatory compliance: Some manufacturing sectors are subject to strict regulations, and compliance requirements may vary depending on the geographic location of the data. Manufacturers need to ensure that their cloud service provider meets the necessary compliance requirements.

Integration issues: Migrating to the cloud may require significant changes to existing IT systems



and processes, which can lead to integration issues and disruptions in operations.

Vendor lock-in: Switching cloud service providers can be difficult and may result in vendor lock-in, making it challenging for manufacturers to switch to a different provider or migrate to a different cloud platform.

Overall, while cloud computing can bring many benefits to the manufacturing sector, it's important to consider these potential drawbacks and carefully evaluate the costs and benefits before making the decision to migrate to the cloud.

Education:

Six cloud computing benefits for education:

In this fast-paced digital era, businesses and organisations across all major industries must embrace contemporary methods and technologies to stay relevant. Similarly, digital transformation in the education sector has the potential to significantly enrich the learning process and place education within the reach of millions of children in remote regions. One of the most accessible modern tools for enterprises of all types and sizes is the cloud. We will explore the various benefits of using cloud computing in education to understand its importance better.

Accessible educational tools

Hosting educational resources on a cloud platform eliminates physical textbooks and makes them easily accessible to students and teachers alike. Teachers can upload course material remotely and students can acquire all the necessary books and modules with a simple device and an internet connection. With every coursework being hosted online, students need not worry about digital storage space for all their learning materials.

Improved collaboration

Enabling real-time collaboration from anywhere in the world, cloud computing for education allows students to work together on assignments without physically present in the same classroom. This lets students who can't attend classes to keep up with their peers, even from home. Teachers can also collaborate online to efficiently share lesson plans or feedback with faculty members across departments and schools.

Better learning facilities

Cloud computing in education enhances connectivity in education for marginalised students who suffer without access to traditional educational facilities. Students from rural villages can rely on a cloud-based learning system to gain education and acquire the tools needed to succeed in today's world. While it is slightly far-fetched to assume that the connectivity and equipment can indeed reach such regions, it is worth the investment. Modernising the education system will also benefit working professionals who lack time to attend regular classes and enable them to participate in online classes at convenient times.

Flexible and efficient processes

Incorporating cloud computing in the education sector can save time and effort for both the faculty and the students due to greater flexibility. Processes that once required both parties to be present at the school, college, or university can now be completed remotely and in a fraction of the time. Students can leverage modern cloud-based applications and platforms to learn at home and at their own pace, and several students spread across various locations can learn from a single teacher. Similarly, teachers aren't restricted to school working hours and have more flexibility when performing their duties.

Cost efficiency

Properly implemented, cloud computing education systems can cost less than a traditional education from a renowned institution. Faculty, institutions, and students also need not spend on the latest hardware as cloud-based and cloud-native applications use the cloud for processing power and can be easily run on even the most basic devices. Minimising paper use also saves resources, eliminating printers, photocopiers, and physical storage space.

Data security



With cloud services comes the added benefit of using cloud servers to store information extremely securely. Not only does this eliminate the need for maintaining physical in-house servers for data storage, but the information stored in the cloud is always backed-up and safe in any unforeseen circumstances, from power outages to natural disasters.

A third-party cloud services provider can effectively manage an educational institution's data and IT resources on off-site cloud servers. It also affords faculty and students on-demand data access at any time and across multiple locations. More and more educational institutions are taking note and are directing time and resources towards modernising their systems. As a result, global cloud computing in the education industry held a market value of close to \$2.2 billion in 2020 and could hit \$8.7 billion by 2027. With the future of cloud computing in the education sector looking up, now is clearly the time for educational institutions to employ the modern tools at their disposal.

For organisations on the digital transformation journey, agility is key in responding to a rapidly changing technology and business landscape. Now more than ever, it is crucial to deliver and exceed organisational expectations with a robust digital mindset backed by innovation. Enabling businesses to sense, learn, respond, and evolve like living organisms will be imperative for business excellence. A comprehensive yet modular suite of services is doing precisely that. Equipping organisations with intuitive decision-making automatically at scale, actionable insights based on real-time solutions, anytime/anywhere experience, and in-depth data visibility across functions leading to hyper-productivity, [Live Enterprise](#) is building connected organisations that are innovating collaboratively for the future.

Government:

1. Indian Railways
2. Kisan Suvidha
3. DigiLocker
4. eHospital

Cloud computing has ensured the success of national initiatives and schemas such as Swachh Bharat Mission, National Scholarship, My-Gov and e-Transport.

Cloud computing increases effectiveness and efficiency by improving employee satisfaction and retention.

Cloud computing is a distributed computing paradigm that focuses on providing a wide range of users with distributed access to virtualized hardware and software infrastructure over the internet.

Cloud computing is one of the latest ICT innovations, offering many advantages and it is a fundamental part of government strategy.

Government at all levels, from federal to state and local, are shifting to flexible hybrid, multi-cloud computing to provide better and faster services to citizens with flexible, scalable and cost-effective technology.

Cloud computing is useful in government such as:

Provides on-line services to citizens (agriculture land records, rural services, municipal services etc).

Deals with businesses using ICTs (for taxation, policy enforcement).

Examples of crucial government services on cloud

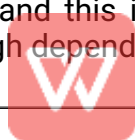
- IT consolidation
- Shared services

Advantages:

Flexibility: Cloud computing offers government agencies more flexibility than traditional IT infrastructures. Using the cloud makes it easy to add and change services without hassle of adding or removing digital space. This flexibility is useful during risk management procedures.

Disadvantages:

Downtime: Cloud computing services only through the internet. When there is an internet outage, services get interrupted and this increases downtime. Therefore, one of the major criticisms of cloud computing is its high dependency on the internet.



Security and Privacy: Inadequate cloud security measures lead to data leakage over cloud networks which can result in intellectual property theft, contract breaches and malware attacks.

Mobile Communication:

Mobile cloud computing technology consists of cloud computing and mobile computing combined on a wireless network. This innovative technology makes it possible to deliver and execute high-quality applications on mobile devices, regardless of the latter's operating system, storage capability, and computing tasks.

Mobile cloud computing offers a greater degree of speed and flexibility for end-users and applications developers. The mobile cloud encompasses cloud-based data and the applications and services created for mobile devices. It brings together mobile application development and cloud-based services, facilitating the delivery of cloud services and apps to mobile users. Remote data centers store the relevant data and run the apps.

Mobile cloud computing is a natural progression of mobile and cloud computing technology and has proven to be a game-changer in using our mobile devices, whether for work or leisure. Just stop and consider how many people use their mobile devices on any given day. That heavy usage gives you some idea of how popular mobile cloud computing is and [why it will only become](#) more widespread in the future.

Benefits of Mobile Cloud Computing:

We've lightly touched upon some of the advantages of [cloud computing](#); let's get into some solid details.

- Mobile cloud computing is fast and flexible. As a result, mobile cloud computing makes it easy for developers to create and share mobile app resources with end-users. Therefore, mobile applications can be built and updated faster.
- Mobile cloud computing shares resources. Mobile apps that run off the cloud aren't constrained by any mobile device's processing and storage limitations. All data-intensive processes can run from the cloud. This advantage means that any mobile device with access to a network can use mobile cloud apps, regardless of the operating system. Thus, users can enjoy cloud computing with Android or OS devices.
- Mobile cloud computing uses integrated data. Mobile cloud computing lets users securely and quickly collect and integrate information from many sources, no matter where the data is.

Applications of mobile communication:

Social Media

Let's begin with what is unquestionably the most popular application. It seems like everybody and their extended family is on Facebook, Twitter, Instagram, TikTok, or a host of other social media platforms. Mobile social networking lets users store data and share videos in real-time, so your friends and followers don't have to wait for that new cat video!

Email

Here's another extremely popular mobile cloud computing use. However, it often seems that people are more likely to log onto their social media accounts every morning rather than open their new emails. Blame the endless spam floods for that phenomenon. However, we're probably using apps like Outlook, Gmail, or Yahoo Mail for those of us who do read our emails. So if you're reading your emails on a mobile device such as your smartphone, that's mobile cloud technology in action.

Finance and Commerce

Do you use your phone to monitor your checking account balances with a banking app? Do you make purchases on eBay or Amazon using your tablet? If so, that's mobile cloud computing again, offering scalable processing power.

Healthcare

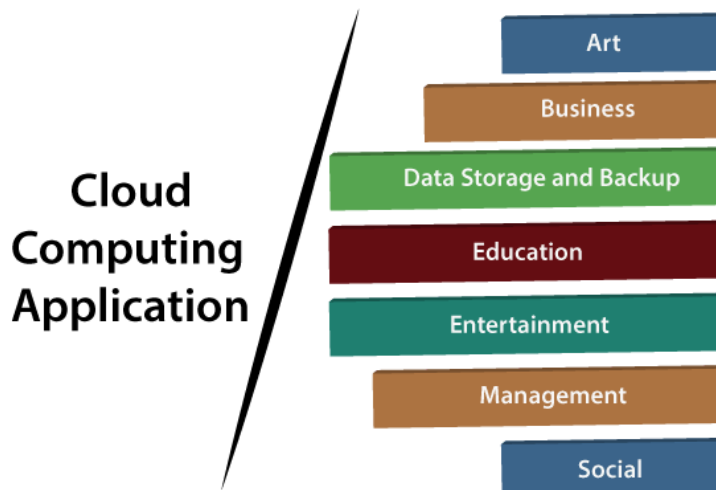
While this application is not as common among the public, it's absolutely essential for healthcare workers. Thanks to mobile cloud computing, huge amounts of data can be instantaneously stored or accessed with a mobile device. Healthcare professionals can instantly access patient records, an especially important feature for referrals. A specialist needs to see a patient's medical history and why they were referred in the first place.

Application Development

Cloud service providers provide various applications in the field of art, business, data storage and backup services, education, entertainment, management, social networking, etc.



The most widely used cloud computing applications are given below -



1. Art Applications

Cloud computing offers various art applications for quickly and easily design **attractive cards, booklets, and images**. Some most commonly used cloud art applications are given below:

MOO:

i. Moo is one of the best cloud art applications. It is used for designing and printing business cards, postcards, and mini cards.

ii. Vistaprint

Vistaprint allows us to easily design various printed marketing products such as business cards, Postcards, Booklets, and wedding invitations cards.

iii. Adobe Creative Cloud

Adobe creative cloud is made for designers, artists, filmmakers, and other creative professionals. It is a suite of apps which includes PhotoShop image editing programming, Illustrator, InDesign, TypeKit, Dreamweaver, XD, and Audition.

2. Business Applications

Business applications are based on cloud service providers. Today, every organization requires the cloud business application to grow their business. It also ensures that business applications are 24*7 available to users.

There are the following business applications of cloud computing -

i. MailChimp

MailChimp is an **email publishing platform** which provides various options to **design, send, and save** templates for emails.

ii. Salesforce

Salesforce platform provides tools for sales, service, marketing, e-commerce, and more. It also provides a cloud development platform.

iii. Chatter

Chatter helps us to **share important information** about the organization in real time

iv. Bitrix24

Bitrix24 is a **collaboration** platform which provides communication, management, and social collaboration tools.

v. Paypal

Paypal offers the simplest and easiest **online payment** mode using a secure internet account. Paypal accepts the payment through debit cards, credit cards, and also from Paypal account holders.

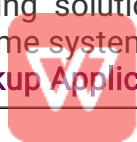
vi. Slack

Slack stands for **Searchable Log of all Conversation and Knowledge**. It provides a **user-friendly** interface that helps us to create public and private channels for communication.

vii. Quickbooks

Quickbooks works on the terminology "**Run Enterprise anytime, anywhere, on any device.**" It provides online accounting solutions for the business. It allows more than 20 users to work simultaneously on the same system.

3. Data Storage and Backup Applications



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Cloud computing allows us to store information (data, files, images, audios, and videos) on the cloud and access this information using an internet connection. As the cloud provider is responsible for providing security, so they offer various backup recovery application for retrieving the lost data.

A list of data storage and backup applications in the cloud are given below -

i. Box.com

Box provides an online environment for **secure content management, workflow, and collaboration**. It allows us to store different files such as Excel, Word, PDF, and images on the cloud. The main advantage of using box is that it provides drag & drop service for files and easily integrates with Office 365, G Suite, Salesforce, and more than 1400 tools.

ii. Mozy

Mozy provides powerful **online backup solutions** for our personal and business data. It schedules automatically back up for each day at a specific time.

iii. Joukuu

Joukuu provides the simplest way to **share and track cloud-based backup files**. Many users use joukuu to search files, folders, and collaborate on documents.

iv. Google G Suite

Google G Suite is one of the best **cloud storage and backup** application. It includes Google Calendar, Docs, Forms, Google+, Hangouts, as well as cloud storage and tools for managing cloud apps. The most popular app in the Google G Suite is Gmail. Gmail offers free email services to users.

4. Education Applications

Cloud computing in the education sector becomes very popular. It offers various **online distance learning platforms** and **student information portals** to the students. The advantage of using cloud in the field of education is that it offers strong virtual classroom environments, Ease of accessibility, secure data storage, scalability, greater reach for the students, and minimal hardware requirements for the applications.

There are the following education applications offered by the cloud -

i. Google Apps for Education

Google Apps for Education is the most widely used platform for free web-based email, calendar, documents, and collaborative study.

ii. Chromebooks for Education

Chromebook for Education is one of the most important Google's projects. It is designed for the purpose that it enhances education innovation.

iii. Tablets with Google Play for Education

It allows educators to quickly implement the latest technology solutions into the classroom and make it available to their students.

iv. AWS in Education

AWS cloud provides an education-friendly environment to universities, community colleges, and schools.

5. Entertainment Applications

Entertainment industries use a **multi-cloud strategy** to interact with the target audience. Cloud computing offers various entertainment applications such as online games and video conferencing.

i. Online games

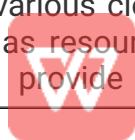
Today, cloud gaming becomes one of the most important entertainment media. It offers various online games that run remotely from the cloud. The best cloud gaming services are Shaow, GeForce Now, Vortex, Project xCloud, and PlayStation Now.

ii. Video Conferencing Apps

Video conferencing apps provides a simple and instant connected experience. It allows us to communicate with our business partners, friends, and relatives using a cloud-based video conferencing. The benefits of using video conferencing are that it reduces cost, increases efficiency, and removes interoperability.

6. Management Applications

Cloud computing offers various cloud management tools which help admins to manage all types of cloud activities, such as **resource deployment, data integration, and disaster recovery**. These management tools also provide administrative control over the platforms, applications, and



infrastructure.

Some important management applications are -

i. Toggl

Toggl helps users to track allocated time period for a particular project.

ii. Evernote

Evernote allows you to sync and save your recorded notes, typed notes, and other notes in one convenient place. It is available for both free as well as a paid version.

It uses platforms like Windows, macOS, Android, iOS, Browser, and Unix.

iii. Outright

Outright is used by management users for the purpose of accounts. It helps to track income, expenses, profits, and losses in real-time environment.

iv. GoToMeeting

GoToMeeting provides **Video Conferencing** and **online meeting apps**, which allows you to start a meeting with your business partners from anytime, anywhere using mobile phones or tablets. Using GoToMeeting app, you can perform the tasks related to the management such as join meetings in seconds, view presentations on the shared screen, get alerts for upcoming meetings, etc.

7. Social Applications

Social cloud applications allow a large number of users to connect with each other using social networking applications such as **Facebook, Twitter, LinkedIn**, etc.

There are the following cloud based social applications -

i. Facebook

Facebook is a **social networking website** which allows active users to share files, photos, videos, status, more to their friends, relatives, and business partners using the cloud storage system. On Facebook, we will always get notifications when our friends like and comment on the posts.

ii. Twitter

Twitter is a **social networking** site. It is a **microblogging** system. It allows users to follow high profile celebrities, friends, relatives, and receive news. It sends and receives short posts called tweets.

iii. Yammer

Yammer is the **best team collaboration** tool that allows a team of employees to chat, share images, documents, and videos.

iv. LinkedIn

LinkedIn is a **social network** for students, freshers, and professionals.



UNIT-II

CLOUD COMPUTING ARCHITECTURE

INTRODUCTION:

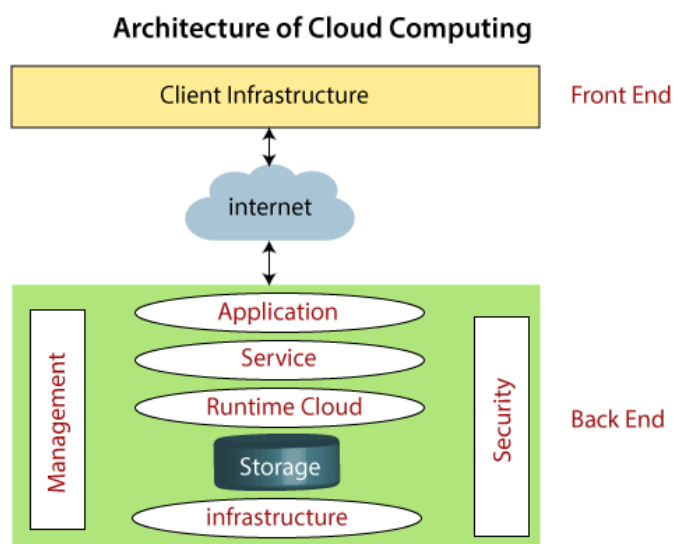
As we know, cloud computing technology is used by both small and large organizations to store the information in cloud and access it from anywhere at anytime using the internet connection.

Cloud computing architecture is a combination of service-oriented architecture and event-driven architecture.

Cloud computing architecture is divided into the following two parts -

- o Front End
- o Back End

The below diagram shows the architecture of cloud computing -

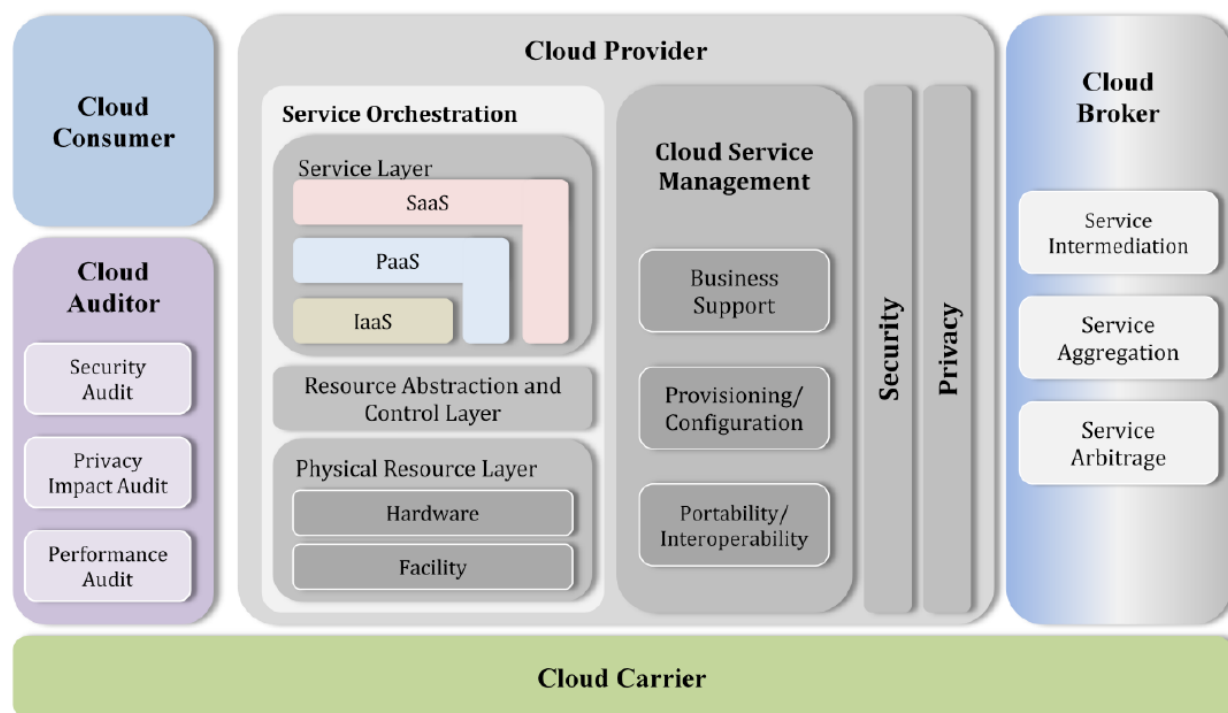


Front End

The front end is used by the client. It contains client-side interfaces and applications that are required to access the cloud computing platforms. The front end includes web servers (including Chrome, Firefox, internet explorer, etc.), thin & fat clients, tablets, and mobile devices.

Back End

The back end is used by the service provider. It manages all the resources that are required to provide cloud computing services. It includes a huge amount of data storage, security mechanism, virtual machines, deploying models, servers, traffic control mechanisms, etc.



1. Cloud Consumer: is a person or organization that maintains a business relationship with, and uses service from, cloud providers. Anyone who purchases a cloud service is a consumer, and within the consumer there could be an array of roles responsible for configuring and managing the resources from the cloud provider depending on the services obtained.

2. Cloud Provider: is a person, organization, or entity responsible for making a [service available](#) to interested parties. A [cloud provider](#) would have a significant number of roles responsible for the management of its cloud resources including those responsible for selling, onboarding, configuring and supporting cloud services for its consumers.

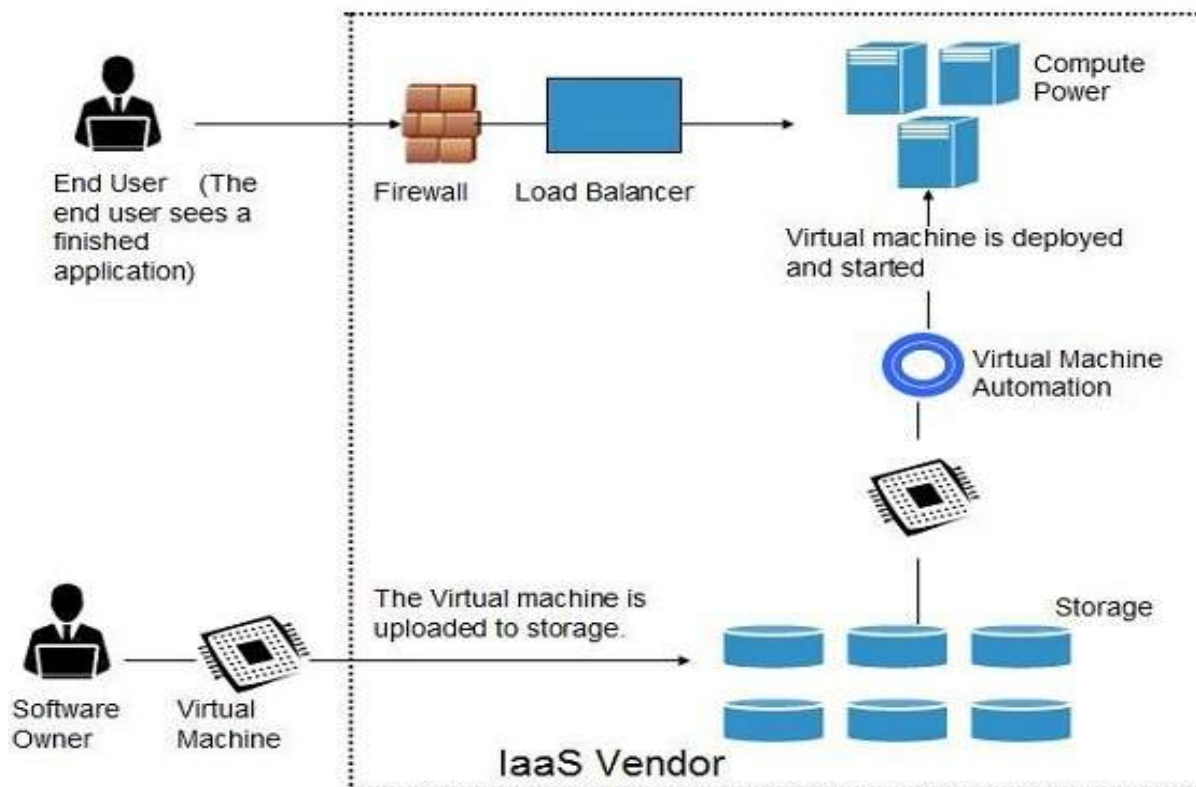
3. Cloud Auditor: is a party that can conduct independent assessment of [cloud services](#), information system operations, performance, and security of the cloud implementation. Generally, cloud auditors are categorized based on intent. For the most part, their focus is on risk and compliance, especially around [information security](#). Other auditors can provide advisory services especially to consumers looking to cut down their bills or raise the level of efficiency in the resources consumed.

4. Cloud Broker: is any entity that manages the use, performance, and delivery of cloud services, and negotiates relationships between Cloud Providers and Cloud Consumers. Cloud brokers support consumers to get value for money by playing the advisory role especially for consumers who have a hybrid mix of resources from multiple providers.

5. Cloud Carrier: is an intermediary that provides connectivity and transport of cloud services from Cloud Providers to Cloud Consumers. Most ISPs have taken the role of cloud carriers as they provide the requisite bandwidth needed to connect consumers with providers as well as capabilities that support the connectivity.

INFRASTRUCTURE / HARDWARE AS A SERVICE:





IaaS is also known as Hardware as a Service (HaaS). It is one of the layers of the cloud computing platform. It allows customers to outsource their IT infrastructures such as servers, networking, processing, storage, virtual machines, and other resources. Customers access these resources on the Internet using a pay-as-per use model.

In traditional hosting services, IT infrastructure was rented out for a specific period of time, with pre-determined hardware configuration. The client paid for the configuration and time, regardless of the actual use. With the help of the IaaS cloud computing platform layer, clients can dynamically scale the configuration to meet changing requirements and are billed only for the services actually used.

IaaS cloud computing platform layer eliminates the need for every organization to maintain the IT infrastructure.

IaaS is offered in three models: public, private, and hybrid cloud. The private cloud implies that the infrastructure resides at the customer-premise. In the case of public cloud, it is located at the cloud computing platform vendor's data center, and the hybrid cloud is a combination of the two in which the customer selects the best of both public cloud or private cloud.

IaaS provider provides the following services -

1. Compute: Computing as a Service includes virtual central processing units and virtual main memory for the Vms that is provisioned to the end- users.
2. Storage: IaaS provider provides back-end storage for storing files.
3. Network: Network as a Service (NaaS) provides networking components such as routers, switches, and bridges for the Vms.
4. Load balancers: It provides load balancing capability at the infrastructure layer.





Advantages of IaaS cloud computing layer

There are the following advantages of IaaS computing layer -

1. Shared infrastructure

IaaS allows multiple users to share the same physical infrastructure.

2. Web access to the resources

IaaS allows IT users to access resources over the internet.

3. Pay-as-per-use model

IaaS providers provide services based on the pay-as-per-use basis. The users are required to pay for what they have used.

4. Focus on the core business

IaaS providers focus on the organization's core business rather than on IT infrastructure.

5. On-demand scalability

On-demand scalability is one of the biggest advantages of IaaS. Using IaaS, users do not worry about to upgrade software and troubleshoot the issues related to hardware components.

Disadvantages of IaaS cloud computing layer

1. Security

Security is one of the biggest issues in IaaS. Most of the IaaS providers are not able to provide 100% security.

2. Maintenance & Upgrade

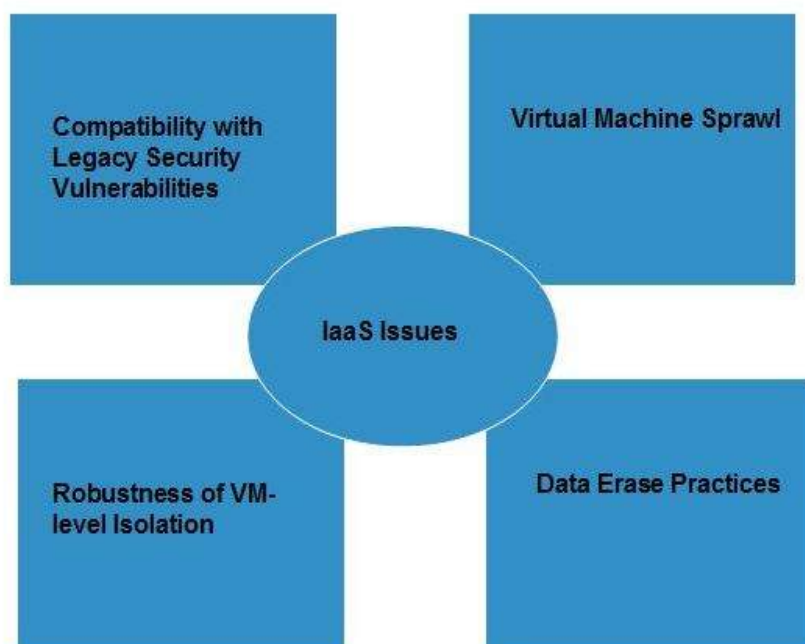
Although IaaS service providers maintain the software, but they do not upgrade the software for some organizations.

3. Interoperability issues



It is difficult to migrate VM from one IaaS provider to the other, so the customers might face problem related to vendor lock-in.

ISSUES OF IAAS:



- IaaS provides access to fundamental resources such as physical machines, virtual machines, virtual storage, etc., a part from these resources.
- The IaaS also offer
 1. Virtual Machine Disk Storage
 2. Virtual Local Area Network
 3. Load Balancers
 4. IP Addresses
 5. Software Bundles.
- All the above resources are made available to end user via Server Virtualization.

CHARACTERISTICS OF IAAS:

1. Automated administrative tasks
2. Dynamic scaling
3. Platform virtualization technology
4. GUI and API-based access
5. Internet connectivity

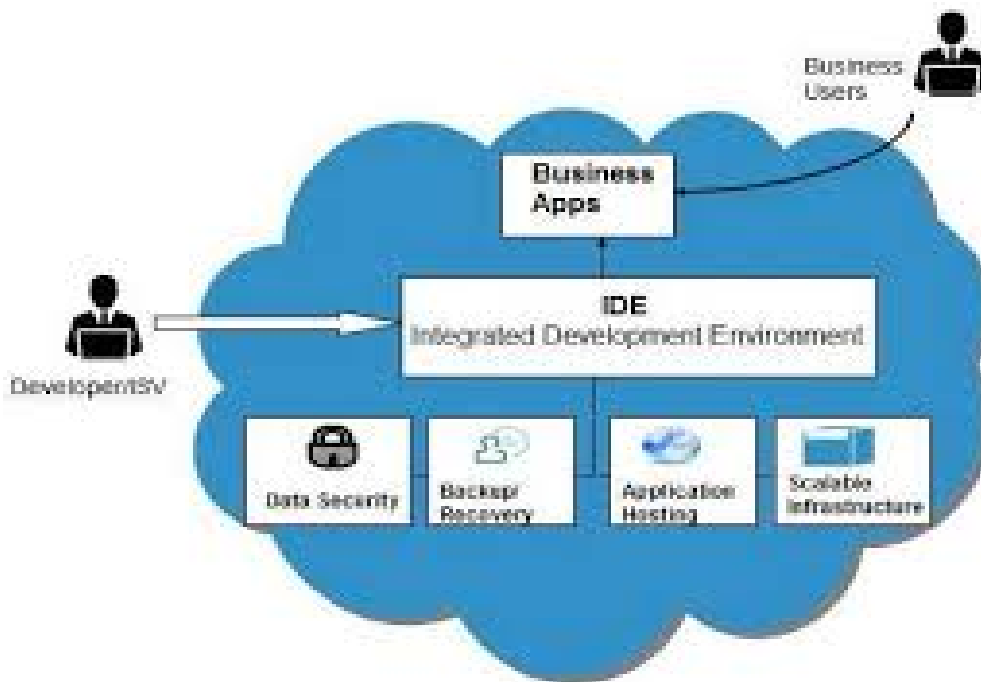
BENEFITS OF IAAS:

- IaaS allow the cloud provider to freely locate the infrastructure over the internet in a cost effective manner.
- Some of the Keybenefits of IaaS as list below.
 1. Full Control of the computing resources through administrative access to virtual machine.
 2. Flexible and Effectient remoting of computer hardware.
 3. Portability, Interoperability with legacy application.

PLATFORM AS A SERVICE:



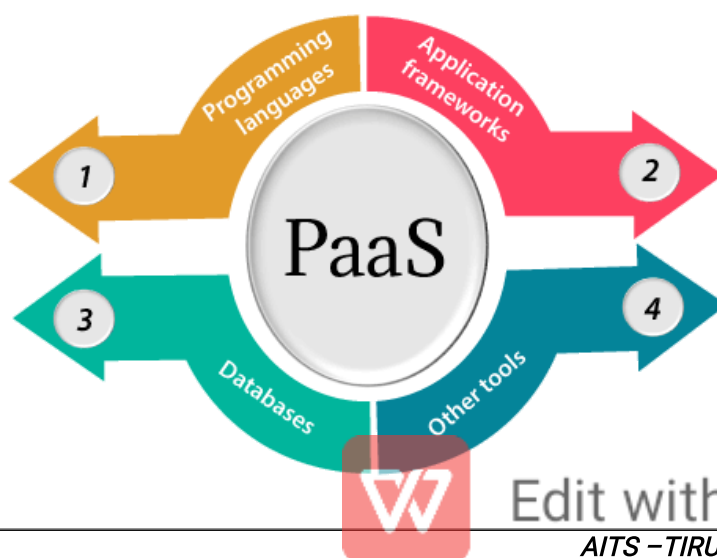
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- Platform as a Service (PaaS) provides a runtime environment. It allows programmers to easily create, test, run, and deploy web applications. You can purchase these applications from a cloud service provider on a pay-as-per use basis and access them using the Internet connection. In PaaS, back end scalability is managed by the cloud service provider, so end-users do not need to worry about managing the infrastructure.
- PaaS includes infrastructure (servers, storage, and networking) and platform (middleware, development tools, database management systems, business intelligence, and more) to support the web application life cycle.
- It offers the runtime environments for applications. It refers to pass. It also offers development and deployment tools required to develop applications.
- PaaS has as feature of point and click tools that enables not developers to create web applications.
- App Engine of google and force.com are example of paas offering vendors.
- Developer may login to this websites and use the built in api to create web based applications.

Example: Google App Engine, Force.com, Joyent, Azure.

PaaS providers provide the Programming languages, Application frameworks, Databases, and Other tools:



1. Programming languages

PaaS providers provide various programming languages for the developers to develop the applications. Some popular programming languages provided by PaaS providers are Java, PHP, Ruby, Perl, and Go.

2. Application frameworks

PaaS providers provide application frameworks to easily understand the application development. Some popular application frameworks provided by PaaS providers are Node.js, Drupal, Joomla, WordPress, Spring, Play, Rack, and Zend.

3. Databases

PaaS providers provide various databases such as ClearDB, PostgreSQL, MongoDB, and Redis to communicate with the applications.

4. Other tools

PaaS providers provide various other tools that are required to develop, test, and deploy the applications.

Advantages of PaaS

There are the following advantages of PaaS -

1) Simplified Development

PaaS allows developers to focus on development and innovation without worrying about infrastructure management.

2) Lower risk

No need for up-front investment in hardware and software. Developers only need a PC and an internet connection to start building applications.

3) Prebuilt business functionality

Some PaaS vendors also provide already defined business functionality so that users can avoid building everything from very scratch and hence can directly start the projects only.

4) Instant community

PaaS vendors frequently provide online communities where the developer can get the ideas to share experiences and seek advice from others.

5) Scalability

Applications deployed can scale from one to thousands of users without any changes to the applications.

Disadvantages of PaaS cloud computing layer

1) Vendor lock-in



One has to write the applications according to the platform provided by the PaaS vendor, so the migration of an application to another PaaS vendor would be a problem.

2) Data Privacy

Corporate data, whether it can be critical or not, will be private, so if it is not located within the walls of the company, there can be a risk in terms of privacy of data.

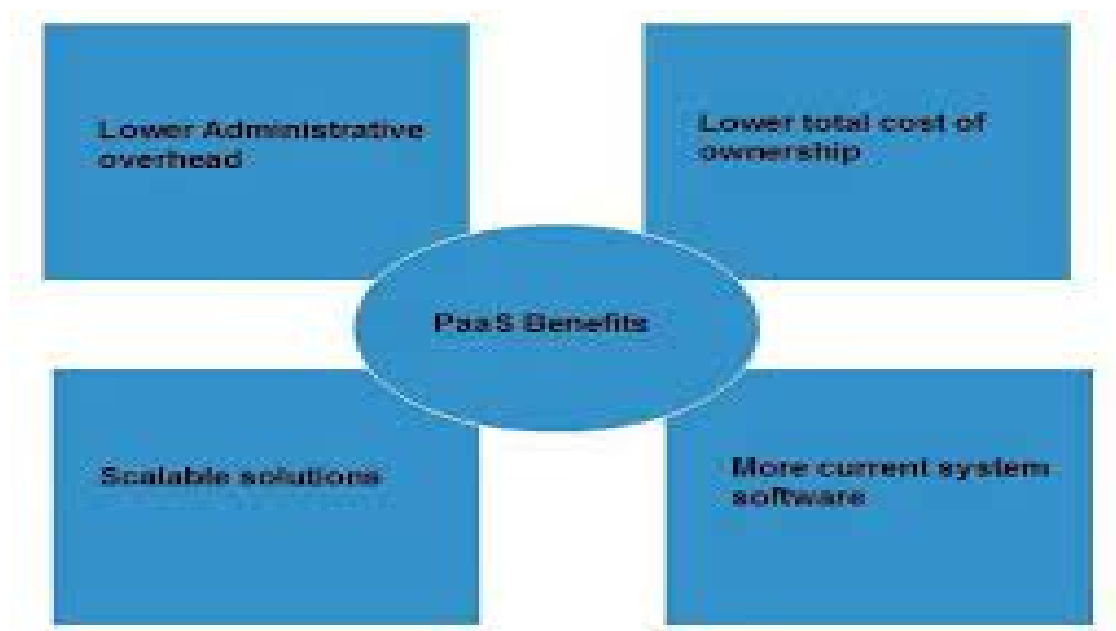
3) Integration with the rest of the systems applications

It may happen that some applications are local, and some are in the cloud. So there will be chances of increased complexity when we want to use data which in the cloud with the local data.

CHARACTERISTICS OF PaaS:

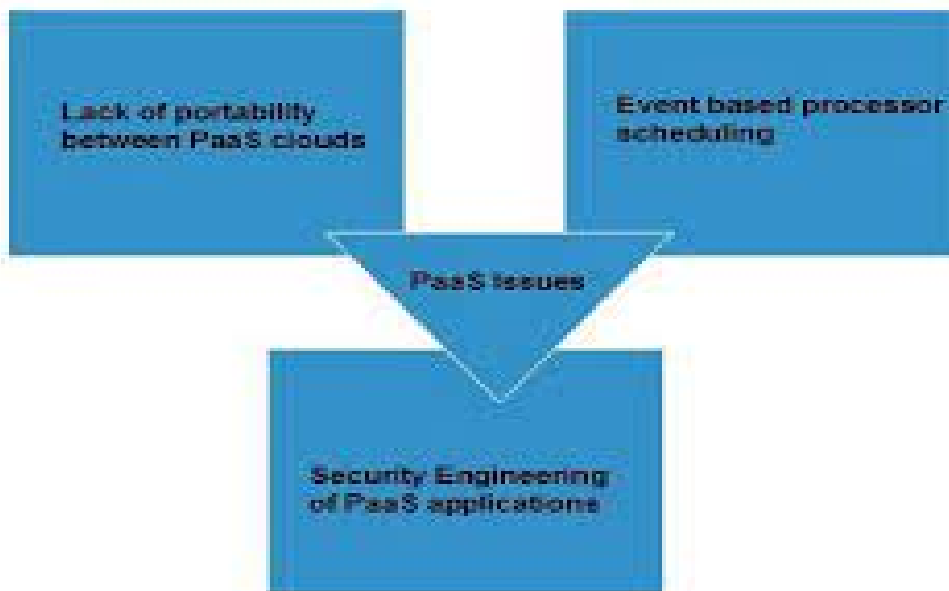
- PaaS offers browser based development environment. It allows the developer to create database and edit the application code either via application programming interface or point and click tools.
- It provides built in security, scalability and web service interfaces.
- It also provides built in tools for defining work flow, approval processes and business rules.
- It is easy to integrate PaaS with other applications on the same platform.
- PaaS also provides web service interfaces that allows us to connect the applications outside the platform.

BENEFITS OF PaaS:



ISSUES OF PaaS:





SOFTWARE AS A SERVICE:

- These Model allows us to provide software application as a service to the end users.
- It refers to a software that it deployed on a high accessible via internet.

APPLICATIONS OF SAAS:

1. Billing and invoicing system
 2. Customer Relationship Management application.
 3. Help Desk application.
 4. Human Resource Solutions.
- Some of the SAAS application are not customizable such as Microsoft Office suite but SAAS provides us API. Which allows the developer to develop a customized application.

CHARACTERISTICS OF SAAS:

1. It make the software available over the internet.
2. The software application are maintained by the vendor.
3. The license to the software may be subscription based or usage based and it is build on recurring bases.
4. SAAS application are cost effective as they don't require any maintaince at end user side.
5. They are available on demand.
6. They can be scaled up are down on demand.
7. They are automatically upgraded and updated.
8. All users run the same version on the software.
9. It offers shared data model.

Therefore multiple users can share single instance of infrastructure.It is not required to hardcore the functionalities for individual users.

BENEFITS OF SAAS;



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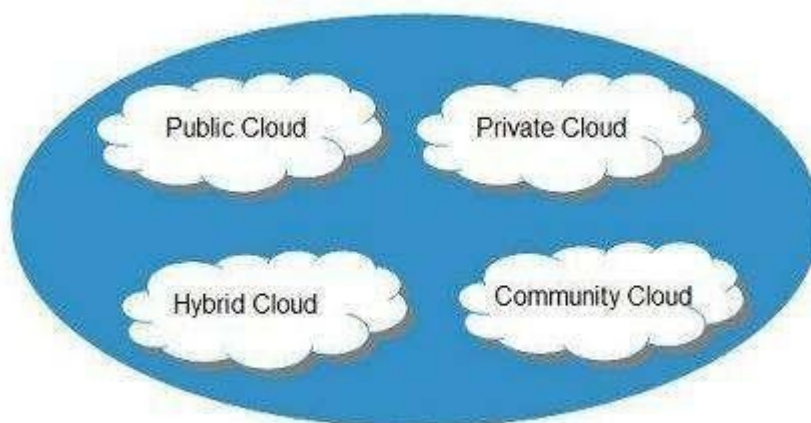
1. Efficient use of software licences.
2. Centralized management and data.
3. Platform responsibilities managed by provider.
4. Multi Tenant solutions.

ISSUES OF SAAS:

1. Browser based risks.
2. Network Dependency.
3. Lack of Portability between SAAS cloud.

TYPES OF CLOUD:

Cloud can have any of the four types of access: Public, Private, Hybrid, and Community.



1. PUBLIC CLOUD

The **public cloud** allows systems and services to be easily accessible to the general public. Public cloud may be less secure because of its openness.

2. PRIVATE CLOUD

The **private cloud** allows systems and services to be accessible within an organization. It is more secured because of its private nature.

3. COMMUNITY CLOUD

The **community cloud** allows systems and services to be accessible by a group of organizations.

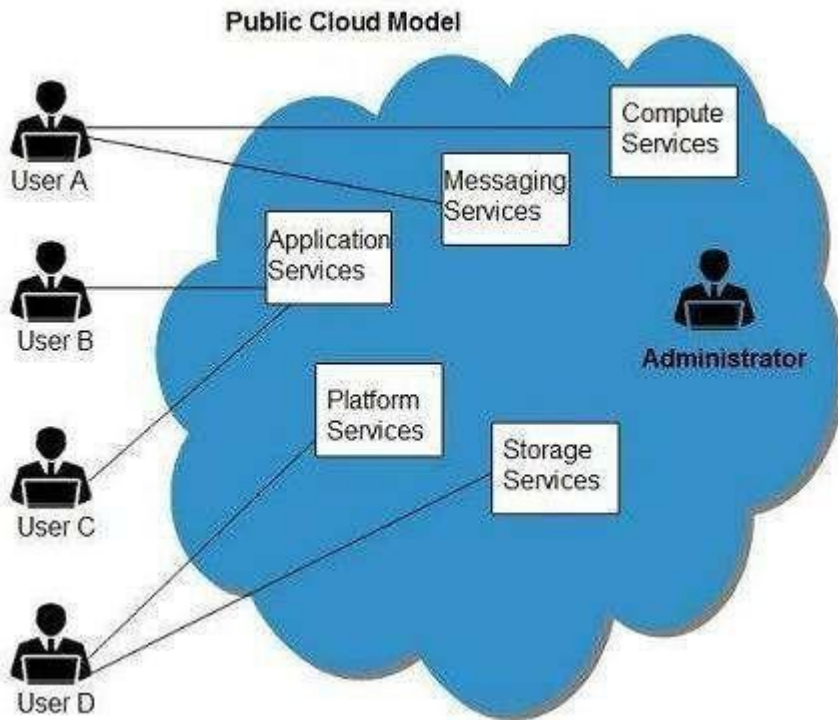
4. HYBRID CLOUD

The **hybrid cloud** is a mixture of public and private cloud, in which the critical activities are performed using private cloud while the non-critical activities are performed using public cloud.

Public cloud

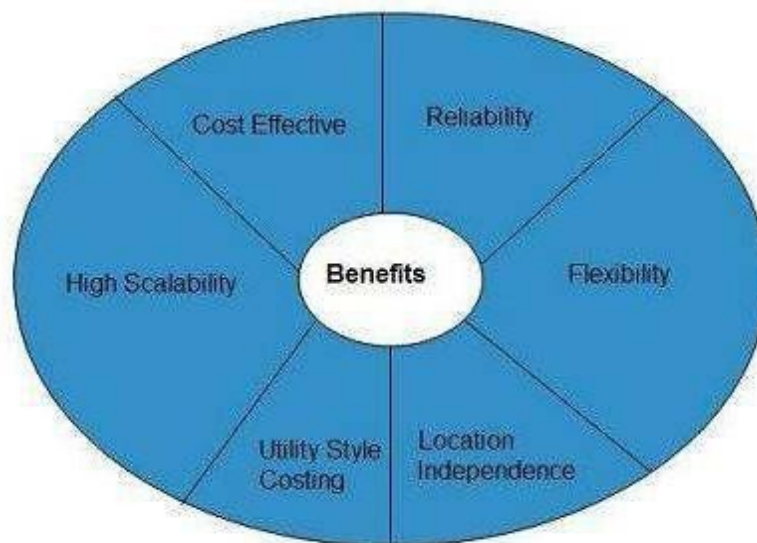
Public Cloud allows systems and services to be easily accessible to general public. The IT giants such as **Google**, **Amazon** and **Microsoft** offer cloud services via Internet. The Public Cloud Model is shown in the diagram below.





Benefits

There are many benefits of deploying cloud as public cloud model. The following diagram shows some of those benefits:



Cost Effective

Since **public cloud** shares same resources with large number of customers it turns out inexpensive.

Reliability

The **public cloud** employs large number of resources from different locations. If any of the resources fails, public cloud can employ another one.

Flexibility

The public cloud can smoothly integrate with private cloud, which gives customers a flexible approach.

Location Independence

Public cloud services are delivered through Internet, ensuring location independence.

Utility Style Costing

Public cloud is also based on **pay-per-use** model and resources are accessible whenever customer needs them.

High Scalability

Cloud resources are made available on demand from a pool of resources, i.e., they can be scaled up or down according to the requirement.

Disadvantages

Here are some disadvantages of public cloud model:

Low Security

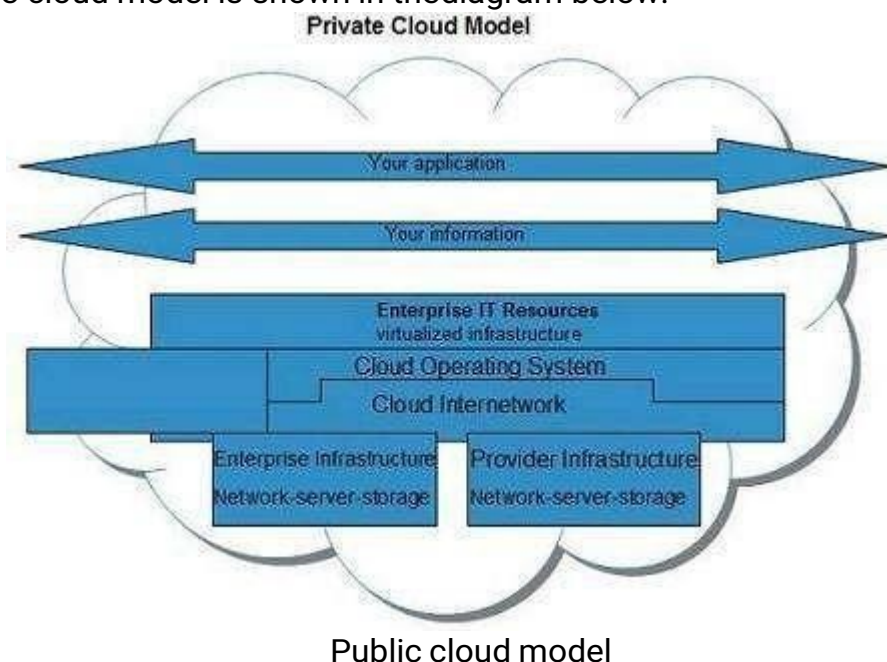
In **public cloud model**, data is hosted off-site and resources are shared publicly, therefore it does not ensure a higher level of security.

Less Customizable

It is comparatively less customizable than private cloud.

3.3. Private cloud

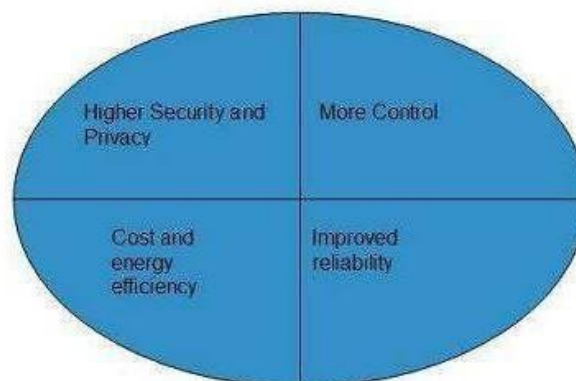
Private Cloud allows systems and services to be accessible within an organization. The Private Cloud is operated only within a single organization. However, it may be managed internally by the organization itself or by a third party. The private cloud model is shown in the diagram below.



Public cloud model

Benefits

There are many benefits of deploying cloud as private cloud model. The following diagram shows some of those benefits:



High Security and Privacy

Private cloud operations are not available to general public and resources are shared from a distinct pool of resources. Therefore, it ensures high **security** and **privacy**.

More Control

The **private cloud** has more control on its resources and hardware than public



cloud because it is accessed only within an organization.

Cost and Energy Efficiency

The **private cloud** resources are not as cost effective as resources in public clouds but they offer more efficiency than public cloud resources.

Disadvantages

Here are the disadvantages of using private cloud model:

Restricted Area of Operation

The private cloud is only accessible locally and is very difficult to deploy globally.

High Priced

Purchasing new hardware in order to fulfill the demand is a costly transaction.

Limited Scalability

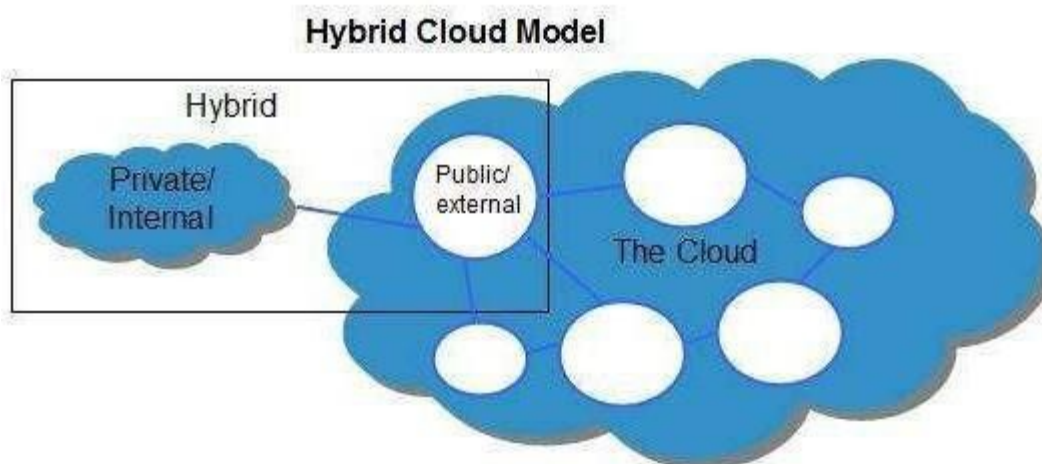
The private cloud can be scaled only within capacity of internal hosted resources.

Additional Skills

In order to maintain cloud deployment, organization requires skilled expertise.

Hybrid cloud

Hybrid Cloud is a mixture of **public** and **private cloud**. Non-critical activities are performed using public cloud while the critical activities are performed using private cloud. The Hybrid Cloud Model is shown in the diagram below.



Hybrid cloud model

Benefits

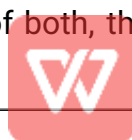
There are many benefits of deploying cloud as hybrid cloud model. The following diagram shows some of those benefits:



Scalability

It offers features of both, the public cloud scalability and the private cloud scalability.

Flexibility



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It offers secure resources and scalable public resources.

Cost Efficiency

Public clouds are more cost effective than private ones. Therefore, hybrid clouds can be cost saving.

Security

The private cloud in hybrid cloud ensures higher degree of security.

Disadvantages

Networking Issues

Networking becomes complex due to presence of private and public cloud.

Security Compliance

It is necessary to ensure that cloud services are compliant with security policies of the organization.

Infrastructure Dependency

The hybrid cloud model is dependent on internal IT infrastructure, therefore it is necessary to ensure redundancy across data centers.

Community cloud

Community Cloud allows system and services to be accessible by group of organizations. It shares the infrastructure between several organizations from a specific community. It may be managed internally by organizations or by the third-party. The Community Cloud Model is shown in the diagram below.

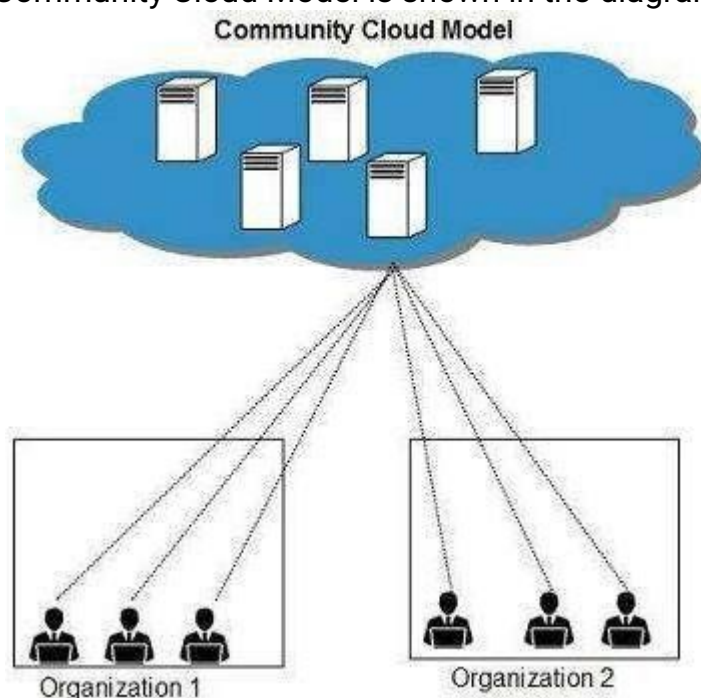


Fig: Community cloud model

Benefits

There are many benefits of deploying cloud as community cloud model.





Cost Effective

Community cloud offers same advantages as that of private cloud at low cost.

Sharing Among Organizations

Community cloud provides an infrastructure to share cloud resources and capabilities among several organizations.

Security

The community cloud is comparatively more secure than the public cloud but less secured than the private cloud.

Issues

Since all data is located at one place, one must be careful in storing data in community cloud because it might be accessible to others.

It is also challenging to allocate responsibilities of governance, security and cost among organizations.

ECONOMICS OF THE CLOUD:

Economics of [Cloud Computing](#) is based on the PAY AS YOU GO method. Users/Customers must have to pay only for their way of the usage of the cloud services. It is definitely beneficial for the users. So the Cloud is economically very convenient for all. Another side is to eliminate some indirect costs which is generated by assets such as license of the software and their support. In the cloud, users can use software applications on a subscription basis without any cost because the property of the software providing service remains to the cloud provider.

Economical background of the cloud is more useful for developers in the following ways:

- Pay as you go model offered by cloud providers.
- Scalable and Simple.

Cloud Computing Allows:

- Reduces the capital costs of infrastructure.
- Removes the maintenance cost.
- Removes the administrative cost.

What is Capital Cost?

It is cost occurred in the purchasing infrastructure or the assets that is important in the production of goods. It takes a long time to generate profit.

In the case of start-ups, there is no extra budget for the infrastructure and its maintenance. So cloud can minimize expenses of any small organization in terms of economy. It leads to the developers can only focus on the development logic and not on the maintenance of the infrastructure.



There are three different **Pricing Strategies** that are introduced by Cloud Computing: Tiered Pricing, Per-unit Pricing, and Subscription-based Pricing. These are explained as following below.

1. **Tiered Pricing:** Cloud Services are offered in the various tiers. Each tier offers to fix service agreements at a specific cost. Amazon EC2 uses this kind of pricing.
2. **Per-unit Pricing:** The model is based upon the unit-specific service concept. Data transfer and memory allocation include in this model for specific units. GoGrid uses this kind of pricing in terms of RAM/hour.
3. **Subscription-based Pricing:** In this model, users are paying periodic subscription fees for the usage of the software.

So these models give more flexible solutions to the cloud economy.

OPEN CHALLENGES IN CLOUD:

Cloud computing, an emergent technology, has placed many challenges in different aspects of data and information handling. Some of these are shown in the following diagram:



Security and Privacy

Security and Privacy of information is the biggest challenge to cloud computing. Security and privacy issues can be overcome by employing encryption, security hardware and security applications.

Portability

This is another challenge to cloud computing that applications should easily be migrated from one cloud provider to another. There must not be vendor lock-in. However, it is not yet made possible because each of the cloud provider uses different standard languages for their platforms.

Interoperability

It means the application on one platform should be able to incorporate services from the other platforms. It is made possible via web services, but developing such web services is very complex.

Computing Performance

Data intensive applications on cloud requires high network bandwidth, which results in high cost. Low bandwidth does not meet the desired computing performance of cloud application.

Reliability and Availability

It is necessary for cloud systems to be reliable and robust because most of the businesses are now becoming dependent on services provided by third party.

CLOUD INTEROPERABILITY AND STANDARDS:

Cloud interoperability refers to the ability of customers to use the same management tools, server images and other software with a variety of cloud computing providers and platforms.

Standards are important in cloud computing for a variety of reasons. Standards for interoperability and data and application portability can ensure an open competitive market in cloud computing because customers are not locked-in to cloud providers and can easily transfer data or applications between cloud providers.

Why cloud interoperability and standards?

Vendor lock-in can prevent a customer from switching to another competitor's solution. If switching is possible, it happens at considerable conversion cost and requires significant amounts of time. Switching happens because a customer may want to find a more suitable solution for customer needs. Or vendor may not be able to provide the service required. So, the presence of standards that are actually implemented and adopted in the cloud computing community gives power for interoperability and then lessens the risks resulting from vendor lock-in.

CLOUD STANDARDS:

1. CCIF - Cloud Computing Interoperability forum.
2. Open cloud Consortium.
3. DMTF Cloud Standard Incubator.
4. Open Cloud Manifesto.

SCALABILITY AND FAULT TOLERANCE:

1. Cloud Scalability

In cloud computing, cloud scalability refers to the ability to increase or reduce IT resources as required to meet evolving demands. One of the hallmarks of the cloud and the key factor of its burgeoning popularity with companies is scalability.

Using existing cloud computing technology, data storage space, processing power and networking can all be escalated. Better still, scaling, usually with little or no interruption or downtime, can be achieved rapidly and easily. Third-party cloud providers now have the entire infrastructure in place; in the past, the process could take weeks or months to scale with on-site physical infrastructure and entail enormous costs.

1.1 How to achieve cloud scalability?

To set up a personalized, scalable cloud solution via a public cloud, private cloud, or hybrid cloud, businesses have several options.

In cloud computing, two specific forms of scalability exist: vertical and horizontal scaling.

We can add or subtract power to an existing cloud server: memory upgrade, storage, or computing power with vertical scaling, also known as "scaling up" or "scaling down". This generally indicates that scaling has an upper limit based on the scaling capability of the server or machine; scaling above that also includes downtime.

We can add more resources like servers to our system using horizontal scalability to spread the workload across computers, which in turn improves efficiency and storage space. For companies with high-availability services that need limited downtime, horizontal scaling is essential.



2. Cloud Fault Tolerance

In cloud computing, fault tolerance is conceptually the same as in private or hosted environments. In other words, it means the infrastructure's ability to continue to provide service/services to underlying applications even when one or more component fails. To continue to work through failure or repair, we do not need to configure certain facilities for our infrastructure to use.

2.1 Objectives of Fault Tolerance in Cloud Computing

The fault-tolerant system uses backup components that take the place of failed components automatically, ensuring no service loss. They include:

- **Hardware systems**
Hardware systems can be backed up using identical or equivalent systems. For instance, using an identical server running in parallel, with all operations mirrored to the backup server, a server can be made fault-tolerant.
- **Software systems**
Software systems can be backed up using software instances. For example, it is possible to continuously replicate a database with customer information on another machine and operations can be mechanically redirected to another database in case a primary database goes down.
- **Power sources**
Power sources use alternative sources using fault-tolerant. In many instances, organizations have power generators that can be used in case the electricity fails. Similarly, using redundancy, any system or component that is a single point of failure can be made fault-tolerant.
- **Security Breach Occurrences**
Owing to security failures, there are many explanations about why fault tolerance exists. The server's hacking adversely affects the server and results in a leak of data. Ransomware, phishing, virus attack, etc. are other explanations for the need for fault tolerance in the form of security violations.

Key principles behind Cloud Computing Device Fault Tolerance

- **Replication**
For every operation, the fault-tolerant system operates on the principle of running many other replicates. Therefore, if one aspect of the device goes wrong, it has other instances that can be put to keep it going instead. For example, a database of clusters that has 3 servers with the same information on each of them. All the acts are written on each of them, such as adding data, upgrading, and deleting. The redundant servers will be in inactive mode unless and until the availability of them is requested by any fault tolerance scheme.
- **Redundancy**
If any part of the system fails or moves to a downstate, then it is necessary to have backup systems. For example, due to some hardware faults, a website programmer that has MS SQL as its database can fail in between. In the redundancy principle, a server works with an emergency database comprising many backup resources.



UNIT-III

VIRTUALIZATION

Introduction:

Virtualization is a technique, which allows to share single physical instance of an application or resource among multiple organizations or tenants (customers). It does so by **assigning a logical name** to a physical resource and providing a **pointer to that physical resource** on demand

Virtualization Concept

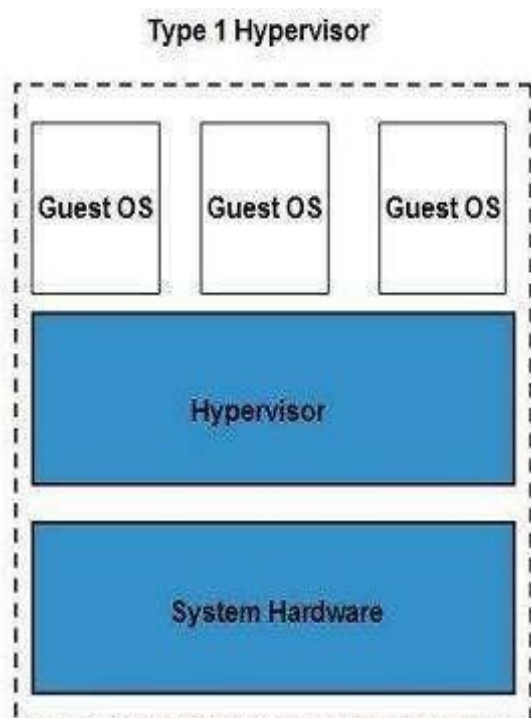
Creating a virtual machine over existing operating system and hardware is referred as Hardware Virtualization. Virtual Machines provide an environment that is logically separated from the underlying hardware.

The machine on which the virtual machine is created is known as **host machine** and **virtual machine** is referred as a **guest machine**. This virtual machine is managed by a software or firmware, which is known as **hypervisor**.

Hypervisor

The **hypervisor** is a firmware or low-level program that acts as a Virtual Machine Manager. There are two types of hypervisor:

Type 1 hypervisor executes on bare system. Lynx Secure, RTS Hypervisor, Oracle VM, Sun xVM Server, Virtual Logic VLX are examples of Type 1 hypervisor. The



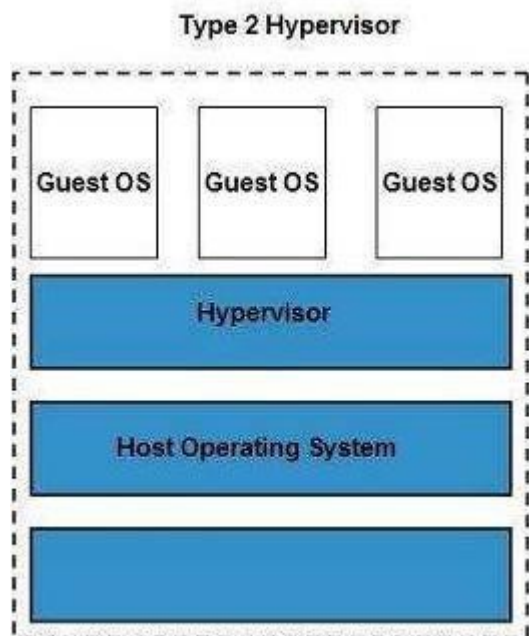
following diagram shows the Type 1 hypervisor.



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The **type1 hypervisor** does not have any host operating system because they are installed on a bare system.

Type 2 hypervisor is a software interface that emulates the devices with which a system normally interacts. Containers, KVM, Microsoft Hyper V, VMWare Fusion, Virtual Server 2005 R2, Windows Virtual PC and **VMW are workstation 6.0** are examples of Type 2 hypervisor. The following diagram shows the Type 2 hypervisor.



Pros of virtualization

Virtualization is being pursued with great attention by numerous IT businesses. One of the main benefits of virtualization for platforms that support remote working is their integration with the cloud.

1. Cheap

IT infrastructures find virtualization to be a more affordable implementation option because it doesn't require the use or installation of actual hardware components. Dedicating substantial amounts of space and money to create an on-site resource is no longer required. We need a licence or access from a third-party vendor to begin using the hardware, just as if it were locally produced.

2. Efficient

By downloading the new versions of the software and hardware from a third-party supplier, efficient virtualization also enables automatic upgrades of both. By handling the problem themselves and saving money, IT specialists are able to avoid having to hire specialists. Virtualization also lessens the difficulty of managing resources to increase the effectiveness of virtual environments.

3. Disaster recovery

When servers are virtualized, disaster recovery is relatively simple thanks to fast backup restoration and current snapshots of your virtual machines. Organizations were better able



to create a low-cost replication location thanks to virtualization. If a disaster occurs in the data centre or server room itself, you can still relocate such virtual machines to a cloud provider. Having the flexibility level guarantees that the disaster recovery plan will be simpler to implement and will have a 99% success rate.

4. Deployment

Resources may be deployed much more quickly when employing virtualization technology. It is feasible to significantly reduce the amount of time required for setting up physical devices or creating local networks. As a result, users really need is at least one connection to the virtual world. Additionally, the implementation of virtual machines is frequently simpler than the installation of actual models.

5. Encourages digital entrepreneurship

Prior to widespread virtualization, the average person found it nearly impossible to start a digital business. Thanks to the multiple networks, servers, and storage devices that are now accessible, almost anyone can start their own side business or turn into a business owner. Everyone can hang out their shingle and start looking for employment.

6. Saves energy

Both individuals and businesses can save energy by using virtualization. The rate of energy consumption can be reduced because no local hardware or software alternatives are being employed. To boost the total ROI of virtualization, monies can be used over time for other operational expenses rather than paying for a data centre's cooling costs and equipment operation costs.

7. Improved uptime

Virtualization technologies have increased uptime dramatically. An uptime of 99.9999% is offered by some providers. Even low-cost carriers now offer uptime at a rate of 99.99%.

8. Consistent cost

People and corporations can have predictable expenses for their IT requirements because third-party vendors frequently offer choices for virtualization.

Cons of virtualization

Numerous complex dimensions that digital technology had to explore have been resolved through virtualization. However, virtualization still shows signs of minor but significant problems.

As a result, virtualization has a lot of drawbacks, which are listed below:

1. Exorbitant costs of implementation

Virtualization would result in very low costs for the common person or business. In a virtualization environment, the suppliers, however, may incur very significant implementation expenses. It follows that devices must either be created, made, or purchased for implementation when hardware and software are eventually required.

2. Restraints

Virtualization is hampered by a number of issues. Virtualization cannot be used with every server and application currently in existence. Therefore, certain firms' IT infrastructures would not be able to support the virtualized solutions. They no longer receive support from a number of vendors as well. The demands of both individuals and organisations must be served using a hybrid approach.

3. Problems with availability

The accessibility of a company is another important factor. Long-term data linking is required. If not, the business would become less competitive in the market. Because every document from and for the client is essential to the service provider, availability difficulties might be seen as one of the drawbacks of virtualization. It seems as though the virtualization servers are taken offline. Additionally, hosted websites would be useless. The user has no control over this; it is completely the responsibility of the third-party providers.

4. Time-intensive

In comparison to local systems, virtualization takes less time to implement, but it ultimately costs users time. This is due to the fact that there are additional procedures that need to be completed in order to attain the desired result.

5. Threats to security



Information is our current currency. Having money allows you to make money. Without it, people will forget about you. The success of a corporation depends on information, hence it is frequently targeted.

6. Problems with scalability

People can grow a business or opportunity quickly owing to virtualization, but won't be able to grow it as large as they would like. In a virtualization network, growth generates latency since multiple firms share the same resources. There wouldn't be much that could be done to stop it, but one powerful presence could syphon money away from other, smaller businesses.

7. A Number of links must interact

If users have access to local equipment, they have complete control over their options. With virtualization, people lose control because numerous ties are required to cooperate in order to complete the same task. We can take the example of saving a document file. Using a local storage device like a flash drive or HDD, users can instantly save the content and even create a backup. In order to use virtualization, the ISP connection must be reliable.

What is a Virtual Machine?

A virtual machine (VM) is a digital version of a physical computer. Virtual machine software can run programs and operating systems, store data, connect to networks, and do other computing functions, and requires maintenance such as updates and system monitoring. Virtual machines run on a physical machine and access computing resources from software called a hypervisor. The hypervisor abstracts the physical machine's resources into a pool that can be provisioned and distributed as needed, enabling multiple VMs to run on a single physical machine.

Implementation levels of virtualization

It is not easy to set up virtualization. Your computer works on an operating system that gets configured on some specific hardware.

Thus, it is not feasible or straightforward to run a different operating system using the corresponding hardware.

To do this, one will need a hypervisor. Hence, the role of the hypervisor is to bridge between the hardware and the virtual operating system, which enables smooth functioning.

Meanwhile, talking of the Implementation levels of virtualization in cloud computing, five levels are commonly used. Let us now look firmly at each of these levels of virtualization implementation in cloud computing.

Instruction Set Architecture Level (ISA)

At the ISA level, virtualization can work via emulating a given ISA by the ISA of the host machine. For instance, MIPS binary code can operate on an x86-based host machine with the help of ISA emulation.

Thus, this strategy makes it possible to run a large volume of legacy binary code written for several processors on any provided different hardware host machine.

The first emulation method is through code interpretation. Therefore, an interpreter program defines the source instructions to target instructions one by one.

Activity happenings within the Instruction Set Architecture Level (ISA):

Instruction set

The hardware on the physical server in cloud computing has its own instruction set that it will process.

Hence, these instructions act as an interface between hardware and software. Therefore, by the instruction set, hardware immediately assigns its services to its upper layers.

Emulator

The emulator serves as an interpreter, and the fundamental role of this is to set communication between the virtualization and hardware layers in cloud computing.

Mapping of the instruction

Instructions that the emulator gets for the resources from the virtual machine can get mapped to the hardware instruction presented in the host machine.

Therefore, after the processing emulator got the result back from the host machine, it sent services to the user with the help of the virtual machine.



Hardware Abstraction Level (HAL)

Hardware-Abstraction level virtualization is available right on top of the bare hardware. On the one side, this approach produces a virtual hardware environment for a VM.

On the other side, the process controls the underlying hardware through virtualization.

Therefore, the purpose is to virtualize computer resources, such as processors, memory, and I/O devices. Also, the aim is to upgrade the hardware utilization rate by various users concurrently.

Activity happenings within the Hardware-Abstraction level (HAL):

Hardware resources in the host machine get mapped using a virtualization layer to the virtual resources present on the guest operating system. Thus, the virtual machine will create for all of them.

Hence, virtualized computing system holds thousands of resources. All of them need some directions for processing, which was not an easy task. And due to, this instruction was classified into two primary forms to make the processing smooth.

Non-privileged instruction– These instructions execute directly without interfering with other tasks.

Privileged instruction– These instructions require some modification before it executes.

- **Operating System Level**

The virtualization model can create an abstract layer between the operating system and the user application at this operating system level.

It is an isolated container on the operating system and the physical server, which uses the software and hardware. Thus, each of these then operates in the form of a server.

Therefore, when there are numerous users and no one wants to share the hardware, then at this point, the virtualization level comes into use.

Each user will get one virtual environment using a virtual hardware resource that is dedicated. Hence, in this manner, there is no issue of any conflict.

- Activity happenings within the Operating System Level:

Any hardware which is in a virtualized environment will process within this operating system. And the essential requirement at the operating system level is that all the user systems on the virtualized environment will hold the same family operating system. Otherwise, we can't transfer the service to the users.

- **Library Support Level**

Most utmost applications use APIs exported by user-level libraries rather than practicing lengthy system calls by the OS.

Since most of the systems provide well-documented APIs, such an interface becomes another applicant for virtualization.

Therefore, virtualization with library interfaces is possible by checking the communication link between applications and the system through API hooks.

- Activity happenings within the Library Support Level:

Use of the application programming interface (API).

At this level, the emulator's idea worked as a tool and provided the guest operating system to practice the resources they want. In short, users use the emulator to run different applications of the other operating systems.

- **User-Application Level**

The application-level virtualization works where there is a desire to virtualize only one application and is the last of the implementation levels of virtualization in cloud computing.

One does not require to virtualize the complete environment of the platform.

Therefore, it generally works when you run virtual machines that practice high-level languages.

Also, it lets the high-level language programs compiled be of use in the application level of the virtual machine that runs seamlessly.



- Activity happenings within the Library User-Application Level:
Virtual machine as an application operates at the user system with the help of the virtualization layer.
Also, the users access the services if the environment in which user and host were of a different type.

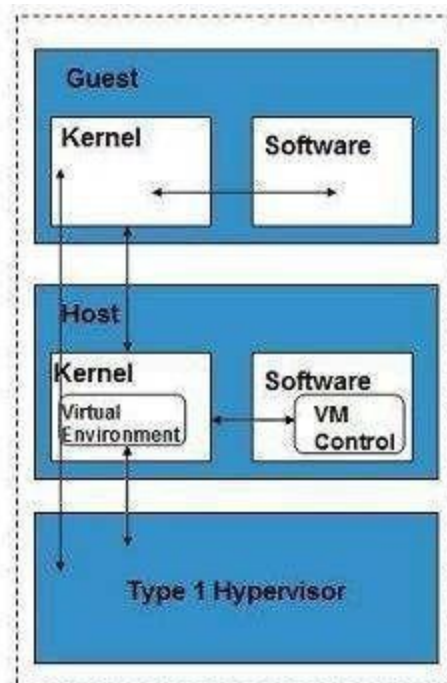
Types of Hardware Virtualization

Here are the three types of hardware virtualization:

- I. Full Virtualization
- II. Emulation Virtualization
- III. Par virtualization

Full Virtualization

In **full virtualization**, the underlying hardware is completely simulated.

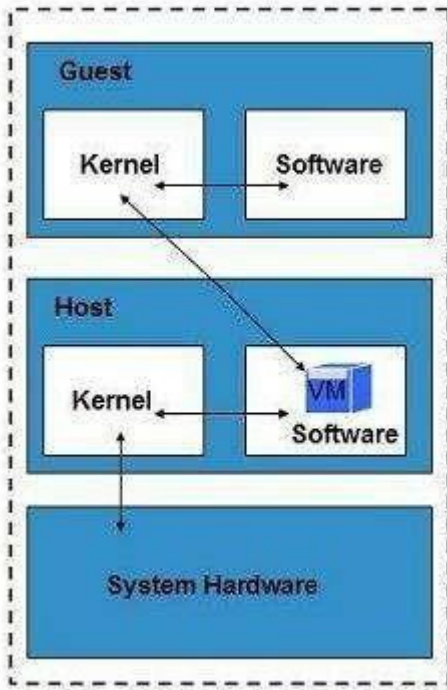


Guest software does not require any modification to run.

Emulation Virtualization

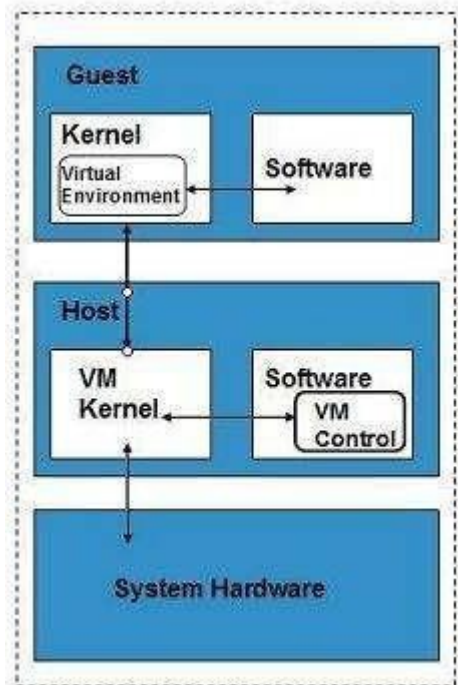
In **Emulation**, the virtual machine simulates the hardware and hence becomes independent of it. In this, the guest operating system does not require modification.





Par virtualization

In **Par virtualization**, the hardware is not simulated. The guest software run their own isolated domains.



VMware vSphere is highly developed infrastructure that offers a management infrastructure framework for virtualization. It virtualizes the system, storage and networking hardware.

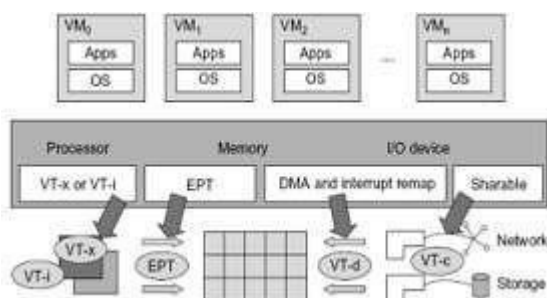
Virtualization of CPU, Memory and I/O devices

1. Hardware Support for Virtualization
2. CPU Virtualization
3. Memory Virtualization
4. I/O Virtualization
5. Virtualization in Multi-Core Processors

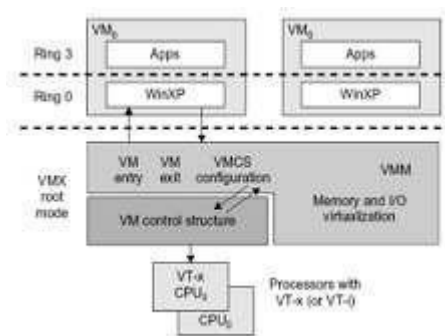


Hardware Support for Virtualization:

EPT (Extended Page Table); VT-x (Intel's Virtualization Technology)



VMCS (Virtual Machine Control Structure):



3.14 Virtualization of CPU:

Intel Hardware-assisted CPU Virtualization:

A VM is a duplicate of an existing computer system in which a majority of the VM instructions are executed on the host processor in native mode. Thus, unprivileged instructions of VMs run directly on the host machine for higher efficiency. Other critical instructions should be handled carefully for correctness and stability. The critical instructions are divided into three categories: privileged instructions, control-sensitive instructions, and behavior-sensitive instructions.

Privileged instructions execute in a privileged mode and will be trapped if executed outside this mode. Control-sensitive instructions attempt to change the configuration of resources used. Behavior-sensitive instructions have

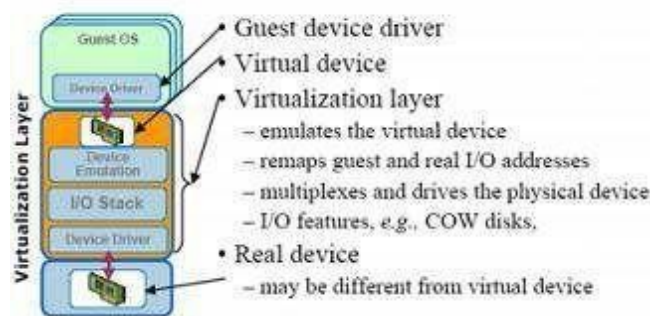


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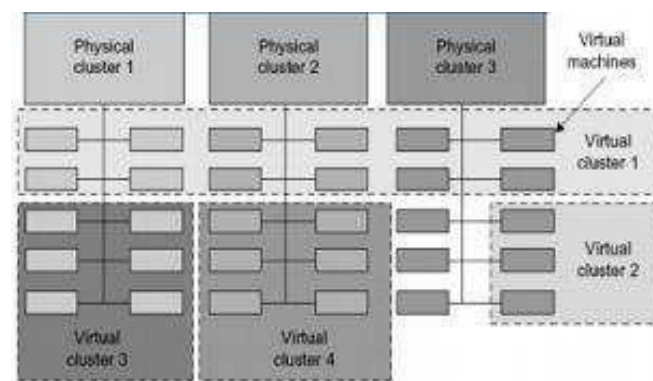
different behaviors depending on the configuration of resources, including the load and store operations over the virtual memory.

A CPU architecture is virtualizable if it supports the ability to run the VM's privileged and unprivileged instructions in the CPU's user mode while the VMM runs in supervisor mode. When the privileged instructions including control- and behavior-sensitive instructions of a VM are executed, they are trapped in the VMM. In this case, the VMM acts as a unified mediator for hardware access from different VMs to guarantee the correctness and stability of the whole system.

Current virtual I/O devices



Virtual Clusters vs. Physical Clusters:



3.15. Virtualization of memory and I/O devices

Memory Virtualization

Virtual memory virtualization is similar to the virtual memory support



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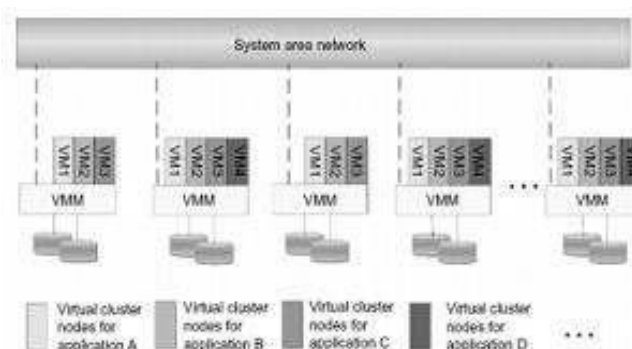
provided by modern operating systems. In a traditional execution environment, the operating system maintains mappings of virtual memory to machine memory using page tables, which is a one-stage mapping from virtual memory to machine memory. All modern x86 CPUs include a memory management unit (MMU) and a translation look aside buffer (TLB) to optimize virtual memory performance.

However, in a virtual execution environment, virtual memory virtualization involves sharing the physical system memory in RAM and dynamically allocating it to the physical memory of the VMs. That means a two-stage mapping process should be maintained by the guest OS and the VMM, respectively: virtual memory to physical memory and physical memory to machine memory.

Furthermore, MMU virtualization should be supported, which is transparent to the guest OS. The guest OS continues to control the mapping of virtual addresses to the physical memory addresses of VMs. But the guest OS cannot directly access the actual machine memory. The VMM is responsible for mapping the guest physical memory to the actual machine memory, shows the two-level memory mapping procedure.

Two-level memory mapping procedure. Courtesy of R. Rblig, et al. Since each page table of the guest OSes has a separate page table in the VMM corresponding to it, the VMM page table is called the shadow page table. Nested page tables add another layer of indirection to virtual memory.

A Virtual Clusters based on Application Partitioning:



Virtual Clusters Projects:



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AITIS – TIRUPATI, Dept Of

Project Name	Design Objectives	Reported Results and References
Cluster-on-Demand at Duke Univ.	Dynamic resource allocation with a virtual cluster management system	Sharing of VMs by multiple virtual clusters using Sun GridEngine [12]
Cellular Disco at Stanford Univ.	To deploy a virtual cluster on a shared-memory multiprocessor	VMs deployed on multiple processors under a VMM called Cellular Disco [8]
VIOLIN at Purdue Univ.	Multiple VM clustering to prove the advantage of dynamic adaptation	Reduce execution time of applications running VIOLIN with adaptation [25,55]
GRAAL Project at INRIA in France	Performance of parallel algorithms in Xen-enabled virtual clusters	75% of max. performance achieved with 30% resource stacks over VM clusters

The MMU already handles virtual-to-physical translations as defined by the OS. Then the physical memory addresses are translated to machine addresses using another set of page tables defined by the hypervisor. Since modern operating systems maintain a set of page tables for every process, the shadow page tables will get flooded.

Consequently, the performance overhead and cost of memory will be very high. VMware uses shadow page tables to perform virtual memory- to-machine-memory address translation. Processors use TLB hardware to map the virtual memory directly to the machine memory to avoid the two levels of translation on every access. When the guest OS changes the virtual memory to a physical memory mapping, the VMM updates the shadow page tables to enable a direct lookup.

The AMD Barcelona processor has featured hardware-assisted memory virtualization since 2007. It provides hardware assistance to the two-stage address translation in a virtual execution environment by using a technology called nested paging.

I/O Virtualization

I/O virtualization involves managing the routing of I/O requests between virtual devices and the shared physical hardware. At the time of this writing, there are three ways to implement I/O virtualization: full device emulation, para-virtualization, and direct I/O. Full device emulation is the first approach for I/O virtualization. Generally, this approach emulates well-known, real-world devices. All the functions of a device or bus infrastructure, such as device enumeration, identification, interrupts, and DMA, are replicated in software. This software is located in the VMM and acts as a virtual device. The I/O access requests of the guest OS are trapped in the VMM which interacts with the I/O devices. Device emulation for I/O virtualization implemented inside the middle layer that maps



real I/O devices into the virtual devices for the guest device driver to use. Courtesy of V. Chadha, et al. and Y. Dong, et al. A single hardware device can be shared by multiple VMs that run concurrently. However, software emulation runs much slower than the hardware it emulates.

The para-virtualization method of I/O virtualization is typically used in Xen. It is also known as the split driver model consisting of a frontend driver and a backend driver. The frontend driver is running in Domain U and the backend driver is running in Domain 0. They interact with each other via a block of shared memory.

The frontend driver manages the I/O requests of the guest OSes and the backend driver is responsible for managing the real I/O devices and multiplexing the I/O data of different VMs. Although para-I/O-virtualization achieves better device performance than full device emulation, it comes with a higher CPU overhead.

Direct I/O virtualization lets the VM access devices directly. It can achieve close-to- native performance without high CPU costs. However, current direct I/O virtualization implementations focus on networking for mainframes. There are a lot of challenges for commodity hardware devices.



For example, when a physical device is reclaimed (required by workload migration) for later reassignment, it may have been set to an arbitrary state (e.g., DMA to some arbitrary memory locations) that can function incorrectly or even crash the whole system.

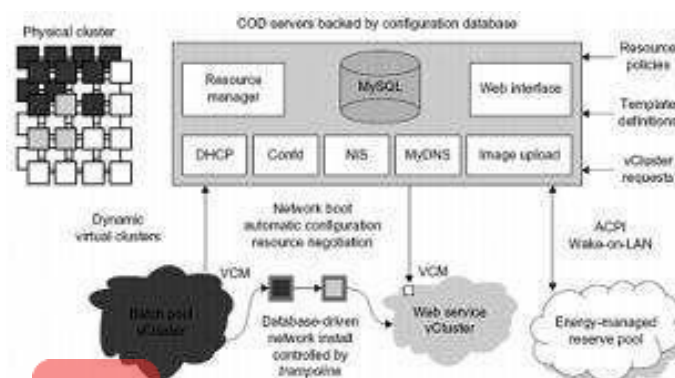
Since software based I/O virtualization requires a very high overhead of device emulation, hardware-assisted I/O virtualization is critical. Intel VT-d supports the remapping of I/O DMA transfers and device-generated interrupts. The architecture of VT-d provides the flexibility to support multiple usage models that may run unmodified, special-purpose, or –virtualization-aware|| guest OSes.

Another way to help I/O virtualization is via self-virtualized I/O (SV-IO) . The key idea of SVIO is to harness the rich resources of a multicore processor. All tasks associated with virtualizing an I/O device are encapsulated in SV-IO. It provides virtual devices and an associated access API to VMs and a management API to the VMM. SV-IO defines one virtual interface (VIF) for every kind of virtualized I/O device, such as virtual network interfaces, virtual block devices (disk), virtual camera devices, and others.

The guest OS interacts with the VIFs via VIF device drivers. Each VIF consists of two message queues. One is for outgoing messages to the devices and the other is for incoming messages from the devices. In addition, each VIF has a unique ID for identifying it in SV-IO.

COD (Cluster-on-Demand) Project at Duke University:

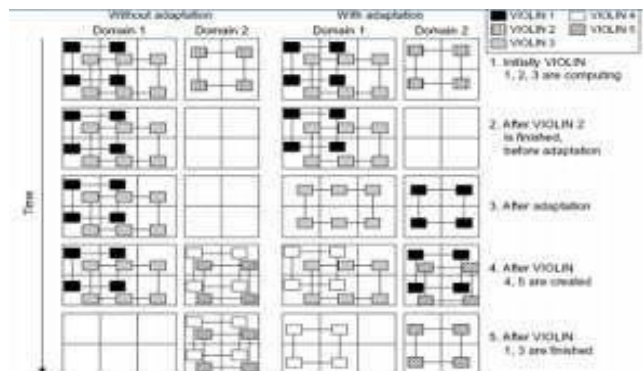
DHCP (Dynamic Host Configuration Protocol); VCM (configuration Manager); NIS (Network Information Service)



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VIOLIN Project at Purdue University

- Live VM migration to reconfigure a virtual cluster environment, Five concurrent virtual environment, labeled VIOLIN 1-5, sharing two physical clusters.



Virtualization Support at Intel

Virtual clusters and Resource Management

A physical cluster is a collection of servers (physical machines) interconnected by a physical network such as a LAN.

Let us study three critical design issues of virtual clusters: live migration of VMs, memory and file migrations, and dynamic deployment of virtual clusters.

When a traditional VM is initialized, the administrator needs to manually write configuration information or specify the configuration sources. When more VMs join a network, an inefficient configuration always causes problems with overloading or underutilization.

Amazon's Elastic Compute Cloud (EC2) is a good example of a web service that provides elastic computing power in a cloud. EC2 permits customers to create VMs and to manage user accounts over the time of their use.

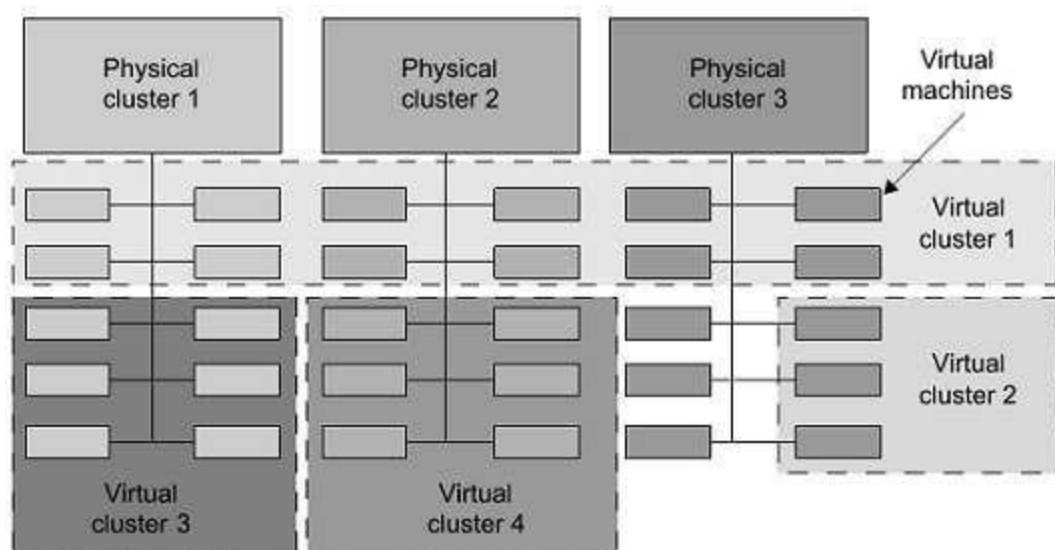
Most virtualization platforms, including XenServer and VMware ESX Server, support a bridging mode which allows all domains to appear on the network as individual hosts. By using this mode, VMs can communicate with one another freely through the virtual network interface card and configure the network automatically.

Virtual clusters are built with VMs installed at distributed servers from one or more physical clusters. The VMs in a virtual cluster are interconnected logically by a virtual network across several physical networks.

The below figure illustrates the concepts of virtual clusters and physical clusters. Each virtual cluster is formed with physical machines or a VM hosted by multiple physical clusters. The virtual cluster boundaries are shown as distinct boundaries.

The provisioning of VMs to a virtual cluster is done dynamically to have the following interesting properties:

- The virtual cluster nodes can be either physical or virtual machines. Multiple VMs running with different OSes can be deployed on the same physical node.
- A VM runs with a guest OS, which is often different from the host OS, that manages the resources in the physical machine, where the VM is implemented.
- The purpose of using VMs is to consolidate multiple functionalities on the



same server. This will greatly enhance server utilization and application flexibility.

A cloud platform with four virtual clusters over three physical clusters shaded differently

- VMs can be colonized (replicated) in multiple servers for the purpose of promoting distributed parallelism, fault tolerance, and disaster recovery.
- The size of a virtual cluster can grow or shrink dynamically, similar to the way an overlay network varies in size in a peer-to-peer (P2P) network.
- The failure of any physical nodes may disable some VMs installed on the failing nodes. But the failure of VMs will not pull down the host system.

Since system virtualization has been widely used, it is necessary to



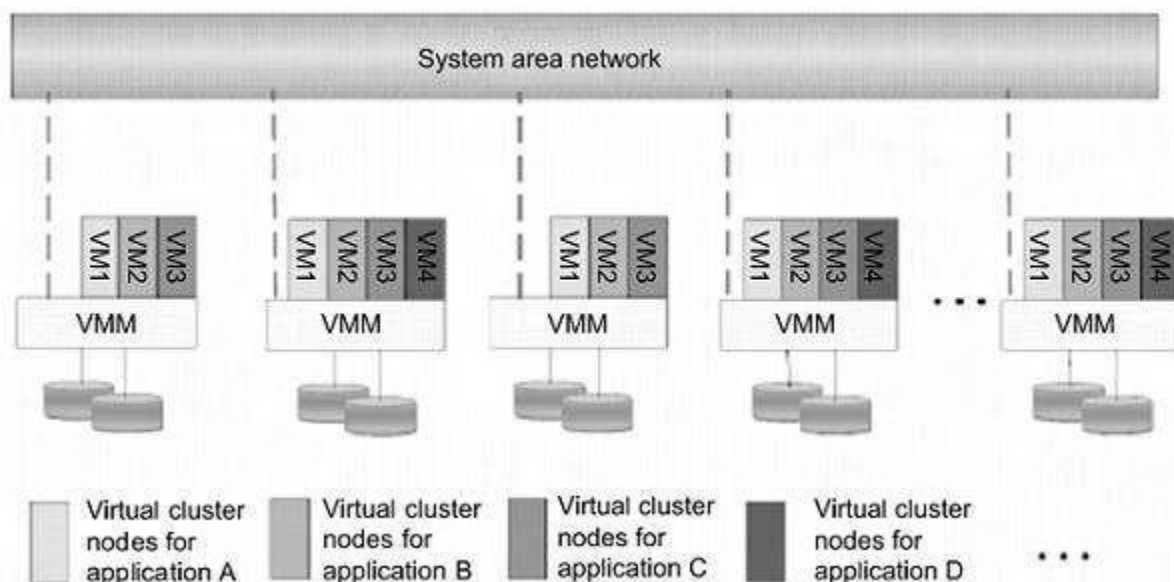
effectively manage VMs running on a mass of physical computing nodes (also called virtual clusters) and consequently build a high-performance virtualized computing environment. This involves virtual cluster deployment, monitoring and management over large-scale clusters, as well as resource scheduling, load balancing, server consolidation, fault tolerance, and other techniques.

The different node colors in the figure above refer to different virtual clusters. In a virtual cluster system, it is quite important to store the large number of VM images efficiently.

The below figure shows the concept of a virtual cluster based on application partitioning or customization. The different colors in the figure represent the nodes in different virtual clusters. As a large number of VM images might be present, the most important thing is to determine how to store those images in the system efficiently. There are common installations for most users or applications, such as operating systems or user-level programming libraries. These software packages can be preinstalled as templates.

With these templates, users can build their own software stacks. New OS instances can be copied from the template VM. User-specific components such as programming libraries and applications can be installed to those instances.

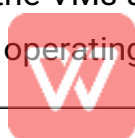
Three physical clusters are shown on the left side of figure below. Four virtual clusters are created on the right, over the physical clusters. The physical



machines are also called host systems.

Concept of a virtual cluster based on application partitioning

In contrast, the VMs are guest systems. The host and guest systems may run with different operating systems. Each VM can be installed on a remote



server or replicated on multiple servers belonging to the same or different physical clusters. The boundary of a virtual cluster can change as VM nodes are added, removed, or migrated dynamically over time.



Fast Deployment and Effective Scheduling

The system should have the capability of fast deployment. Here, deployment means two things: to construct and distribute software stacks (OS, libraries, applications) to a physical node inside clusters as fast as possible, and to quickly switch runtime environments from one user's virtual cluster to another user's virtual cluster.

If one user finishes using his system, the corresponding virtual cluster should shut down or suspend quickly to save the resources to run other VMs for other users.

The concept of "green computing" has attracted much attention recently. However, previous approaches have focused on saving the energy cost of components in a single workstation without a global vision.

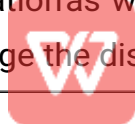
Consequently, they do not necessarily reduce the power consumption of the whole cluster. Other cluster-wide energy-efficient techniques can only be applied to homogeneous workstations and specific applications. The live migration of VMs allows workloads of one node to transfer to another node. However, it does not guarantee that VMs can randomly migrate among themselves.

In fact, the potential overhead caused by live migrations of VMs cannot be ignored. The overhead may have serious negative effects on cluster utilization, throughput and QoS issues. Therefore, the challenge is to determine how to design migration strategies to implement green computing without influencing the performance of clusters. Another advantage of virtualization is load balancing of applications in a virtual cluster. Load balancing can be achieved using the load index and frequency of user logins. The automatic scale-up and scale-down mechanism of a virtual cluster can be implemented based on this model.

Consequently, we can increase the resource utilization of nodes and shorten the response time of systems. Mapping VMs onto the most appropriate physical node should promote performance. Dynamically adjusting loads among nodes by live migration of VMs is desired, when the loads on cluster nodes become quite unbalanced.

High-Performance Virtual Storage

The template VM can be distributed to several physical hosts in the cluster to customize the VMs. In addition, existing software packages reduce the time for customization as well as switching virtual environments. It is important to efficiently manage the disk spaces occupied by template software packages.



Some storage architecture design can be applied to reduce duplicated blocks in a distributed file system of virtual clusters. Hash values are used to compare the contents of data blocks. Users have their own profiles which store the identification of the data blocks for corresponding VMs in a user-specific virtual cluster. New blocks are created when users modify the corresponding data.

Newly created blocks are identified in the users' profiles. Basically, there are four steps to deploy a group of VMs onto a target cluster: preparing the disk image, configuring the VMs, choosing the destination nodes and executing the VM deployment command on every host. Many systems use templates to simplify the disk image preparation process.

A template is a disk image that includes a preinstalled operating system with or without certain application software. Users choose a proper template according to their requirements and make a duplicate of it as their own disk image. Templates could implement the COW (Copy on Write) format.

A new COW backup file is very small and easy to create and transfer. Therefore, it definitely reduces disk space consumption. In addition, VM deployment time is much shorter than that of copying the whole raw image file.

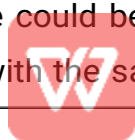
Every VM is configured with a name, disk image, network setting and allocated CPU and memory. One needs to record each VM configuration into a file. However, this method is inefficient when managing a large group of VMs. VMs with the same configurations could use predicted profiles to simplify the process.

In this scenario, the system configures the VMs according to the chosen profile. Most configuration items use the same settings, while some of them, such as UUID, VM name and IP address, are assigned with automatically calculated values. Normally, users do not care which host is running their VM.

A strategy to choose the proper destination host for any VM is needed. The deployment principle is to fulfil the VM requirement and to balance workloads among the whole host network.

Live VM Migration Steps and Performance Effects

In a cluster built with mixed nodes of host and guest systems, the normal method of operation is to run everything on the physical machine. When a VM fails, its role could be replaced by another VM on a different node, as long as they both run with the same guest OS. In other words, a physical node can



fail over to a VM on another host. This is different from physical-to-physical failover in a traditional physical cluster. The advantage is enhanced failover flexibility. The potential drawback is that a VM must stop playing its role if its residing host node fails. However, this problem can be mitigated with VM life migration.

The figure below shows the process of life migration of a VM from host A to host B. The migration copies the VM state file from the storage area to the host machine. There are four ways to manage a virtual cluster. First, we can use a guest-based manager, by which the cluster manager resides on a guest system. In this case, multiple VMs form a virtual cluster.

For example, openMosix is an open source Linux cluster running different guest systems on top of the Xen hypervisor. Another example is Sun's cluster Oasis, an experimental Solaris cluster of VMs supported by a VMware VMM. Second, we can build a cluster manager on the host systems. The host-based manager supervises the guest systems and can restart the guest system on another physical machine.

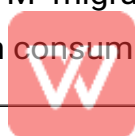
A good example is the VMware HA system that can restart a guest system after failure. These two cluster management systems are either guest-only or host-only, but they do not mix. A third way to manage a virtual cluster is to use an independent cluster manager on both the host and guest systems. This will make infrastructure management more complex, however.

Finally, we can use an integrated cluster on the guest and host systems. This means the manager must be designed to distinguish between virtualized resources and physical resources. Various cluster management schemes can be greatly enhanced when VM life migration is enabled with minimal overhead.

VMs can be live-migrated from one physical machine to another; in case of failure, one VM can be replaced by another VM. Virtual clusters can be applied in computational grids, cloud platforms and high-performance computing (HPC) systems.

The major attraction of this scenario is that virtual clustering provides dynamic resources that can be quickly put together upon user demand or after a node failure.

In particular, virtual clustering plays a key role in cloud computing. When a VM runs a live service, it is necessary to make a trade-off to ensure that the migration occurs in a manner that minimizes all three metrics. The motivation is to design a live VM migration scheme with negligible downtime, the lowest network bandwidth consumption possible and a reasonable total migration time.



Furthermore, we should ensure that the migration will not disrupt other active services residing in the same host through resource contention (e.g., CPU, network bandwidth). A VM can be in one of the following four states. An inactive state is defined by the virtualization platform, under which the VM is not enabled.

An active state refers to a VM that has been instantiated at the virtualization platform to perform a real task. A paused state corresponds to a VM that has been instantiated but disabled to process a task or paused in a waiting state. A VM enters the suspended state if its machine file and virtual resources are stored back to the disk.



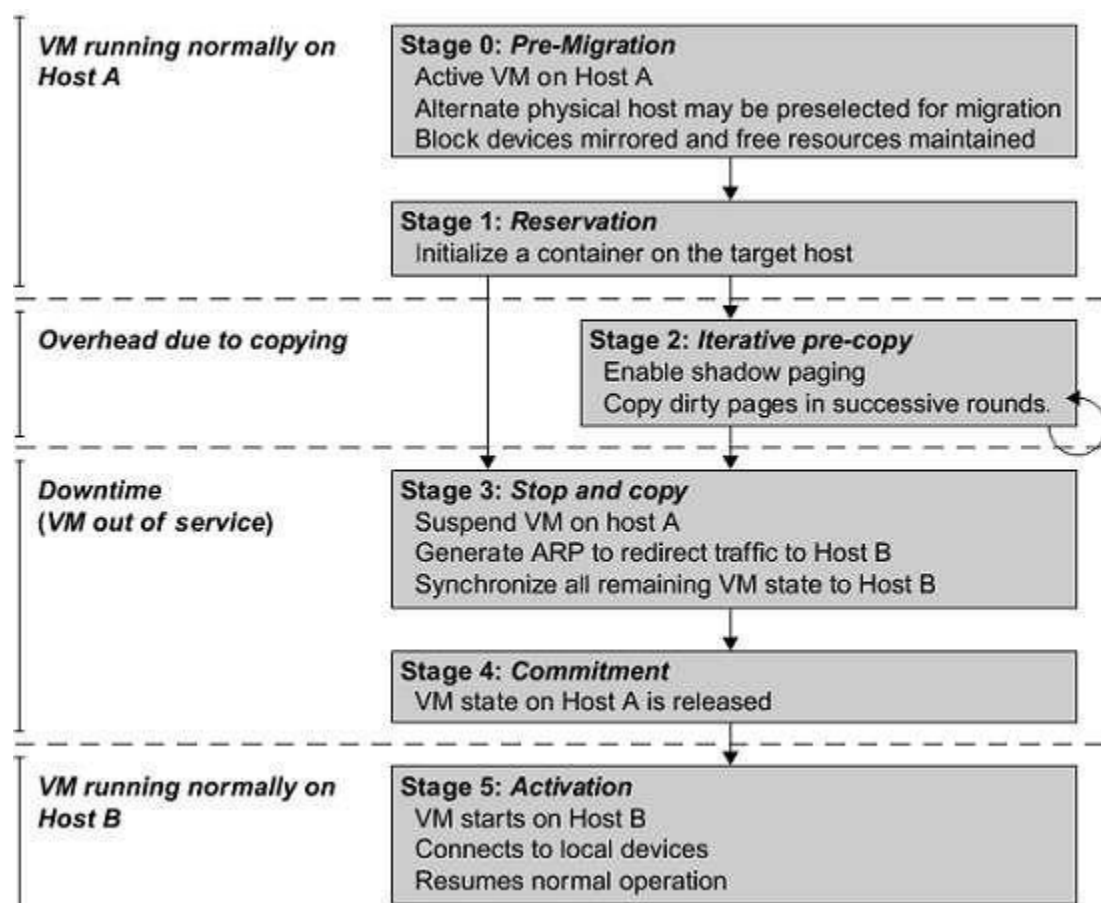
As shown in the figure below, live migration of a VM consists of the following six

steps:

Steps 0 and 1: Start migration. This step makes preparations for the migration, including determining the migrating VM and the destination host. Although users could manually make a VM migrate to an appointed host, in most circumstances, the migration is automatically started by strategies such as load balancing and server consolidation.

Steps 2: Transfer memory. Since the whole execution state of the VM is stored in memory, sending the VM's memory to the destination node ensures continuity of the service provided by the VM. All of the memory data is transferred in the first round, and then the migration controller recopies the memory data which is changed in the last round.

These steps keep iterating until the dirty portion of the memory is small enough to handle the final copy. Although precopying memory is performed iteratively,



the execution of programs is not obviously interrupted.

Live migration process of a VM from one host to another

Step 3: Suspend the VM and copy the last portion of the data. The migrating VM's execution is suspended when the last round's memory data is transferred. Other non-memory data such as CPU and network states should be sent as well.

During this step, the VM is stopped and its applications will no longer run. This “service unavailable” time is called the “downtime” of migration, which should be as short as possible so that it can be negligible to users.

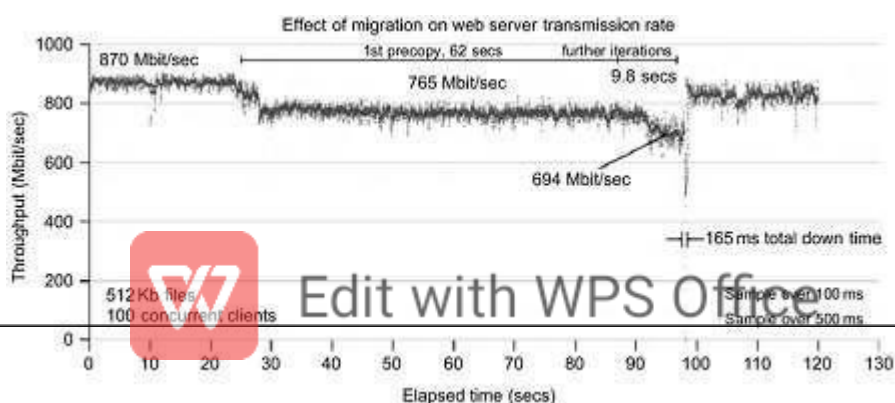
Steps 4 and 5: Commit and activate the new host. After all the needed data is copied, on the destination host, the VM reloads the states and recovers the execution of programs in it and the service provided by this VM continues. Then the network connection is redirected to the new VM and the dependency to the source host is cleared. The whole migration process finishes by removing the original VM from the source host.

The bellow figure shows the effect on the data transmission rate (Mbit/second) of live migration of a VM from one host to another. Before copying the VM with 512 KB files for 100 clients, the data throughput was 870 MB/second.

The first precopy takes 63 seconds, during which the rate is reduced to 765 MB/second. Then the data rate reduces to 694 MB/second in 9.8 seconds for more iterations of the copying process. The system experiences only 165 ms of downtime, before the VM is restored at the destination host. This experimental result shows a very small migration overhead in live transfer of a VM between host nodes. This is critical to achieve dynamic cluster reconfiguration and disaster recovery as needed in cloud computing.

With the emergence of widespread cluster computing more than a decade ago, many cluster configuration and management systems have been developed to achieve a range of goals. These goals naturally influence individual approaches to cluster management. VM technology has become a popular method for simplifying management and sharing of physical.

computing resources. Platforms such as VMware and Xen allow multiple VMs with different operating systems and configurations to coexist on the same physical host in mutual isolation. Clustering inexpensive computers is an effective way to obtain reliable, scalable computing power for network services



and compute-intensive applications.

Effect on data transmission rate of a VM migrated from one failing web server to another



Migration of Memory, Files, and Network Resources

Since clusters have a high initial cost of ownership, including space, power conditioning, and cooling equipment, leasing or sharing access to a common cluster is an attractive solution when demands vary over time. Shared clusters offer economies of scale and more effective utilization of resources by multiplexing.

Early configuration and management systems focus on expressive and scalable mechanisms for defining clusters for specific types of service, and physically partition cluster nodes among those types. When one system migrates to another physical node, we should consider the following issues.

1. Memory Migration
2. File System Migration
3. Network Migration
4. Live Migration of VM Using Xen

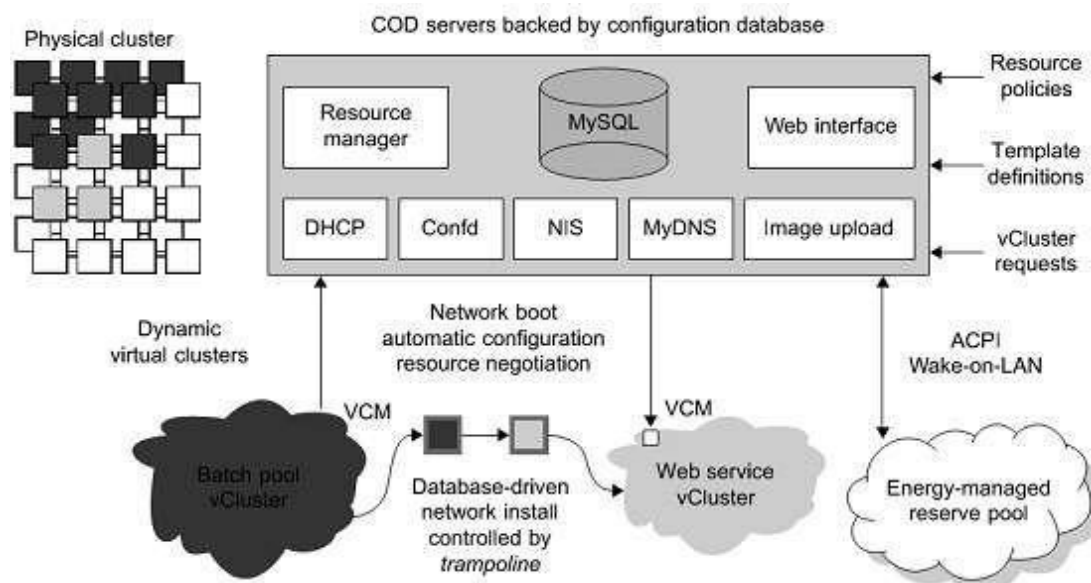
Dynamic Deployment of Virtual Clusters

The below table summarizes four virtual cluster research projects. We briefly introduce them here just to identify their design objectives and reported results. The Cellular Disco at Stanford is a virtual cluster built in a shared-memory multiprocessor system. The INRIA virtual cluster was built to test parallel algorithm performance. The COD and VIOLIN clusters are studied in forthcoming examples.

Experimental Results on Four Research Virtual Clusters

Project Name	Design Objectives	Reported Results and References
Cluster-on-Demand at Duke Univ.	Dynamic resource allocation with a virtual cluster management system	Sharing of VMs by multiple virtual clusters using Sun GridEngine [12]
Cellular Disco at Stanford Univ.	To deploy a virtual cluster on a shared-memory multiprocessor	VMs deployed on multiple processors under a VMM called Cellular Disco [8]
VIOLIN at Purdue Univ.	Multiple VM clustering to prove the advantage of dynamic adaptation	Reduce execution time of applications running VIOLIN with adaptation [25,55]
GRAAL Project at INRIA in France	Performance of parallel algorithms in Xen-enabled virtual clusters	75% of max. performance achieved with 30% resource slacks over VM clusters



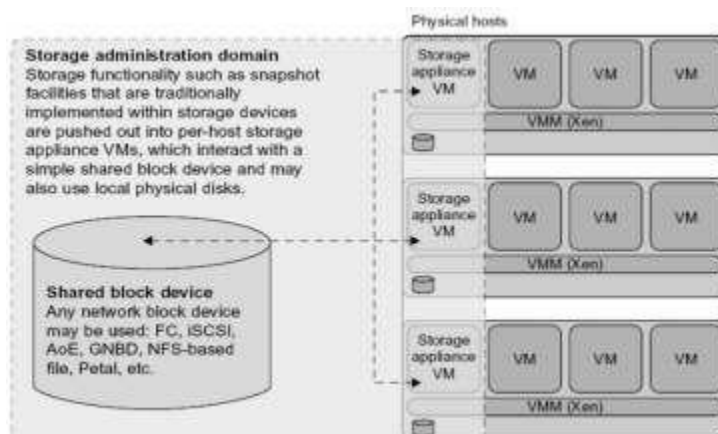


COD partitioning a physical cluster into multiple virtual clusters

Virtualization for data center automation

1. Virtual Storage Management
2. Cloud OS for Virtualized Data Centers
3. Trust Management in Virtualized Data Centers
4. VM-based Intrusion Detection

Parallax Providing Virtual Disks to Clients VMs from a Large Common Shared PhysicalDisk.



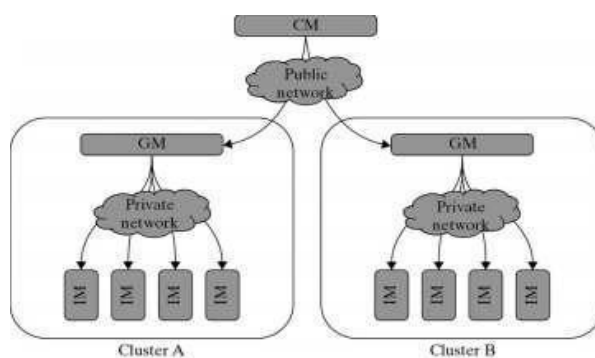
ComputeCloud).

Manager/ OS, Platforms, License	Resources Being Virtualized, Web Link	Client API, Language	Hypervisors Used	Public Cloud Interface	Special Features
Nimbus Linux, Apache v2	VM creation, virtual cluster, www.nimbusproject.org/	EC2 WS, WSRF, CLI	Xen, KVM	EC2	Virtual networks
Eucalyptus Linux, BSD	Virtual networking (Example 3.12 and [41]), www.eucalyptus.com/	EC2 WS, CLI	Xen, KVM	EC2	Virtual networks
OpenNebula Linux, Apache v2	Management of VM, host, virtual network, and scheduling tools, www.opennebula.org/	XML-RPC, CLI, Java	Xen, KVM	EC2, Elastic Host	Virtual networks, dynamic provisioning
vSphere 4 Linux, Windows, proprietary	Virtualizing OS for data centers (Example 3.13), www.vmware.com/products/vsphere/ [66]	CLI, GUI, Portal, WS	VMware ESX, ESXi	VMware vCloud partners	Data protection, vStorage, VMFS, DRM, HA

Eucalptus: An Open-Source OS for Setting Up and Managing Private Clouds (IaaS)

Three Resource Managers: CM (Cloud Manager), GM (Group Manager), and IM (InstanceManager)

Works like AWS APIs

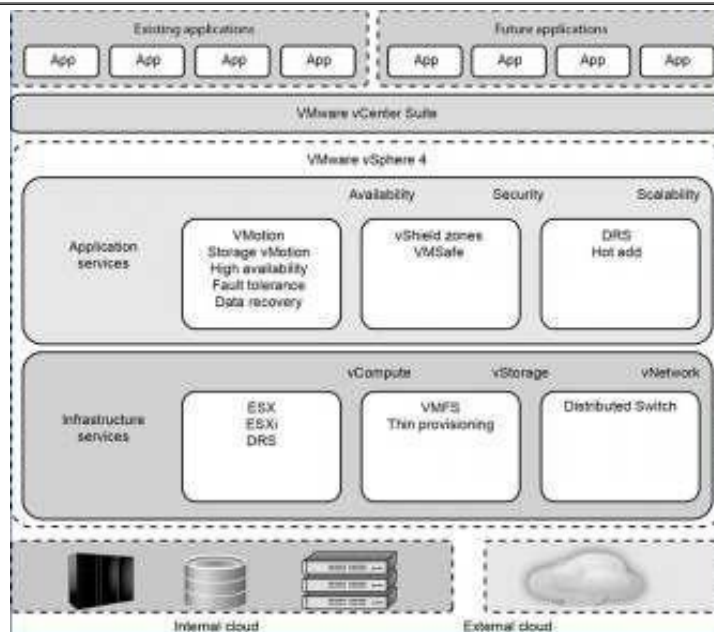


VMware vSphere 4 – A Commercial Cloud OS.

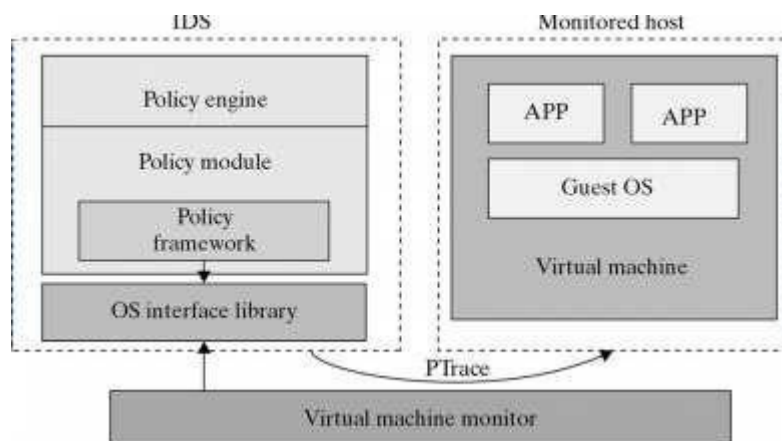


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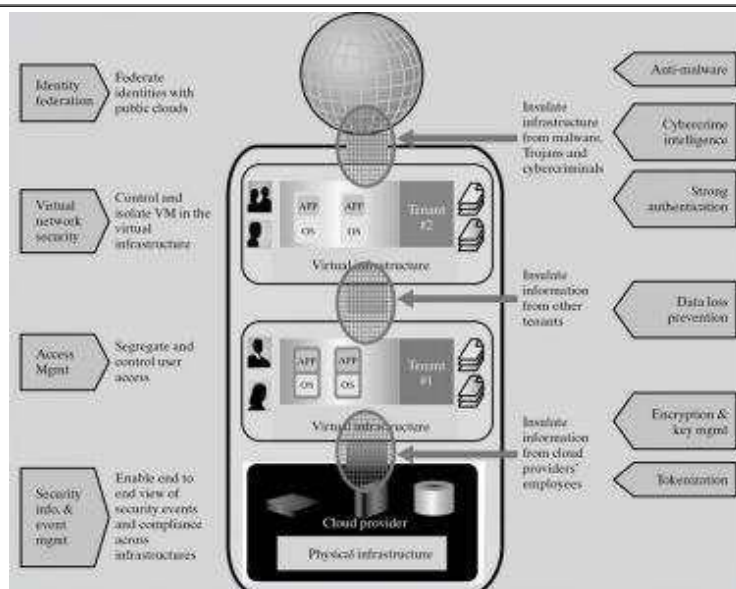


VM-based Intrusion Detection.



Techniques for establishing trusted zones for virtual cluster insulation and VM isolation.





UNIT-IV

PROGRAMMING MODEL



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Introduction to Hadoop Framework

Apache Hadoop and the Hadoop

Ecosystem

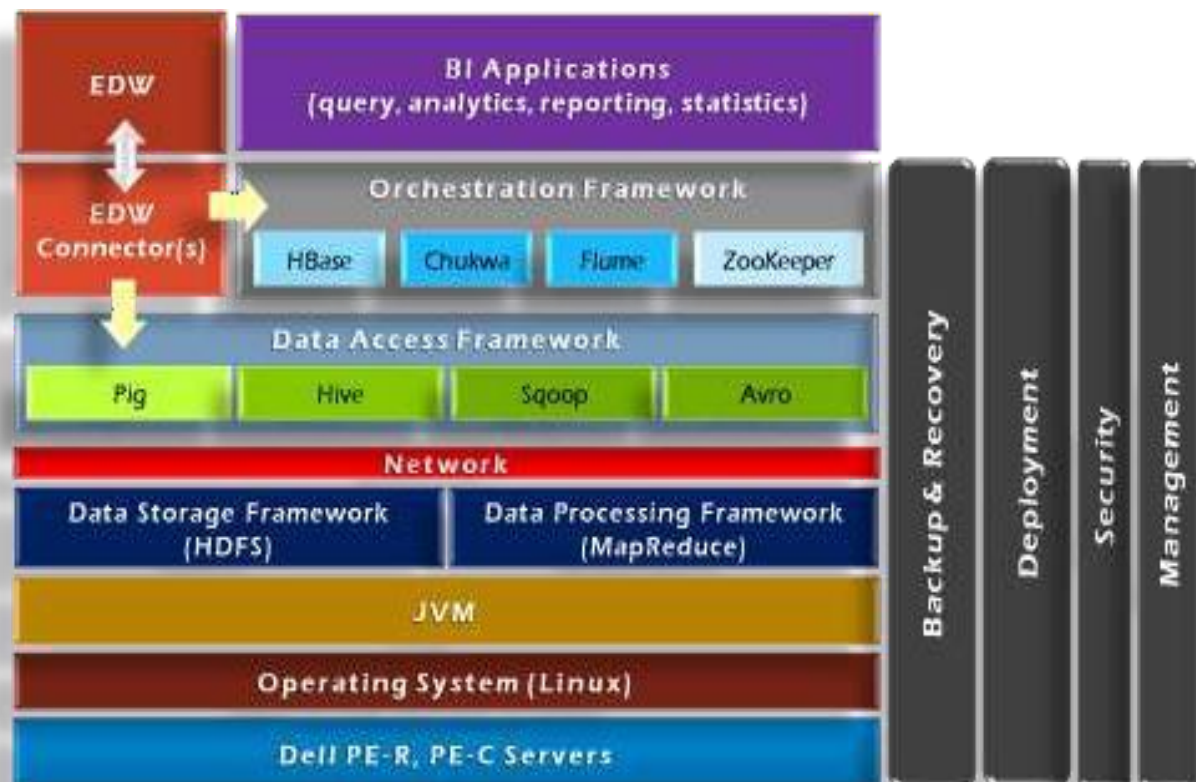
Although Hadoop is best known for Map Reduce and its distributed file system (HDFS, Renamed from NDFS),

Map Reduce is a programming model for data processing. The model is simple, yet not

Too simple to express useful programs in. Hadoop can run Map Reduce programs written in various languages; in this chapter, we shall look at the same program expressed in Java, Ruby, Python, and C++. Most important, Map Reduce programs are inherently parallel, thus putting very large-scale data analysis into the hands of anyone with enough machines at their disposal. Map Reduce comes into its own for large datasets, so let's start by looking at one.

Analyzing the Data with Hadoop

To take advantage of the parallel processing that Hadoop provides, we need to express our query as a Map Reduce job. After some local, small-scale testing, we will be able to run it on a cluster of machines.



- Distributed, with some centralization
- Main nodes of cluster are where most of the computational power and storage of the system lies
- Main nodes run Task Tracker to accept and reply to Map Reduce tasks, and also DataNode to store needed blocks closely as possible
- Central control node runs Name Node to keep track of HDFS directories & files, and Job Tracker to dispatch compute tasks to Task Tracker
- Written in Java, also supports Python and Ruby

Map and Reduce

Map Reduce works by breaking the processing into two phases: the map phase and the reduce phase. Each phase has key-value pairs as input and output, the types of which may be chosen by the programmer. The programmer also specifies two functions: the map function and the reduce function

Anatomy of a Map Reduce Job Run

You can run a Map Reduce job with a single line of code: `Job Client .run Job(conf)`. It's very short, but it conceals a great deal of processing behind the scenes. This section uncovers the steps Hadoop takes to run a job.

The whole process is illustrated in [Figure](#). At the highest level, there are four independent entities:

- The client, which submits the Map Reduce job.
- The job tracker, which coordinates the job run. The job tracker is a Java application whose main class is Job Tracker.
- The task trackers, which run the tasks that the job has been split into. Task trackers are Java applications whose main class is Task Tracker.
- The distributed file system, which is used for sharing job files between the other entities.

Job Submission

The `run Job ()` method on Job Client is a convenience method that creates a new Job Client instance and calls `submit Job()` on it (step 1). Having submitted the job, `run Job()` polls the job's progress once a second and reports the progress to the console if it has changed since the last report. When the job is complete, if it was successful, the job counters are displayed. Otherwise, the error that caused the job to fail is logged to the console.

The job submission process implemented by Job Client's `submit Job()` method does the following:

- Asks the job tracker for a new job ID (by calling `get New Job Id ()` on Job Tracker) (step 2).
- Checks the output specification of the job. For example, if the output directory has not been specified or it already exists, the job is not submitted and an error is thrown to the Map Reduce program.
- Computes the input splits for the job. If the splits cannot be computed, because the input paths don't exist, for example, then the job is not submitted and an error is thrown to the MapReduce program.



- Copies the resources needed to run the job, including the job JAR file, the configuration file, and the computed input splits, to the job tracker's file system in a directory named after the job ID. The job JAR is copied with a high replication factor (controlled by the mapped. submit .replication property, which defaults to 10) so that there are lots of copies across the cluster for the task trackers to access when they run tasks for the job (step 3).
- Tells the job tracker that the job is ready for execution (by calling submit Job() on Job Tracker) (step 4).

Job Initialization

When the Job Tracker receives a call to its submit Job() method, it puts it into an internal queue from where the job scheduler will pick it up and initialize it. Initialization involves creating an object to represent the job being run, which encapsulates its tasks, and bookkeeping information to keep track of the tasks' status and progress (step 5).

To create the list of tasks to run, the job scheduler first retrieves the input splits computed by the Job Client from the shared file system (step 6). It then creates one map task for each split. The number of reduce tasks to create is determined by the mapped. Reduce. Tasks property in the Job Conf, which is set by the set Num Reduce Tasks() method, and the scheduler simply creates this number of reduce tasks to be run. Tasks are given IDs at this point.

- Tells the job tracker that the job is ready for execution (by calling submit Job() on Job Tracker) (step 4).

Input Splitting

Task Assignment

Task trackers run a simple loop that periodically sends heartbeat method calls to the job tracker. Heartbeats tell the job tracker that a task tracker is alive, but they also double as a channel for messages. As a part of the heartbeat, a task tracker will indicate whether it is ready to run a new task, and if it is, the job tracker will allocate it a task, which it communicates to the task tracker using the heartbeat return value (step 7).

Before it can choose a task for the task tracker, the job tracker must choose a job to select the task from.

Task Execution

Now that the task tracker has been assigned a task, the next step is for it to run the task. First, it localizes the job JAR by copying it from the shared file system to the task tracker's file system. It also copies any files needed from the distributed cache by the application to the local disk. Second, it creates a local working directory for the task, and un-jars the contents of the JAR into this directory. Third, it creates an instance of Task Runner to run the task. Task Runner launches a new Java Virtual Machine (step 9) to run each task in (step 10), so that any bugs in the user-defined map and reduce functions don't affect the task tracker (by causing it to crash or hang, for example). It is, however, possible to reuse the JVM between tasks.

The child process communicates with its parent through the umbilical interface. This way it informs the parent of the task's progress every few seconds until the

task is complete.

Job Completion

When the job tracker receives a notification that the last task for a job is complete, it changes the status for the job to “successful.” Then, when the Job Client polls for status, it learns that the job has completed successfully, so it prints a message to tell the user and then returns from the run Job() method.

Java Map Reduce

Having run through how the Map Reduce program works, the next step is to express it in code. We need three things: a map function, a reduce function, and some code to run the job. The map function is represented by an implementation of the Mapper interface, which declares a map() method.

Analyzing the Data with Hadoop

To take advantage of the parallel processing that Hadoop provides, we need to express our query as a Map Reduce job. After some local, small-scale testing, we will be able to run it on a cluster of machines.

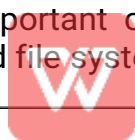
HADOOP LIBRARY FROM APACHE

Hadoop is an open source implementation of Map Reduce coded and released in Java (rather than C) by Apache. The Hadoop implementation of Map Reduce uses the Hadoop Distributed File System (HDFS) as its underlying layer rather than GFS. The Hadoop core is divided into two fundamental layers: the Map Reduce engine and HDFS. The Map Reduce engine is the computation engine running on top of HDFS as its data storage manager. The following two sections cover the details of these two fundamental layers.

HDFS: HDFS is a distributed file system inspired by GFS that organizes files and stores their data on a distributed computing system.

HDFS Architecture: HDFS has a master/slave architecture containing a single Name Node as the master and a number of Data Nodes as workers (slaves). To store a file in this architecture, HDFS splits the file into fixed-size blocks (e.g., 64 MB) and stores them on workers (Data Nodes). The mapping of blocks to Data Nodes is determined by the Name Node. The Name Node (master) also manages the file system’s metadata and namespace. In such systems, the namespace is the area maintaining the metadata, and metadata refers to all the information stored by a file system that is needed for overall management of all files. For example, Name Node in the metadata stores all information regarding the location of input splits/blocks in all Data Nodes. Each Data Node, usually one per node in a cluster, manages the storage attached to the node. Each Data Node is responsible for storing and retrieving its file blocks.

HDFS Features: Distributed file systems have special requirements, such as performance, scalability, concurrency control, fault tolerance, and security requirements, to operate efficiently. However, because HDFS is not a general-purpose file system, as it only executes specific types of applications, it does not need all the requirements of a general distributed file system. For example, security has never been supported for HDFS systems. The following discussion highlights two important characteristics of HDFS to distinguish it from other generic distributed file systems.



HDFS Fault Tolerance: One of the main aspects of HDFS is its fault tolerance characteristic. Since Hadoop is designed to be deployed on low-cost hardware by default, a hardware failure in this system is considered to be common rather than an exception. Therefore, Hadoop considers the following issues to fulfill reliability requirements of the filesystem:

- **Block replication** to reliably store data in HDFS, file blocks are replicated in this system. In other words, HDFS stores a file as a set of blocks and each block is replicated and distributed across the whole cluster. The replication factor is set by the user and is three by default.

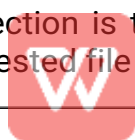
- **Replica placement** the placement of replicas is another factor to fulfill the desired fault tolerance in HDFS. Although storing replicas on different nodes (Data Nodes) located in different racks across the whole cluster provides more reliability, it is sometimes ignored as the cost of communication between two nodes in different racks is relatively high in comparison with that of different nodes located in the same rack. Therefore, sometimes HDFS compromises its reliability to achieve lower communication costs. For example, for the default replication factor of three, HDFS stores one replica in the same node the original data is stored, one replica on a different node but in the same rack, and one replica on a different node in a different rack to provide three copies of the data .

- **Heartbeat and Block report messages** Heartbeats and Block reports are periodic messages sent to the Name Node by each Data Node in a cluster. Receipt of a Heartbeat implies that the Data Node is functioning properly, while each Block report contains a list of all blocks on a Data Node. The Name Node receives such messages because it is the sole decision maker of all replicas in the system.

HDFS High-Throughput Access to Large Data Sets (Files): Because HDFS is primarily designed for batch processing rather than interactive processing, data access throughput in HDFS is more important than latency. Also, because applications run on HDFS typically have large data sets, individual files are broken into large blocks (e.g., 64 MB) to allow HDFS to decrease the amount of metadata storage required per file. This provides two advantages: The list of blocks per file will shrink as the size of individual blocks increases, and by keeping large amounts of data sequentially within a block, HDFS provides fast streaming reads of data.

HDFS Operation: The control flow of HDFS operations such as write and read can properly highlight roles of the Name Node and Data Nodes in the managing operations. In this section, the control flow of the main operations of HDFS on files is further described to manifest the interaction between the user, the Name Node, and the Data Nodes in such systems.

- **Reading a file** to read a file in HDFS, a user sends an `—open||` request to the Name Node to get the location of file blocks. For each file block, the Name Node returns the address of a set of Data Nodes containing replica information for the requested file. The number of addresses depends on the number of block replicas. Upon receiving such information, the user calls the read function to connect to the closest Data Node containing the first block of the file. After the first block is streamed from the respective Data Node to the user, the established connection is terminated and the same process is repeated for all blocks of the requested file until the whole file is streamed to the user.



- **Writing to a file** To write a file in HDFS, a user sends a `—create` request to the Name Node to create a new file in the file system namespace. If the file does not exist, the Name Node notifies the user and allows him to start writing data to the file by calling the write function. The first block of the file is written to an internal queue termed the data queue while a data streamer monitors its writing into a Data Node. Since each file block needs to be replicated by a predefined factor, the data streamer first sends a request to the Name Node to get a list of suitable Data Nodes to store replicas of the first block.

The steamer then stores the block in the first allocated Data Node. Afterward, the block is forwarded to the second Data Node by the first Data Node. The process continues until all allocated Data Nodes receive a replica of the first block from the previous Data Node.

Once this replication process is finalized, the same process starts for the second block and continues until all blocks of the file are stored and replicated on the file system.

Architecture of Map Reduce in Hadoop

The topmost layer of Hadoop is the Map Reduce engine that manages the data flow and control flow of Map Reduce jobs over distributed computing systems shows the Map Reduce engine architecture cooperating with HDFS. Similar to HDFS, the Map Reduce engine also has a master/slave architecture consisting of a single Job Tracker as the master and a number of Task Trackers as the slaves (workers). The Job Tracker manages the Map Reduce job over a cluster and is responsible for monitoring jobs and assigning tasks to Task Trackers. The Task Tracker manages the execution of the map and/or reduce tasks on a single computation node in the cluster. HDFS and Map Reduce architecture in Hadoop where boxes with different shadings refer to different functional nodes applied to different blocks of data.

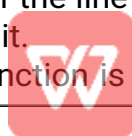
Each Task Tracker node has a number of simultaneous execution slots, each executing either a map or a reduce task. Slots are defined as the number of simultaneous threads supported by CPUs of the Task Tracker node. For example, a Task Tracker node with N CPUs, each supporting M threads, has $M * N$ simultaneous execution slots. It is worth noting that each data block is processed by one map task running on a single slot. Therefore, there is a one-to-one correspondence between map tasks in a Task Tracker and data blocks in the respective Data Node.

Map and Reduce functions

Map Reduce works by breaking the processing into two phases: the map phase and the reduce phase. Each phase has key-value pairs as input and output, the types of which may be chosen by the programmer. The programmer also specifies two functions: the map function and the reduce function.

The input to our map phase is the raw NCDC data. We choose a text input format that gives us each line in the dataset as a text value. The key is the offset of the beginning of the line from the beginning of the file, but as we have no need for this, we ignore it.

Our map function is simple. We pull out the year and the air temperature,



since these are the only fields we are interested in. In this case, the map function is just a data preparation phase, setting up the data in such a way that the reducer function can do its work on it: finding the maximum temperature for each year. The map function is also a good place to drop bad records: here we filter out temperatures that are missing, suspect, or erroneous.

These lines are presented to the map function as the key-value pairs:

The keys are the line offsets within the file, which we ignore in our map function. The map function merely extracts the year and the air temperature (indicated in bold text), and emits them as its output (the temperature values have been interpreted as integers):

The output from the map function is processed by the Map Reduce framework before being sent to the reduce function. This processing sorts and groups the key-value pairs by key. So, continuing the example, our reduce function sees the following input:

Java Map Reduce

Having run through how the Map Reduce program works, the next step is to express it in code. We need three things: a map function, a reduce function, and some code to run the job. The map function is represented by an implementation of the Mapper interface, which declares a map() method.

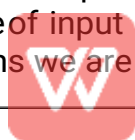
The implementation of four map function.

```
public class Max Temperature Mapper extends Map Reduce
Baseimplements Mapper<Long Writable, Text, Text, Int
Writable> { private static final int MISSING = 9999;
public void map(Long Writable key, Text value,
Output Collector<Text, Int Writable> output, Reporter reporter)
throws IOException {
String line = value.toString();
String year = line.substring(15,
19);intairTemperature;
if (line.charAt(87) == '+') { // parseInt doesn't like leading plus signs
airTemperature = Integer.parseInt(line.substring(88, 92));
} else {
airTemperature = Integer.parseInt(line.substring(87, 92));
}
String quality = line.substring(92, 93);
if (airTemperature != MISSING && quality.matches("[01459]"))
{output.collect(new Text(year), new IntWritable(airTemperature));
}
}
}
```

The Mapper interface is a generic type, with four formal type parameters that specify the input key, input value, output key, and output value types of the map function.

Hadoop provides its own set of basic types that are optimized for network serialization. These are found in the org.apache.hadoop.io package. Here we use Long Writable, which corresponds to a Java Long, Text (like Java String), and Int Writable (like Java Integer).

The map() method is passed a key and a value. We convert the Text value containing the line of input into a Java String, then use its substring() method to extract the columns we are interested in.



The `map()` method also provides an instance of Output Collector to write the output to. In this case, we write the year as a Text object (since we are just using it as a key), and the temperature is wrapped in an Int Writable. We write an output record only if the temperature is present and the quality code indicates the temperature reading is OK. The reduce function is similarly defined using a Reducer, as illustrated in [Example 2-4](#).

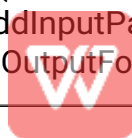
Example 2-4. Reducer for maximum temperature example

```
import java.io.IOException;
import java.util.Iterator;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;
public class MaxTemperatureReducer extends
MapReduceBase implements Reducer<Text, IntWritable,
Text, IntWritable> { public void reduce(Text key,
Iterator<IntWritable> values, OutputCollector<Text,
IntWritable> output, Reporter reporter)
throws IOException {
int maxVal = Integer.MIN_VALUE;
while (values.hasNext()) {
maxVal = Math.max(maxVal, values.next().get());
}
output.collect(key, new IntWritable(maxVal));}}
```

Again, four formal type parameters are used to specify the input and output types, this time for the reduce function. The input types of the reduce function must match the output types of the map function: Text and IntWritable. And in this case, the output types of the reduce function are Text and IntWritable, for a year and its maximum. The third piece of code runs the Map Reduce job (see [Example 2-5](#)).

Example 2-5. Application to find the maximum temperature in the weather dataset

```
import java.io.IOException;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
public class MaxTemperature {
public static void main(String[] args) throws
IOException {if (args.length != 2) {
System.err.println("Usage: MaxTemperature<input path><output
path>"); System.exit(-1);
}
JobConf conf = new
JobConf(MaxTemperature.class);
conf.setJobName("Max temperature");
FileInputFormat.addInputPath(conf, new
Path(args[0])); FileOutputFormat.setOutputPath(conf,
```



```

new Path(args[1]));
conf.setMapperClass(MaxTemperatureMapper.class);
conf.setReducerClass(MaxTemperatureReducer.class);
conf.setOutputKeyClass(Text.class);
conf.setOutputValueClass(IntWritable.class);
JobClient.runJob(conf);
}
}

```

A Job Conf object forms the specification of the job. It gives you control over how the job is run. When we run this job on a Hadoop cluster, we will package the code into a JAR file (which Hadoop will distribute around the cluster). Rather than explicitly specify the name of the JAR file, we can pass a class in the Job Conf constructor, which Hadoop will use to locate the relevant JAR file by looking for the JAR file containing this class.

Having constructed a Job Conf object, we specify the input and output paths. An input path is specified by calling the static add Input Path() method on File Input Format, and it can be a single file, a directory (in which case, the input forms all the files in that directory), or a file pattern. As the name suggests, add Input Path() can be called more than once to use input from multiple paths.

The output path (of which there is only one) is specified by the static set Output Path() method on File Output Format. It specifies a directory where the output files from the reducer functions are written. The directory shouldn't exist before running the job, as Hadoop will complain and not run the job. This precaution is to prevent data loss (it can be very annoying to accidentally overwrite the output of a long job with another).

Next, we specify the map and reduce types to use via the set Mapper Class() and Set Reducer Class() methods.

The set Output Key Class() and set Output Value Class() methods control the output types for the map and the reduce functions, which are often the same, as they are in our case. If they are different, then the map output types can be set using the methods set Map Output Key Class() and set Map Output Value Class().

The input types are controlled via the input format, which we have not explicitly set since we are using the default Text Input Format.

After setting the classes that define the map and reduce functions, we are ready to run the job. The static run Job() method on Job Client submits the job and waits for it to finish, writing information about its progress to the console.

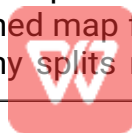
Data Flow

First, some terminology. A Map Reduce job is a unit of work that the client wants to be performed: it consists of the input data, the Map Reduce program, and configuration information. Hadoop runs the job by dividing it into tasks, of which there are two types: map tasks and reduce tasks.

There are two types of nodes that control the job execution process: a job tracker and a number of task trackers. The job tracker coordinates all the jobs run on the system by scheduling tasks to run on task trackers. Task trackers run tasks and send progress reports to the job tracker, which keeps a record of the overall progress of each job. If a task fails, the job tracker can reschedule it on a different task tracker.

Hadoop divides the input to a Map Reduce job into fixed-size pieces called input splits, or just splits. Hadoop creates one map task for each split, which runs the user defined map function for each record in the split.

Having many splits means the time taken to process each split is small



compared to the time to process the whole input. So if we are processing the splits in parallel, the processing is better load-balanced if the splits are small, since a faster machine will be able to process proportionally more splits over the course of the job than a slower machine. Even if the machines are identical, failed processes or other jobs running concurrently make load balancing desirable, and the quality of the load balancing increases as the splits become more fine-grained.

On the other hand, if splits are too small, then the overhead of managing the splits and of map task creation begins to dominate the total job execution time. For most jobs, a good split size tends to be the size of an HDFS block, 64 MB by default, although this can be changed for the cluster (for all newly created files), or specified when each file is created.

Hadoop does its best to run the map task on a node where the input data resides in HDFS. This is called the data locality optimization. It should now be clear why the optimal split size is the same as the block size: it is the largest size of input that can be guaranteed to be stored on a single node. If the split spanned two blocks, it would be unlikely that any HDFS node stored both blocks, so some of the split would have to be transferred across the network to the node running the map task, which is clearly less efficient than running the whole map task using local data.

Map tasks write their output to the local disk, not to HDFS. Why is this? Map output is intermediate output: it's processed by reduce tasks to produce the final output, and once the job is complete the map output can be thrown away. So storing it in HDFS, with replication, would be overkill. If the node running the map task fails before the map output has been consumed by the reduce task, then Hadoop will automatically run.

Specifying Input Output Parameters, Map Parameters

A record emitted from a map will be serialized into a buffer and metadata will be stored into accounting buffers. As described in the following options, when either the serialization buffer or the metadata exceed a threshold, the contents of the buffers will be sorted and written to disk in the background while the map continues to output records. If either buffer fills completely while the spill is in progress, the map thread will block. When the map is finished, any remaining records are written to disk and all on-disk segments are merged into a single file. Minimizing the number of spills to disk can decrease map time, but a larger buffer also decreases the memory available to the mapper.

Name	Type	Description
io.sort.mb	int	The cumulative size of the serialization and accounting buffers storing records emitted from the map, in megabytes.
io.sort.record.percent	float	The ratio of serialization to accounting space can be adjusted. Each serialized record requires 16 bytes of accounting information in addition to its serialized size to effect the sort. This percentage of space allocated from io.sort.mb affects the probability of a spill to disk being caused by either exhaustion of the serialization buffer or the accounting space. Clearly, for a map outputting small records, a higher value than the default will likely decrease the number of spills to disk.



io.sort.spill.percent	float	This is the threshold for the accounting and serialization buffers. When this percentage of either buffer has filled, their contents will be spilled to disk in the background. Let io.sort.record.percent be r , io.sort.mb be x , and this value be q . The maximum number of records collected before the collection thread will spill is $r * x * q * 2^{16}$. Note that a higher value may decrease the number of- or even eliminate- merges, but will also increase the probability of the map task getting blocked. The lowest average map times are usually obtained by accurately estimating the size of the map output and preventing multiple spills.
-----------------------	-------	---

Other notes

- If either spill threshold is exceeded while a spill is in progress, collection will continue until the spill is finished. For example, if io.sort.buffer.spill.percent is set to 0.33, and the remainder of the buffer is filled while the spill runs, the next spill will include all the collected records, or 0.66 of the buffer, and will not generate additional spills. In other words, the thresholds are defining triggers, not blocking.
- A record larger than the serialization buffer will first trigger a spill, then be spilled to a separate file. It is undefined whether or not this record will first pass through the combiner.

Shuffle/Reduce Parameters

As described previously, each reduce fetches the output assigned to it by the Partitioner via HTTP into memory and periodically merges these outputs to disk. If intermediate compression of map outputs is turned on, each output is decompressed into memory. The following options affect the frequency of these merges to disk prior to the reduce and the memory allocated to map output during the reduce.

Name	Type	Description
io.sort.factor	int	Specifies the number of segments on disk to be merged at the same time. It limits the number of open files and compression codecs during the merge. If the number of files exceeds this limit, the merge will proceed in several passes. Though this limit also applies to the map, most jobs should be configured so that hitting this limit is unlikely there.



mapred.inmem.merge.threshold	int	The number of sorted map outputs fetched into memory before being merged to disk. Like the spill thresholds in the preceding note, this is not defining a unit of partition, but a trigger. In practice, this is usually set very high (1000) or disabled (0), since merging in-memory segments is often less expensive than merging from disk (see notes following this table). This threshold influences only the frequency of in-memory merges during the shuffle.
mapred.job.shuffle.merge.percent	float	The memory threshold for fetched map outputs before an in-memory merge is started, expressed as a percentage of memory allocated to storing map outputs in memory. Since map outputs that can't fit in memory can be stalled, setting this high may decrease parallelism between the fetch and merge. Conversely, values as high as 1.0 have been effective for reduces whose input can fit entirely in memory. This parameter influences only the frequency of in-memory merges during the shuffle.
mapred.job.shuffle.input.buffer.percent	float	The percentage of memory- relative to the maximum heap size as typically specified in <code>mapred.reduce.child.java.opts-</code> that can be allocated to storing map outputs during the shuffle. Though some memory should be set aside for the framework, in general it is advantageous to set this high enough to store large and numerous map outputs.
mapred.job.reduce.input.buffer.percent	float	The percentage of memory relative to the maximum heap size in which map outputs may be retained during the reduce. When the reduce begins, map outputs will be merged to disk until those that remain are under the resource limit this defines. By default, all map outputs are merged to disk before the reduce begins to maximize the memory available to the reduce. For less memory- intensive reduces, this should be increased to avoid trips to disk.



Configuring and Running a Job

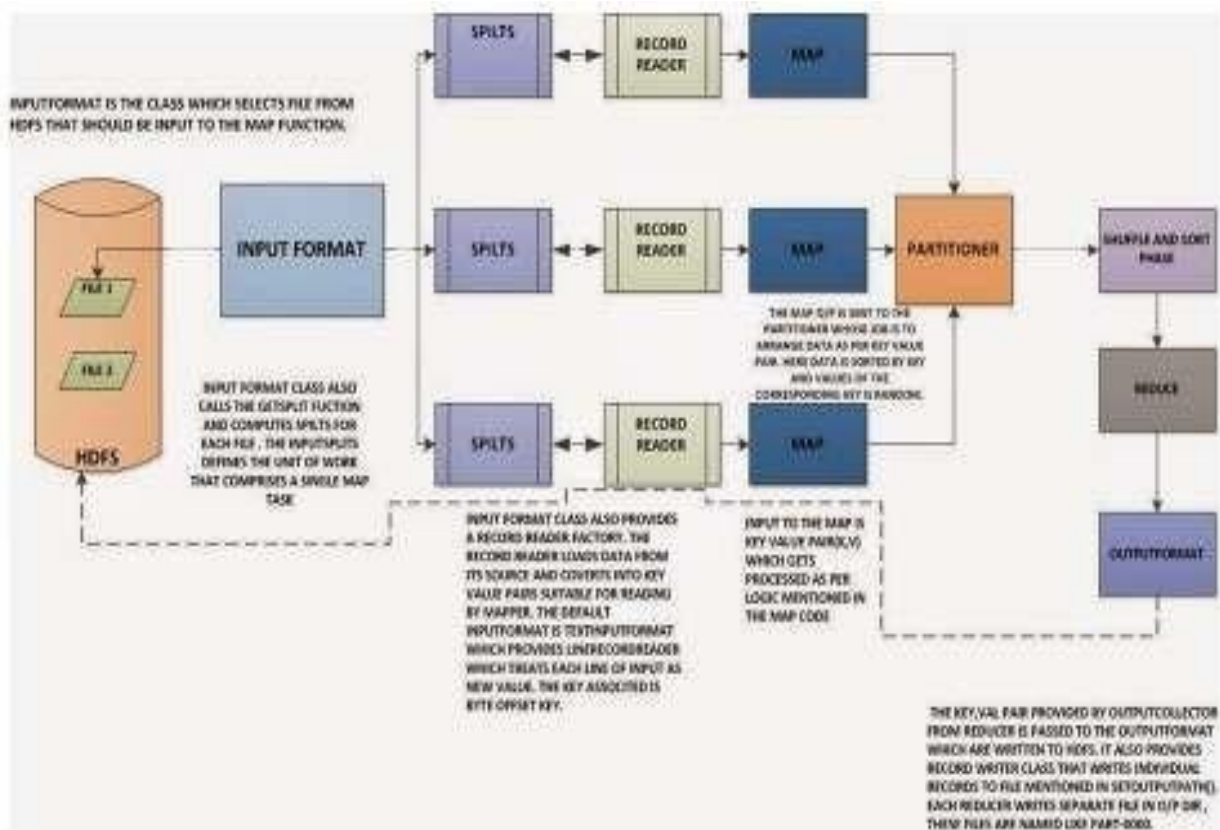
Running a Job in Hadoop

Three components contribute in running a job in this system: a user node, a Job Tracker, and several Task Trackers. The data flow starts by calling the `runJob(conf)` function inside a user program running on the user node, in which `conf` is an object containing some tuning parameters for the Map Reduce framework and HDFS.

The `runJob(conf)` function and `conf` are comparable to the `Map Reduce(Spec, &Results)` function and `Spec` in the first implementation of Map Reduce by Google, depicts the data flow of running a Map Reduce job in Hadoop. Data flow in running a Map Reduce job at various task trackers using the Hadoop library.

- **Job Submission** Each job is submitted from a user node to the Job Tracker node that might be situated in a different node within the cluster through the following procedure:
 - A user node asks for a new job ID from the Job Tracker and computes input file splits.
 - The user node copies some resources, such as the job's JAR file, configuration file, and computed input splits, to the Job Tracker's file system.
 - The user node submits the job to the Job Tracker by calling the `submitJob()` function.
 - **Task assignment** The Job Tracker creates one map task for each computed input split by the user node and assigns the map tasks to the execution slots of the Task Trackers.
- The Job Tracker considers the localization of the data when assigning the map tasks to the Task Trackers. The Job Tracker also creates reduce tasks and assigns them to the Task Trackers. The number of reduce tasks is predetermined by the user, and there is no locality consideration in assigning them.
- **Task execution** The control flow to execute a task (either map or reduce) starts inside the Task Tracker by copying the job JAR file to its file system. Instructions inside the job JAR file are executed after launching a Java Virtual Machine (JVM) to run its map or reduce task.
- **Task running check** A task running check is performed by receiving periodic heartbeat messages to the Job Tracker from the Task Trackers. Each heartbeat notifies the Job Tracker that the sending Task Tracker is alive, and whether the sending Task Tracker is ready to run a new task.





Design of Hadoop file

systemBasic File system

Operations

The file system is ready to be used, and we can do all of the usual file system operations such as reading files, creating directories, moving files, deleting data, and listing directories. You can type `hadoop fs -help` to get detailed help on every command.

Start by copying a file from the local file system to HDFS:

```
% hadoop fs -copy From Local
input/docs/quangle.txt hdfs://localhost/user/tom/quangle.txt
```

This command invokes Hadoop's file system shell command `fs`, which supports a number of subcommands—in this case, we are running `-copy From Local`. The local file `quangle.txt` is copied to the file `/user/tom/quangle.txt` on the HDFS instance running on local host. In fact, we could have omitted the scheme and host of the URI and picked

up the default, `hdfs://localhost`, as specified in `core-site.xml`.

```
% hadoopfs -copyFromLocal input/docs/quangle.txt /user/tom/quangle.txt
```

We could also have used a relative path, and copied the file to our home directory in HDFS, which in this case is `/user/tom`:

```
% hadoopfs -copyFromLocal input/docs/quangle.txt quangle.txt
```

Let's copy the file back to the local filesystem and check whether it's the same:

```
% hadoopfs -copyToLocal quangle.txt quangle.copy.txt
% md5 input/docs/quangle.txt quangle.copy.txt
```

```
MD5 (input/docs/quangle.txt) =
a16f231da6b05e2ba7a339320e7dacd9MD5 (quangle.copy.txt)
= a16f231da6b05e2ba7a339320e7dacd9
```

The MD5 digests are the same, showing that the file survived its trip to HDFS and is back intact.

Finally, let's look at an HDFS file listing. We create a directory first just to see how it is displayed in the listing:

```
% hadoopfs -mkdir books
% hadoopfs -ls .
Found 2 items
drwxr-xr-x - tom supergroup 0 2009-04-02 22:41 /user/tom/books
-rw-r--r-- 1 tom supergroup 118 2009-04-02 22:29 /user/tom/quangle.txt
```

The information returned is very similar to the Unix command `ls -l`, with a few minor differences. The first column shows the file mode. The second column is the replication factor of the file (something a traditional Unix filesystems does not have). Remember we set the default replication factor in the site-wide configuration to be 1, which is why we see the same value here. The entry in this column is empty for directories since the concept of replication does not apply to them—directories are treated as metadata and stored by the namenode, not the datanodes. The third and fourth columns show the file owner and group. The fifth column is the size of the file in bytes, or zero for directories. The six and seventh columns are the last modified date and time. Finally, the eighth column is the absolute name of the file or directory.

HDFS concepts

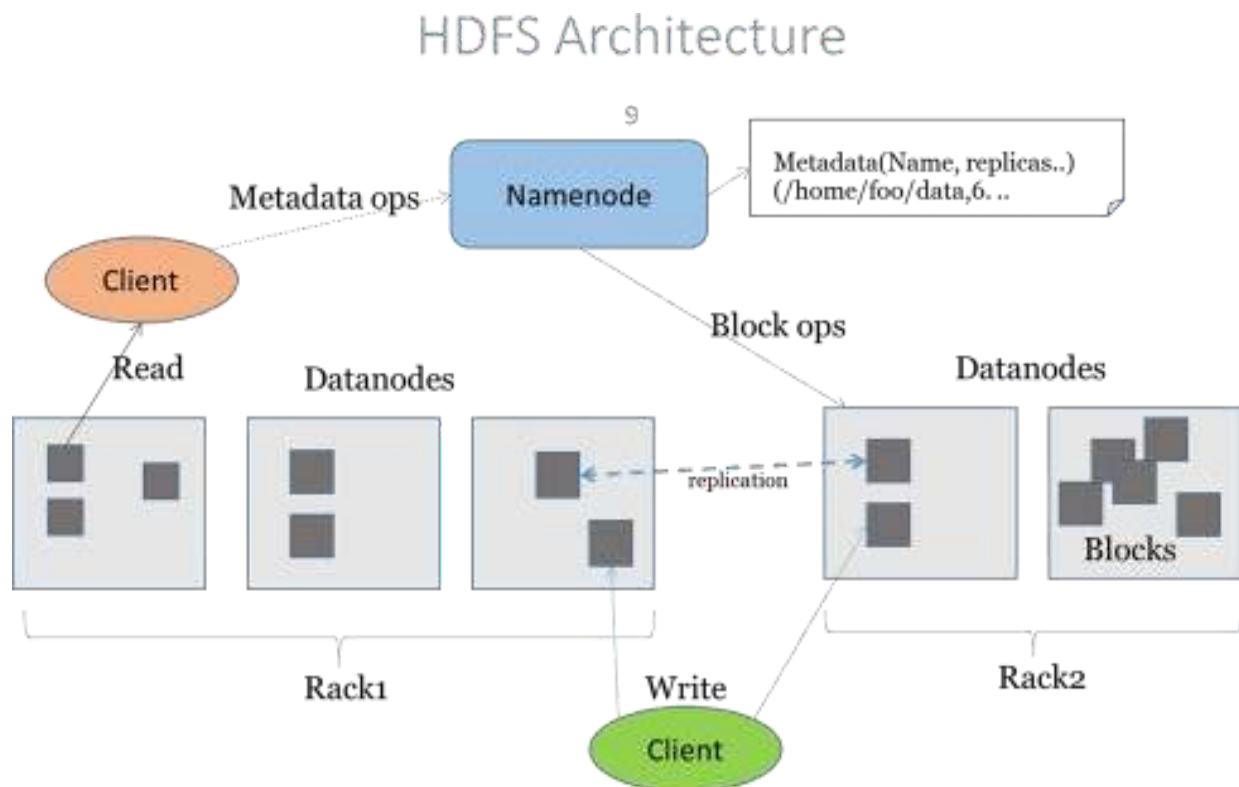
- Master/slave architecture
- HDFS cluster consists of a single Name node, a master server that manages the filesystem namespace and regulates access to files by clients.
- There are a number of Data Nodes usually one per node in a cluster. The Data Nodes manage storage attached to the nodes that they run on.
- HDFS exposes a file system namespace and allows user data to be stored in files. A file is split into one or more blocks and set of blocks are stored in Data Nodes. Data Nodes: serves read, write requests, performs block creation, deletion, and replication upon instruction from Name node.
- Hierarchical file system with directories and files
- Create, remove, move, rename etc.
- Name node maintains the file system
- Any meta information changes to the file system recorded by the Name node.
- An application can specify the number of replicas of the file needed: replication factor of the file. This information is stored in the Name node

Data Replication

- HDFS is designed to store very large files across machines in a large cluster. Each file is a sequence of blocks.
- All blocks in the file except the last are of the same size. Blocks are replicated for fault tolerance.
- Block size and replicas are configurable per file.

The Name node receives a Heartbeat and a Block Report from each Data Node in the cluster.

Block Report contains all the blocks on a Data node.



6/15/2017

Command line interface in HDFS

There are many other interfaces to HDFS, but the command line is one of the simplest, and to many developers the most familiar. We are going to run HDFS on one machine, so first follow the instructions for setting up Hadoop in pseudo-distributed mode

.Later you'll see how to run on a cluster of machines to give us scalability and fault tolerance. There are two properties that we set in the pseudo-distributed configuration that deserve further explanation. The first is `fs.default.name`, set to `hdfs://localhost/`, which is used to set a default file system for Hadoop. File systems are specified by a URI, and here we have used a `hdfs` URI to configure Hadoop to use HDFS by default. The HDFS daemons will use this property to determine the host and port for the HDFS name node. We'll be running it on local host, on the default HDFS port, 8020. And HDFS clients will use this property to work out where the name node is running so they can connect to it.

We set the second property, `dfs.replication`, to one so that HDFS doesn't replicate file system blocks by the usual default of three. When running with a single data node, HDFS can't replicate blocks to three data nodes, so it would perpetually warn about blocks being under-replicated. This setting solves that problem.

The HDFS can be manipulated through a Java API or through a command line interface. It commands for manipulating HDFS through Hadoop's command



Edit with WPS Office

line interface begin with "hadoop", a space, and "fs". This is the file system shell. This is followed by the command name as an argument to "hadoop fs". These commands start with a dash. For example, the "ls" command for listing a directory is a common UNIX command and is preceded with a dash. As on UNIX systems, ls can take a path as an argument.

In this example, the path is the current directory, represented by a single dot. Let's begin looking at these HDFS commands by starting up Hadoop. We'll run the start.sh script to bring up each of the Hadoop nodes: first the Name Node, the Secondary Name Node, a Data Node, the Job Tracker, and a Task Tracker. Now we'll run the -ls command to give us the current directory.

As we saw for the "ls" command, the file system shell commands can take paths as arguments. These paths can be expressed in the form of uniform resource identifiers or URIs. The URI format consists of a scheme, an authority, and path. There are multiple schemes support. The local file system has a scheme of "file". HDFS has a scheme called "hdfs".

For example, let us say you wish to copy a file called "myfile.txt" from your local file system to an HDFS file system on the local host. You can do this by issuing the command shown. The copy From Local command takes a URI for the source and a URI for the destination. The scheme and the authority do not always need to be specified. Instead you may rely on their default values. These defaults can be overridden by specifying them in a file named core-site.xml in the conf directory of your Hadoop installation.

Now let's examine the copy From Local command. We will copy a file named myfile.txt to the HDFS. Now we can just examine the HDFS with the -ls command and see that our file has been copied. And there it is. HDFS supports many POSIX-like commands. HDFS is not a fully POSIX compliant file system, but it supports many of the commands. The HDFS commands are mostly easily-recognized UNIX commands like cat and chmod. There are also a few commands that are specific to HDFS such as copy From Local.

Copy from Local and put are two HDFS-specific commands that do the same thing - copy files from the local file system to a location on another file system. Their opposite is the copy To Local command which can also be referred to as get. This command copies files out of the file system you specify and into the local file system. Get merge is an enhanced form of get that can merge the files from multiple locations into a single local file.

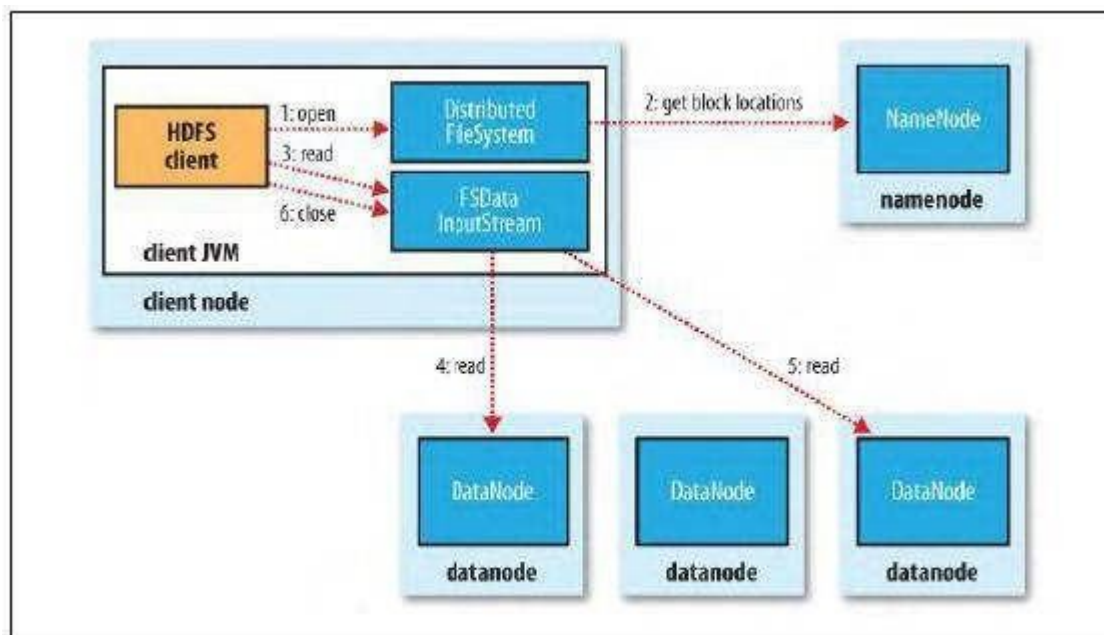
Now let's try out the get merge command. First we will copy the myfile.txt into a second copy on the HDFS called myfile2.txt. Then we will use the get merge command to combine the two and write them as one file in the local file system called myfiles.txt. Now if we cat out the value, we see the text twice. setrep lets you override the default level of replication to a level you specify. You can do this for one file or, with the -R option, to an entire tree. This command returns immediately after requesting the new replication level. If you want the command to block until the job is done, pass the -w option.

Dataflow of file read and write in HDFS

Dataflow of file read in HDFS



To get an idea of how data flows between the client interacting with HDFS, the name node and the data node, consider the below diagram, which shows the main sequence of events when reading a file.



A client reading data from HDFS

The client opens the file it wishes to read by calling `open()` on the File System object, which for HDFS is an instance of Distributed File System (step 1). Distributed File System calls the name node, using RPC, to determine the locations of the blocks for the first few blocks in the file (step 2). For each block, the name node returns the addresses of the data nodes that have a copy of that block. Furthermore, the data nodes are sorted according to their proximity to the client. If the client is itself a data node (in the case of a Map Reduce task, for instance), then it will read from the local data node.

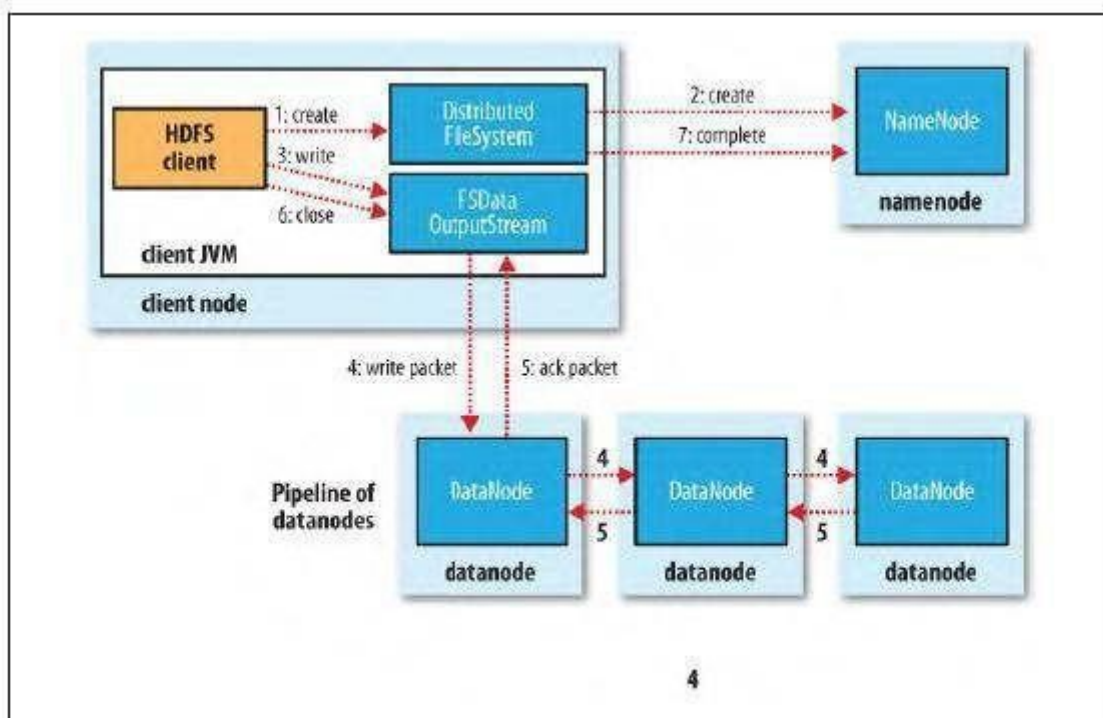
The Distributed File System returns a FS Data Input Stream to the client for it to read data from. FS Data Input Stream in turn wraps a DFS Input Stream, which manages the data node and name node I/O. The client then calls `read()` on the stream (step 3). DFS Input Stream, which has stored the data node addresses for the first few blocks in the file, then connects to the first (closest) data node for the first block in the file. Data is streamed from the data node back to the client, which calls `read()` repeatedly on the stream (step 4). When the end of the block is reached, DFS Input Stream will close the connection to the data node, then find the best data node for the next block (step 5). This happens transparently to the client, which from its point of view is just reading a continuous stream. Blocks are read in order with the DFS Input Stream opening new connections to data nodes as the client reads through the stream. It will also call the name node to retrieve the data node locations for the next batch of blocks as needed. When the client has finished reading, it calls `close()` on the FS Data Input Stream (step 6).



One important aspect of this design is that the client contacts data nodes directly to retrieve data, and is guided by the name node to the best data node for each block. This design allows HDFS to scale to large number of concurrent clients, since the data traffic is spread across all the data nodes in the cluster. The name node meanwhile merely has to service block location requests (which it stores in memory, making them very efficient), and does not, for example, serve data, which would quickly become a bottleneck as the number of clients grew.

Dataflow of file write in HDFS

The case we're going to consider is the case of creating a new file, writing data to it, then closing the file.



A client writing data to HDFS

The client creates the file by calling `create()` on Distributed File System (step 1). Distributed File System makes an RPC call to the name node to create a new file in the file system's namespace, with no blocks associated with it (step 2). The name node performs various checks to make sure the file doesn't already exist, and that the client has the right permissions to create the file. If these checks pass, the name node makes a record of the new file; otherwise, file creation fails and the client is thrown an IO Exception. The Distributed File System returns a FS Data Output Stream for the client to start writing data to. Just as in the read case, FS Data Output Stream wraps a DFS Output Stream, which handles communication with the data nodes and name node.

As the client writes data (step 3), DFS Output Stream splits it into packets, which it writes to an internal queue, called the data queue. The data queue is consumed by the Data Streamer, whose responsibility it is to ask the name node to allocate new blocks by picking a list of suitable data nodes to store the replicas. The list of data nodes forms a pipeline—we'll assume the replication level is 3, so there are three nodes in the pipeline. The Data Streamer streams the packets to the first data node in the pipeline, which stores the packet and forwards it to the second data node in the pipeline. Similarly, the second data node stores the packet and forwards it to the third (and last) data node in the pipeline (step 4). DFS Output Stream also maintains an internal queue of packets

that are waiting to be acknowledged by data nodes, called the ack queue. A packet is removed from the ack queue only when it has been acknowledged by all the data nodes in the pipeline (step 5).



If a data node fails while data is being written to it, then the following actions are taken, which are transparent to the client writing the data. First the pipeline is closed, and any packets in the ack queue are added to the front of the data queue so that data nodes that are downstream from the failed node will not miss any packets. The current block on the good data nodes is given a new identity, which is communicated to the name node, so that the partial block on the failed data node will be deleted if the failed data node recovers later on. The failed data node is removed from the pipeline and the remainder of the block's data is written to the two good data nodes in the pipeline. The name node notices that the block is under-replicated, and it arranges for a further replica to be created on another node. Subsequent blocks are then treated as normal.

When the client has finished writing data it calls `close()` on the stream (step 6). This action flushes all the remaining packets to the data node pipeline and waits for acknowledgments before contacting the name node to signal that the file is complete (step 7). The name node already knows which blocks the file is made up of (via Data Streamer asking for block allocations), so it only has to wait for blocks to be minimally replicated before returning successfully.

Reading Data from a Hadoop URL and Deleting Data

The Hadoop's File System class: the API for interacting with one of Hadoop's file systems. While we focus mainly on the HDFS implementation, Distributed File System, in general you should strive to write your code against the File System abstract class, to retain portability across file systems. This is very useful when testing your program.

One of the simplest ways to read a file from a Hadoop file system is by using a `java.net.URL` object to open a stream to read the data from. The general idiom is:

```
try {
    in = new URL("hdfs://host/path").openStream();
    // process in
} finally
{ IOUtils.closeStream(in);
}
```

There's a little bit more work required to make Java recognize Hadoop's `hdfs` URL scheme. This is achieved by calling the `setURLStreamHandlerFactory` method on `URL` with an instance of `FsUrlStreamHandlerFactory`. This method can only be called once per JVM, so it is typically executed in a static block. This limitation means that if some other part of your program perhaps a third-party component outside your control sets `URLStreamHandlerFactory`, you won't be able to use this approach for reading data from Hadoop. The next section discusses an alternative. A program for displaying files from Hadoop file systems on standard output, like the Unix `cat` command.

We make use of the handy `IOUtils` class that comes with Hadoop for closing the stream in the `finally` clause, and also for copying bytes between the input stream and the output stream (`System.out` in this case). The last two arguments to the `copyBytes` method are the buffer size used for copying, and whether to close the streams when the copy is complete. We close the input stream ourselves, and `System.out` doesn't need to be closed.



Displaying files from a Hadoop filesystem on standard output using a `URLStreamHandler`

```
public class URLLCat {
    static {
        URL.setURLStreamHandlerFactory(new FsUrlStreamHandlerFactory());
    }

    public static void main(String[] args) throws Exception {
        InputStream in = null;
        try {
            in = new URL(args[0]).openStream();
            IOUtils.copyBytes(in, System.out, 4096, false);
        } finally {
            IOUtils.closeStream(in);
        }
    }
}
```

Deleting Data

Use the `delete()` method on File System to permanently remove files or directories: `public Boolean delete(Path f, boolean recursive)` throws `IOException`. If `f` is a file or an empty directory, then the value of `recursive` is ignored. A nonempty directory is only deleted, along with its contents, if `recursive` is `true` (otherwise an `IOException` is thrown).

File pattern in HDFS

It is a common requirement to process sets of files in a single operation. For example, a Map Reduce job for log processing might analyze a month worth of files, contained in a number of directories. Rather than having to enumerate each file and directory to specify the input, it is convenient to use wildcard characters to match multiple files with a single expression, an operation that is known as globbing. Hadoop provides two File System methods for processing globs:

`public FileStatus[] globStatus(Path pathPattern) throws IOException`
`public FileStatus[] globStatus(Path pathPattern, PathFilter filter) throws IOException`
 The `globStatus()` methods returns an array of `FileStatus` objects whose paths match the

supplied pattern, sorted by path. An optional `PathFilter` can be specified to restrict the matches further.

glob	name	matches
*	asterisk	Matches zero or more characters
?	question mark	Matches a single character
[a-b]	character class	Matches a single character in the set {a, b}
[^ab]	negated character class	Matches a single character that is not in the set {a, b}
[a-b]	character range	Matches a single character in the (closed) range [a, b], where a is lexicographically less than or equal to b
[^a-b]	negated character range	Matches a single character that is not in the (closed) range [a, b], where a is lexicographically less than or equal to b
{a,b}	alternation	Matches either expression a or b
\c	escaped character	Matches character c when it is a metacharacter

Hadoop supports the same set of glob characters as Unixbash.

Imagine that logfiles are stored in a directory structure organized

hierarchically by date. So, for example, logfiles for the last day of 2007 would go in a directory named

/2007/12/31. Suppose that the full file listing is:

/2007/12/30

/2007/12/31

/2008/01/01

/2008/01/02

CLOUD PLATFORM IN INDUSTRY:

Amazon Web Services:

AWS provides different wide-ranging cloud IaaS services, which range from virtual compute, storage, and networking to complete computing stacks. AWS is well known for its storage and compute on demand services, named as Elastic Compute Cloud (EC2) and Simple Storage Service (S3). EC2 offers customizable virtual hardware to the end user which can be utilized as the base infrastructure for deploying computing systems on the cloud. It is likely to choose from a large variety of virtual hardware configurations including GPU and cluster instances. Either the AWS console, which is a wide-ranged Web portal for retrieving AWS services, or the web services API available for several programming languages is used to deploy the EC2 instances. EC2 also offers the capability of saving an explicit running instance as image, thus allowing users to create their own templates for deploying systems. S3 stores these templates and delivers persistent storage on demand. S3 is well ordered into buckets which contain objects that are stored in binary form and can be grown with attributes. End users can store objects of any size, from basic file to full disk images and have them retrieved from anywhere. In addition, EC2 and S3, a wide range of services can be leveraged to build virtual computing systems including: networking support, caching system, DNS, database support, and others.

Google AppEngine –

Google AppEngine is a scalable runtime environment frequently dedicated to executing web applications. These utilize benefits of the large computing infrastructure of Google to dynamically scale as per the demand. AppEngine offers both a secure execution environment and a collection of which simplifies the development of scalable and high-performance Web applications. These services include: in-memory caching, scalable data store, job queues, messaging, and cron tasks. Developers and Engineers can build and test applications on their own systems by using the AppEngine SDK, which replicates the production runtime environment, and helps test and profile applications. On completion of development, Developers can easily move their applications to AppEngine, set quotas to containing the cost generated, and make it available to the world. Currently, the supported programming languages are Python, Java, and Go.

Microsoft Azure –

Microsoft Azure is a Cloud operating system and a platform in which user can develop the applications in the cloud. Generally, a scalable runtime environment for web applications and distributed applications is provided. Application in Azure are organized around the fact of roles, which identify a distribution unit for applications and express the application's logic. Azure provides a set of additional services that complement application execution such as support for storage, networking, caching, content delivery, and others.

Hadoop –

Apache Hadoop is an open source framework that is appropriate for processing large data sets on commodity hardware. Hadoop is an implementation of MapReduce, an application programming model which is developed by Google. This model provides two fundamental operations for data processing: map and reduce. Yahoo! is the sponsor of the Apache Hadoop project, and has put considerable effort in transforming the project to an

enterprise-ready cloud computing platform for data processing. Hadoop is an integral part of the Yahoo! Cloud infrastructure and it supports many business processes of the corporates. Currently, Yahoo! Manages the world's largest Hadoop cluster, which is also available to academic institutions.

COMPUTE SERVICES:

Understanding the fundamentals of AWS is critical if you want to deploy services and resources within the AWS Cloud. The Compute category of services are key resources that allow you to carry out computational abilities via a series of instructions used by applications and systems. These resources cover a range of different services and features, these being:

- EC2 - Amazon Elastic Compute Cloud
- ECS - Amazon Elastic Container Service
- ECR - Amazon Elastic Container Registry
- EKS - Amazon Elastic Container Service for Kubernetes
- AWS Elastic Beanstalk
- AWS Lambda
- AWS Batch
- Amazon Lightsail

This course will provide the fundamental elements of all of these Compute services and features that will allow you to select the most appropriate service for your project and implementations. Each have their advantages by providing something of value that's different to the others, which will all be discussed.

Topics covered within this course consist of:

- What is Compute: This lecture explains what 'Compute' is and what is meant by Compute resources and services
- Amazon Elastic Compute Cloud (EC2): This is one of the most common Compute services, as a result this will likely be the longest lecture as you will cover a lot of elements around EC2 to ensure you are aware of how it's put together and how it works
- Amazon ECS (EC2 Container Service): Within this lecture you will gain a high-level overview of what the EC2 Container Service is and how it relates to Docker
- Amazon Elastic Container Registry: In this lecture you will consider how this service links closely with ECS to provide a secure location to store and manage your docker images
- Amazon Elastic Container Service for Kubernetes (EKS): Here you will look at how EKS provides a managed service, allowing you to run Kubernetes across your AWS infrastructure without having to take care of running the Kubernetes control plane
- AWS Elastic Beanstalk: This lecture will provide an overview of the service, showing you how it's used to automatically deploy applications using EC2 and a number of other AWS services
- AWS Lambda: This lecture covers the Lambda 'serverless' service, where you will explore what serverless means and how this service is used to run your own code in response to events
- AWS Batch: Here you will consider a high-level overview of this service that relates to Batch Computing
- Amazon Lightsail: Finally we will look at Amazon Lightsail, a Virtual Private Server solution used for small-scale projects and use cases

If you want to learn the differences between the different Compute services, then this course is for you!



With demonstrations provided, along with links to a number of our labs that allow you to gain hands-on experience in using many of these services, you will gain a solid understanding of the Compute services used within AWS.

STORAGE SERVICES:

Storage Systems in the Cloud:

There are 3 types of storage systems in the Cloud as follows.

- Block-Based Storage System
 - File-Based Storage System
 - Object-Based Storage System
- Let's discuss it one by one as follows.

Type-1:

Block-Based Storage System –

1. Hard drives are block-based storage systems. Your operating system like Windows or Linux actually sees a hard disk drive. So, it sees a drive on which you can create a volume, and then you can partition that volume and format them.

2. For example, If a system has 1000 GB of volume, then we can partition it into 800 GB and 200 GB for local C and local D drive respectively.

3. Remember with a block-based storage system, your computer would see a drive, and then you can create volumes and partitions.

Type-2:

File-Based Storage System –

1. In this, you are actually connecting through a [Network Interface Card \(NIC\)](#). You are going over a network, and then you can access the network-attached storage server (NAS). NAS devices are file-based storage systems.

2. This storage server is another computing device that has another disk in it. It is already created a file system so that it's already formatted its partitions, and it will share its file systems over the network. Here, you can actually map the drive to its network location.

3. In this, like the previous one, there is no need to partition and format the volume by the user. It's already done in file-based storage systems. So, the operating system sees a file system that is mapped to a local drive letter.

Type-3:

Object-Based Storage System –

1. In this, a user uploads objects using a web browser and uploading an object to a container i.e, Object Storage Container. This uses the HTTP Protocols with the rest of the APIs (example: GET, PUT, POST, SELECT, DELETE).

2. For example, when you connect to any website, and you need to download some images, text, or anything that the website contains. For that, it is a code HTTP GET request. If you want to review any product then you can use PUT and POST requests.

3. Also, there is no hierarchy of objects in the container. Every file is on the same level in an Object-Based storage system.

Advantages :

Scalability –

Capacity and storage can be expanded and performance can be enhanced.



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Flexibility –

Data can be manipulated and scaled according to the rules.

Simpler Data Migrations –

As it can add and remove the new and old data when required eliminates disruptive data migrations.

Disadvantages :

1. Data centers require electricity and proper internet facility to operate their work, failing in which system will not work properly.



UNIT-5 CLOUD SECURITY & DISASTER RECOVERY

Cloud Security:

What are the Security Risks of Cloud Computing

Cloud computing provides various advantages, such as improved collaboration, excellent accessibility, Mobility, Storage capacity, etc. But there are also security risks in cloud computing. Some most common Security Risks of Cloud Computing are given below-

Data Loss

Data loss is the most common cloud security risks of cloud computing. It is also known as data leakage. Data loss is the process in which data is being deleted, corrupted, and unreadable by a user, software, or application. In a cloud computing environment, data loss occurs when our sensitive data is somebody else's hands, one or more data elements can not be utilized by the data owner, hard disk is not working properly, and software is not updated.

Hacked Interfaces and Insecure APIs

As we all know, cloud computing is completely depends on Internet, so it is compulsory to protect interfaces and APIs that are used by external users. APIs are the easiest way to communicate with most of the cloud services. In cloud computing, few services are available in the public domain. These services can be accessed by third parties, so there may be a chance that these services easily harmed and hacked by hackers.

Data Breach

Data Breach is the process in which the confidential data is viewed, accessed, or stolen by the third party without any authorization, so organization's data is hacked by the hackers.

Vendor lock-in

Vendor lock-in is the of the biggest security risks in cloud computing. Organizations may face problems when transferring their services from one vendor to another. As different vendors provide different platforms, that can cause difficulty moving one cloud to another.

Increased complexity strains IT staff

Migrating, integrating, and operating the cloud services is complex for the IT staff. IT staff must require the extra capability and skills to manage, integrate, and maintain the data to the cloud.

Spectre & Meltdown

Spectre & Meltdown allows programs to view and steal data which is currently processed on computer. It can run on personal computers, mobile devices, and in the cloud. It can store the password, your personal information such as images, emails, and business documents in the memory of other running programs.

Denial of Service (DoS) attacks

Denial of service (DoS) attacks occur when the system receives too much traffic to buffer the server. Mostly, DoS attackers target web servers of large organizations such as banking sectors, media companies, and government organizations. To recover the lost data, DoS attackers charge a great deal of time and money to handle the data.

Account hijacking

Account hijacking is a serious security risk in cloud computing. It is the process in which individual user's or organization's cloud account (bank account, e-mail account, and social media account) is stolen by hackers. The hackers use the stolen account to perform unauthorized activities.

Privacy:

Cloud computing is a widely well-discussed topic today with interest from all fields, be it research, academia, or the IT industry. It has seen suddenly started to be a hot topic in international



conferences and other opportunities throughout the whole world. The spike in job opportunities is attributed to huge amounts of data being processed and stored on the servers. The cloud paradigm revolves around convenience and easy the provision of a huge pool of shared computing resources.

The rapid development of the cloud has led to more flexibility, cost-cutting, and scalability of products but also faces an enormous amount of privacy and security challenges. Since it is a relatively new concept and is evolving day by day, there are undiscovered security issues that creep up and need to be taken care of as soon as discovered. Here we discuss the top 7 privacy challenges encountered in cloud computing:

1. Data Confidentiality Issues

Confidentiality of the user's data is an important issue to be considered when externalizing and outsourcing extremely delicate and sensitive data to the cloud service provider. Personal data should be made unreachable to users who do not have proper authorization to access it and one way of making sure that confidentiality is by the usage of severe access control policies and regulations. The lack of trust between the users and cloud service providers or the cloud database service provider regarding the data is a major security concern and holds back a lot of people from using cloud services.

2. Data Loss Issues

Data loss or data theft is one of the major security challenges that the cloud providers face. If a cloud vendor has reported data loss or data theft of critical or sensitive material data in the past, more than sixty percent of the users would decline to use the cloud services provided by the vendor. Outages of the cloud services are very frequently visible even from firms such as Dropbox, Microsoft, Amazon, etc., which in turn results in an absence of trust in these services during a critical time. Also, it is quite easy for an attacker to gain access to multiple storage units even if a single one is compromised.

3. Geographical Data Storage Issues

Since the cloud infrastructure is distributed across different geographical locations spread throughout the world, it is often possible that the user's data is stored in a location that is out of the legal jurisdiction which leads to the user's concerns about the legal accessibility of local law enforcement and regulations on data that is stored out of their region. Moreover, the user fears that local laws can be violated due to the dynamic nature of the cloud makes it very difficult to delegate a specific server that is to be used for trans-border data transmission.

4. Multi-Tenancy Security Issues

Multi-tenancy is a paradigm that follows the concept of sharing computational resources, data storage, applications, and services among different tenants. This is then hosted by the same logical or physical platform at the cloud service provider's premises. While following this approach, the provider can maximize profits but puts the customer at a risk. Attackers can take undue advantage of the multi-residence opportunities and can launch various attacks against their co-tenants which can result in several privacy challenges.

5. Transparency Issues

In cloud computing security, transparency means the willingness of a cloud service provider to reveal different details and characteristics on its security preparedness. Some of these details compromise policies and regulations on security, privacy, and service level. In addition to the willingness and disposition, when calculating transparency, it is important to notice how reachable the security readiness data and information actually are. It will not matter the extent to which the security facts about an organization are at hand if they are not presented in an organized and easily understandable way for cloud service users and auditors, the transparency of the organization can then also be rated relatively small.



6. Hypervisor Related Issues

Virtualization means the logical abstraction of computing resources from physical restrictions and constraints. But this poses new challenges for factors like user authentication, accounting, and authorization. The hypervisor manages multiple Virtual Machines and therefore becomes the target of adversaries. Different from the physical devices that are independent of one another, Virtual Machines in the cloud usually reside in a single physical device that is managed by the same hypervisor. The compromise of the hypervisor will hence put various virtual machines at risk. Moreover, the newness of the hypervisor technology, which includes isolation, security hardening, access control, etc. provides adversaries with new ways to exploit the system.

7. Managerial Issues

There are not only technical aspects of cloud privacy challenges but also non-technical and managerial ones. Even on implementing a technical solution to a problem or a product and not managing it properly is eventually bound to introduce vulnerabilities. Some examples are lack of control, security and privacy management for virtualization, developing comprehensive service level agreements, going through cloud service vendors and user negotiations, etc.

Privacy Impact Assessment:

The purpose of the ETA Cloud Platform Services project is to provide a data and computing services environment for the modernized applications and to provide a highly scalable and adaptive virtual utilities; with secure services catalog; to support ETA analysis, classification, categorization, recommendations and prediction; models, reports and graphs.

The first module deployed onto this platform will be:

- Workforce Investment Opportunities Act – Performance Reporting System (eDRVS 2.0): Provides report validation and data element validation for the Workforce Innovation and Opportunity Act (WIOA) program. This software allows the States to upload, review, and validate their WIOA reporting data before final submission to the National Office (NO).

A Privacy Impact Assessment is being conducted since this system stores and transmits Personally Identifiable Information that includes social security numbers.

Characterization of the Information

Personally identifiable information (PII) is collected on members of the public who participate in certain grant programs.

From whom is information to be collected?

Grantees collect PII from participants in the program.

Why is the Information being collected?

The PII is collected to access to wage records from the Wage Record Interchange System (WRIS) and the Federal Employment Data Exchange System (FEDES) and compute the common measures

What is the PII being collected, used, disseminated, or maintained?

- Date of birth
- Social Security Number (SSN)

How is the PII collected?

PII data is included in the Comma Separated Value file containing participant level records uploaded into the system by grantee.

How will the information collected from individuals or derived from the system be checked for accuracy?

At the grantee level, where PII is initially collected, staff review participant identification and validate all PII before entering it into their respective systems.

What specific legal authorities, arrangements, and/or agreements defined allow the collection of PII?

- Workforce Innovation and Opportunity Act (WIOA)
- Section 303(a)(6) of the Social Security Act

Privacy Impact Analysis

The risk to privacy is inappropriate handling or disclosure of PII, especially SSNs. Access controls



mitigate the risk that data will be compromised. In addition, the SSN data is encrypted to ensure the confidentiality of this data element.

Describe the Uses of the PII

Describe all the uses of the PII

DOL/ETA and the Kansas Department of Commerce (Kansas) have a Memorandum of Understanding that allows ETA's national program grantees access to wage records through the Wage Record Interchange System (WRIS) and the Federal Employment Data Exchange System (FEDES) in order to calculate the common measures. This arrangement is the Common Reporting Information System (CRIS).

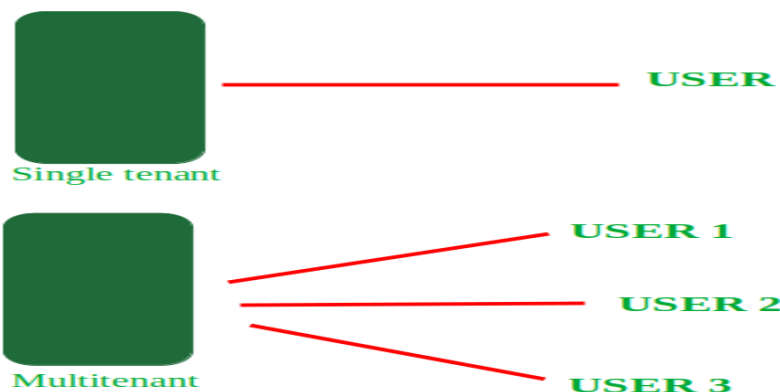
1. National program grantees submit program exit information to DOL's Enterprise Business Support System (EBSS). NFJP grantees submit their information to Social Policy Research Associates (SPRA).
2. DOL sends program exit information, including social security numbers (SSNs), from EBSS to Kansas via secure File Transit Protocol (sFTP). SPRA sends NFJP data directly to Kansas via sFTP.
3. Kansas submits SSNs to the WRIS and FEDES operators. WRIS matches wage data to program exiters covered in the nation's Unemployment Insurance (UI) system. FEDES matches wage records to program exiters employed in the Federal workforce.
4. The WRIS operator transmits SSNs to state UI agencies that have indicated wages exist for that SSN. The state processes the request and returns this information to the WRIS operator. The FEDES operator transmits SSNs to the Office of Personnel Management, the Department of Defense, and the U.S. Postal Service. If these departments have a wage record for the SSN, they return this information to the FEDES operator.
5. The WRIS and FEDES operators return matched wage records to Kansas. Kansas compiles this information and calculates aggregate, program specific common measures reports.
6. Kansas submits these common measures reports to DOL. These reports do NOT contain personally identifiable information (PII).
7. These aggregate, program specific, common measures reports are available on the CRIS portal. Users can select reports based by program and time period. Both national and grantee-specific reports are available.

MULTI-TENANCY ISSUES:

Multitenancy in Cloud computing: Multitenancy is a type of software architecture where a single software instance can serve multiple distinct user groups. It means that multiple customers of cloud vendor are using the same computing resources. As they are sharing the same computing resources but the data of each Cloud customer is kept totally separate and secure. It is very important concept of Cloud Computing.

In cloud computing Multitenancy also refer as shared host where same resources are divided among different customer's.





For Example:

The example of multitenancy is the same as working of Bank. Multiple people can store money in the one same bank. But every customer asset is totally different like one customer cannot have access to the other customer's money and account and different customers are not aware about each other's account balance and details etc.

Multi-tenancy is an architectural approach enabling a single instance of an application to be shared among multiple organizations or users, and is applied only to SaaS (Software as a service). The core principle here is, it is the single instance of the application which is being shared. Hence, multi-instance architectures aren't the same as multi-tenant architectures.

Advantages of Multitenancy:

- Use of Available resources is maximized by sharing resources.
- Customer's Cost of Physical Hardware System is reduces.
- It reduce usage of physical devices and thus power consumption and cooling cost save.
- Save Vendor's cost as it become difficult for cloud vendor to provide separate Physical Services to each individual.
- Provides isolation to user while maximizing resource utilization.

Disadvantages of Multitenancy:

- As data is stored in third party services , this reduces security of our data and put it into vulnerable condition .
- Unauthorized access will cause damage of data.
- Possibility of competition for system resources.
- It has single point of failure since there is only one instance of resource, clients face loss of service in such cases.

Virtualized Security:

The term "**Virtualized Security**," sometimes known as "security virtualization," describes security solutions that are software-based and created to operate in a virtualized IT environment. This is distinct from conventional hardware-based network security, which is static and is supported by equipment like conventional switches, routers, and firewalls.

Virtualized security is flexible and adaptive, in contrast to hardware-based security. It can be deployed anywhere on the network and is frequently cloud-based so it is not bound to a specific device.

In [Cloud Computing](#), where operators construct workloads and applications on-demand, virtualized security enables security services and functions to move around with those on-demand-created workloads. This is crucial for virtual machine security. It's crucial to protect virtualized security in cloud computing technologies such as isolating multitenant setups in public cloud settings. Because data and workloads move around a complex ecosystem including several providers, virtualized security's flexibility is useful for securing hybrid and multi-cloud settings.



Types of Hypervisors

Type-1 Hypervisors

Its functions are on unmanaged systems. Type 1 hypervisors include **Lynx Secure, RTS Hypervisor, Oracle VM, Sun xVM Server, and Virtual Logic VLX**. Since they are placed on bare systems, type 1 hypervisor do not have any host operating systems.

Type-2 Hypervisor

It is a software interface that simulates the hardware that a system typically communicates with. Examples of Type 2 hypervisors include **containers, KVM, Microsoft Hyper V, VMWare Fusion, Virtual Server 2005 R2, Windows Virtual PC, and VMware workstation 6.0**.

Type I Virtualization

In this design, the **Virtual Machine Monitor (VMM)** sits directly above the hardware and eavesdrops on all interactions between the VMs and the hardware. On top of the VMM is a management VM that handles other guest VM management and handles the majority of a hardware connections. The Xen system is a common illustration of this kind of virtualization design.

Type II virtualization

In these architectures, like VMware Player, allow for the operation of the VMM as an application within the host operating system (OS). I/O drivers and guest VM management are the responsibilities of the host OS.

Service Provider Security

The system's virtualization hardware shouldn't be physically accessible to anyone not authorized. Each VM can be given an access control that can only be established through the Hypervisor in order to safeguard it against unwanted access by Cloud administrators. The three fundamental tenets of access control, identity, authentication, and authorization, will prevent unauthorized data and system components from being accessed by administrators.

Hypervisor Security

The Hypervisor's code integrity is protected via a technology called Hyper safe. Securing the write-protected memory pages, expands the hypervisor implementation and prohibits coding changes. By restricting access to its code, it defends the Hypervisor from control-flow hijacking threats. The only way to carry out a VM Escape assault is through a local physical setting. Therefore, insider assaults must be prevented in the physical Cloud environment. Additionally, the host OS and the interaction between the guest machines need to be configured properly.

Virtual Machine Security

The administrator must set up a program or application that prevents virtual machines from consuming additional resources without permission. Additionally, a lightweight process that gathers logs from the VMs and monitors them in real-time to repair any **VM tampering must operate on a Virtual Machine**. Best security procedures must be used to harden the guest OS and any running applications. These procedures include setting up firewalls, host intrusion prevention systems (HIPS), anti-virus and anti-spyware programmers, online application protection, and log monitoring in guest operating systems.

Guest Image Security

A policy to control the creation, use, storage, and deletion of images must be in place for organizations that use virtualization. To find viruses, worms, spyware, and rootkits that hide from security software running in a guest OS, image files must be analyzed.

Benefits of Virtualized Security

Virtualized security is now practically required to meet the intricate security requirements of a virtualized network, and it is also more adaptable and effective than traditional physical security.

- **Cost-Effectiveness:** Cloud computing's virtual machine security enables businesses to keep their networks secure without having to significantly raise their expenditures on pricey

proprietary hardware. Usage-based pricing for cloud-based virtualized security services can result in significant savings for businesses that manage their resources effectively.

- **Flexibility:** It is essential in a virtualized environment that security operations can follow workloads wherever they go. A company is able to profit fully from virtualization while simultaneously maintaining data security thanks to the protection it offers across various data centers, in multi-cloud, and hybrid-cloud environments.
- **Operational Efficiency:** Virtualized security can be deployed more quickly and easily than hardware-based security because it doesn't require IT teams to set up and configure several hardware appliances. Instead, they may quickly scale security systems by setting them up using centralized software. Security-related duties can be automated when security technology is used, which frees up more time for IT employees.
- **Regulatory Compliance:** Virtual machine security in cloud computing is a requirement for enterprises that need to maintain regulatory compliance because traditional hardware-based security is static and unable to keep up with the demands of a virtualized network.

Virtualization Machine Security Challenges

- As we previously covered, buffer overflows are a common component of classical network attacks. **Trojan horses, worms, spyware, rootkits, and DoS attacks** are examples of malware.
- In a cloud context, more recent assaults might be caused via VM rootkits, hypervisor malware, or guest hopping and hijacking. Man-in-the-middle attacks against VM migrations are another form of attack. Typically, passwords or sensitive information are stolen during passive attacks. Active attacks could alter the kernel's data structures, seriously harming cloud servers.
- **HIDS or NIDS** are both types of IDSs. To supervise and check the execution of code, use programmed shepherding. The **RIO dynamic optimization infrastructure**, the v Safe and v Shield tools from VMware, security compliance for hypervisors, and Intel vPro technology are some further protective solutions.

Four Steps to ensure VM Security in Cloud Computing

Protect Hosted Elements by Segregation

To secure virtual machines in cloud computing, the first step is to segregate the newly hosted components. Let's take an example where three features that are now running on an edge device may be placed in the cloud either as part of a private subnetwork that is invisible or as part of the service data plane, with addresses that are accessible to network users.

All Components are Tested and Reviewed

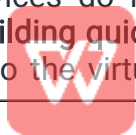
Before allowing virtual features and functions to be implemented, you must confirm that they comply with security standards as step two of cloud-virtual security. Virtual networking is subject to outside attacks, which can be dangerous, but insider attacks can be disastrous. When a feature with a backdoor security flaw is added to a service, it becomes a part of the infrastructure of the service and is far more likely to have unprotected attack paths to other infrastructure pieces.

Separate Management APIs to Protect the Network

The third step is to isolate service from infrastructure management and orchestration. Because they are created to regulate features, functions, and service behaviors, management APIs will always pose a significant risk. All such APIs should be protected, but the ones that keep an eye on infrastructure components that service users should never access must also be protected.

Keep Connections Secure and Separate

The fourth and last aspect of cloud virtual network security is to make sure that connections between tenants or services do not cross over into virtual networks. **Virtual Networking is a fantastic approach to building quick connections to scaled or redeployed features**, but each time a modification is made to the virtual network, it's possible that an accidental connection will be



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made between two distinct services, tenants, or feature/function deployments. A data plane leak, a link between the actual user networks, or a management or control leak could result from this allowing one user to affect the service provided to another.

Disaster Recovery:

HOW GOOD IS YOUR DISASTER RECOVERY PLAN? It's fully documented and you regularly test it by running disaster recovery drills, right?

So far in this book, we have talked about what happens in the event of routine, expected failures. Disaster recovery is the practice of making a system capable of surviving unexpected or extraordinary failures. A disaster recovery plan, for example, will help your IT systems survive a fire in your data center that destroys all of the servers in that data center and the systems they support. Every organization should have a documented disaster recovery process and should test that process at least twice each year. In reality, even well-disciplined companies tend to fall short in their disaster recovery planning. Too many small- and medium-size businesses would simply go out of business in the case of the data center fire scenario I just outlined.

One of the things I personally love about virtualization is the way it lets you automate disaster recovery. Recovery from trivial failures and disaster recovery in the cloud become largely indistinguishable operations. As a result, if your entire cloud infrastructure falls apart, you should have the capabilities in place to restore it on internal servers, at a managed hosting services provider, or at another cloud provider in minutes or hours.

Disaster Recovery Planning

Disaster recovery deals with catastrophic failures that are extremely unlikely to occur during the lifetime of a system. If they are reasonably expected failures, they fall under the auspices of traditional availability planning. Although each single disaster is unexpected over the lifetime of a system, the possibility of some disaster occurring over time is reasonably nonzero.

Through disaster recovery planning, you identify an acceptable recovery state and develop processes and procedures to achieve the recovery state in the event of a disaster. By "acceptable recovery state," I specifically mean how much data you are willing to lose in the event of a disaster.

Defining a disaster recovery plan involves two key metrics:

Recovery Point Objective (RPO)

The recovery point objective identifies how much data you are willing to lose in the event of a disaster. This value is typically specified in a number of hours or days of data. For example, if you determine that it is OK to lose 24 hours of data, you must make sure that the backups you'll use for your disaster recovery plan are never more than 24 hours old.

Recovery Time Objective (RTO)

The recovery time objective identifies how much downtime is acceptable in the event of a disaster. If your RTO is 24 hours, you are saying that up to 24 hours may elapse between the point when your system first goes offline and the point at which you are fully operational again.

In addition, the team putting together a disaster recovery plan should define the criteria that would trigger invocation of the plan. In general, invocation of any plan that results in accepting a loss of data should involve the heads of the business organization—even if the execution of the plan is automated, as I am promoting in this chapter.

Everyone would love a disaster recovery scenario in which no downtime and no loss of data occur,



no matter what the disaster. The nature of a disaster, however, generally requires you to accept some level of loss; anything else will come with a significant price tag. In a citywide disaster like Hurricane Katrina, the cost of surviving with zero downtime and zero data loss could have been having multiple data centers in different geographic locations that were constantly synchronized. In other words, you would need two distinct data centers from different infrastructure providers with dedicated, high-bandwidth connections between the two.

Accomplishing that level of redundancy is expensive. It would also come with a nontrivial performance penalty. The cold reality for most businesses is likely that the cost of losing 24 hours of data is less than the cost of maintaining a zero downtime/zero loss of data infrastructure.

Determining an appropriate RPO and RTO is ultimately a financial calculation: at what point does the cost of data loss and downtime exceed the cost of a backup strategy that will prevent that level of data loss and downtime? The right answer is radically different for different businesses. If you are in a senior IT management role in an organization, you should definitely know the right answer for your business.

The final element of disaster recovery planning is understanding the catastrophic scenario. There's ultimately some level of disaster your IT systems will not survive no matter how much planning and spending you do. A good disaster recovery plan can describe that scenario so that all stakeholders can understand and accept the risk.

The Recovery Point Objective

The easiest place to start is your RPO. The Armageddon scenario results in total loss of all system data and the binaries of all applications required to run the system. Your RPO is somewhere between the application state when you first deployed it and the state at the time of the disaster. You may even define multiple disaster levels with different RPOs.*

Just about any software system should be able to attain an RPO between 24 hours for a simple disaster to one week for a significant disaster without incurring absurd costs. Of course, losing 24 hours of banking transactions would never be acceptable, much less one week.

Your RPO is typically governed by the way in which you save and back up data:

1. Weekly off-site backups will survive the loss of your data center with a week of data loss. Daily off-site backups are even better.
2. Daily on-site backups will survive the loss of your production environment with a day of data loss plus replicating transactions during the recovery period after the loss of the system. Hourly on-site backups are even better.
3. A NAS/SAN will survive the loss of any individual server, except for instances of data corruption with no data loss.
4. A clustered database will survive the loss of any individual data storage device or database node with no data loss.
5. A clustered database across multiple data centers will survive the loss of any individual data center with no data loss.

Later in this chapter, we talk about how the cloud changes your options and makes lower RPOs possible.

The Recovery Time Objective

Having up-to-the-second off-site backups does you no good if you have no environment to which



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you can restore them in the event of failure. The ability to assemble a replacement infrastructure for your disasters—including the data restore time—governs the RTO.

* You might, for example, define a level 1 disaster as involving the loss of a single data center and a level 2 disaster as involving the loss of multiple data centers.

What would happen if your managed services provider closed its doors tomorrow? If you have a number of dedicated servers, it can be days or weeks before you are operational again unless you have an agreement in place for a replacement infrastructure.†

In a traditional infrastructure, a rapid RTO is very expensive. As I already noted, you would have to have an agreement in place with another managed services provider to provide either a backup infrastructure or an SLA for setting up a replacement infrastructure in the event your provider goes out of business. Depending on the nature of that agreement, it can nearly double the costs of your IT infrastructure.

The cloud—even over virtualized data centers—alters the way you look at your RTO. We'll dive into the reasons why later in the chapter.

Disasters in the Cloud

Assuming unlimited budget and capabilities, I focus on three key things in disaster recovery planning:

1. Backups and data retention
2. Geographic redundancy
3. Organizational redundancy

If I can take care of those three items, it's nearly certain I can meet most RPO and RTO needs. But I have never been in a situation in which I had an unlimited budget and capabilities, so I have always had to compromise. As a result, the order of the three items matters. In addition, if your hosting provider is a less-proven organization, organizational redundancy may be more important than geographic redundancy.

Fortunately, the structure of the Amazon cloud makes it very easy to take care of the first and second items. In addition, cloud computing in general makes the third item much easier.

† This scenario also illustrates that simply having off-site backups is not enough. Those backups must reside outside the control of your managed services provider, or you risk losing access to them in the event of that provider going bankrupt.

Backup Management

In Chapter 4, I looked at the technical details of how to manage AMIs and execute backups in a cloud environment. Now it's time to take a step back from the technical details and examine the kinds of data you are planning to back up and how it all fits into your overall disaster recovery plan.

Your ability to recover from a disaster is limited by the quality and frequency of your backups. In a traditional IT infrastructure, companies often make full weekly backups to tape with nightly differentials and then ship the weekly backups off-site. You can do much better in the cloud, and do it much more cheaply, through a layered backup strategy.

BACKUPS, BUSINESS CONTINUITY, AND AWS

In this section, I cover a number of technologies that Amazon Web Services provide to help you manage backups effectively. If you are using a different cloud, they likely have some similar tools as well as some that are completely different. A critical part of any backup strategy, however, is the concept of off-site backups. Whatever your backup strategy, you must not only have a mechanism for moving all data critical for achieving your Recovery Point Objectives out of the cloud, but you must also store that data in a portable format so you can recover into an environment that might be radically different from your current cloud provider.

Kind of data	Description
Fixed data	Fixed data, such as your operating system and common utilities, belong in your AMI. In the cloud, you don't back up your AMI, because it has no value beyond the cloud. ^a
Transient data	File caches and other data that can be lost completely without impacting the integrity of the system. Because your application state is not dependent on this data, don't back it up.
Configuration data	Runtime configuration data necessary to make the system operate properly in a specific context. This data is not transient, since it must survive machine restarts. On the other hand, it should be easily reconfigured from a clean application install. This data should be backed up semi-regularly.
Persistent data	Your application state, including critical customer data such as purchase orders. It changes constantly and a database engine is the best tool for managing it. Your database engine should store its state to a block device, and you should be performing constant backups. Clustering and/or replication are also critical tools in managing the database.

[Table 6-1](#) illustrates the different kinds of data that web applications typically manage.

TABLE 6-1. Backup requirements by data type

a Keep in mind that even if Amazon S3 failed completely and lost your AMIs, as long as EC2 is available and you have EC2 instances running based on the lost AMI, you will be able to quickly rebuild the lost AMI. On the other hand, if EC2 goes down, S3 goes down, and all S3 data is lost completely, you'll have to recover into a different cloud that doesn't recognize your AMI anyway!

In disaster recovery, persistent data is generally the data of greatest concern. We can always



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rebuild the operating system, install all the software, and reconfigure it, but we have no way of manually rebuilding the persistent data.

Fixed data strategy

If you are fixated on the idea of backing up your machine images, you can download the images out of S3 and store them outside of the Amazon cloud. If S3 were to go down and incur data loss or corruption that had an impact on your AMIs, you would be able to upload the images from your off-site backups and reregister them. It's not a bad idea and it is not a lot of trouble, but the utility is limited given the uniqueness of the failure scenario that would make you turn to those backups.

Configuration data strategy

A good backup strategy for configuration information comprises two levels. The first level can be either a regular filesystem dump to your cloud storage or a filesystem snapshot. For most applications, you can back up your configuration data once a day or even once a week and be fine. You should, however, think back to your Recovery Point Objective. If your configuration data changes twice a day and you have a two-hour RPO, you will need to back up your configuration data twice a day. If configuration data changes irregularly, it may be necessary to make hourly backups or specifically tie your backups to changes in application configuration.

An alternate approach is to check your application configuration into a source code repository outside of the cloud and leverage that repository for recovery from even minor losses.

Whether you perform filesystem snapshots or simply zip up the filesystem, that data will hibernate inside S3. Snapshots tend to be the most efficient and least intrusive mechanism for performing backups, but they are also the least portable. You don't have direct access to the EC2 snapshots, and even if you did, they would not be usable outside of the Amazon cloud. At some point, you do need to get that data out of the cloud so that you have off-site backups in a portable format. Here's what I recommend:

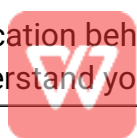
1. Create regular—at a minimum, daily—snapshots of your configuration data.
2. Create semi-regular—at least less than your RPO—filesystem archives in the form of ZIP or TAR files and move those archives into Amazon S3.
3. On a semi-regular basis—again, at least less than your RPO—copy your filesystem archives out of the Amazon cloud into another cloud or physical hosting facility.

Let's say I had a one-day RPO with an application housing less than 10 GB of data whose configuration data could change on a whim:

4. I would make hourly snapshots of the filesystem with the configuration data and archive the whole thing in a portable format to S3 at least once a day.
 1. I would retain a week's worth of full backups in S3 in case of a corruption issue during backup. In addition, I would copy each daily backup out of S3 to another cloud or my internal file servers at some point right after it is made.
 2. Finally, I would keep a week's worth of backups off site for the last week along with one backup a week for the prior month and one backup a month for the prior year.

Depending on the amount of data in question and the data retention issues/requirements, I might do more or less in the archiving of old backups.

Different RPOs and application behaviors mandate different strategies. I have to emphasize again: you must absolutely understand your RPO in order to create a proper backup strategy. Given the



example strategy I just outlined, I will be able to create a functional replication of my application in any data center in the world and lose at most 24 hours of data, except in the event of complete data corruption of backups or simultaneous loss of all of Amazon and wherever my off-site backups are stored.

Persistent data strategy (aka database backups)

I've already recommended using a relational database to store customer information and other persistent data. After all, the purpose of a relational database is to maintain the consistency of complex transactional data. The challenge of setting up database backups is doing them regularly in a manner that does not impact operations while retaining database integrity.

MySQL, like all database engines, provides several convenient tools for backups, but you must use them carefully to avoid data corruption. The techniques are fairly well-documented in the database literature, but they're important enough for me to summarize them here as well as apply them to an Amazon EC2 environment.

If you aren't familiar with the way databases work, you might be wondering, "Why is that so hard? Why don't I do a snapshot or make an archive like I did with the configuration data?" With the configuration data, it is highly unlikely you will be making a backup in between the writing of two different files that must remain consistent or in the middle of writing out a file to the filesystem. With database storage, it is a near certainty that every time you try to copy those files, the database will be in the middle of doing something with them. As a result, you need to get clever with your database backup strategy.

The first line of defense is either *multimaster replication* or *clustering*. A multimaster database is one in which two master servers execute write transactions independently and replicate the transactions to the other master. A clustered database environment contains multiple servers that act as a single logical server. Under both scenarios, when one goes down, the system remains operational and consistent.

Instead, you can perform *master-slave replication*. Master-slave replication involves setting up a *master server* that handles your write operations and replicating transactions over to a *slave server*. Each time something happens on the master, it replicates to the slave.

Replication in itself is not a "first line of defense," since replication is not atomic with respect to the transactions that take place on the master. In other words, a master can crash after a transaction has completed but before it has had time to replicate to the slave. To get around this problem, I generally do the following:

5. Set up a master with its data files stored on a block storage device.
6. Set up a replication slave, storing its data files on a block storage device.
7. Take regular snapshots of the master block storage device based on my RPO.
8. Create regular database dumps of the slave database and store them in S3.
9. Copy the database dumps on a semi-regular basis from S3 to a location outside the Amazon cloud.

Amazon's Elastic Block Storage (EBS) offering has not been around long enough to provide a good feel for how likely you are to see database corruption in the event of your MySQL server failing. I have taken the position, however, that corruption is extremely likely—as it is with many filesystems—and have adjusted my backup strategy accordingly. I could be proven wrong on this issue, but for the moment, I prefer to play it safe.

Actually taking snapshots or creating database dumps for some database engines is actually very tricky in a runtime environment, especially if you want to do it hourly or even more frequently. The challenge in creating your backups for these database engines is the need to stop processing all transactions while the backup is taking place. To complicate the situation, database dumps can take a long time to complete. As a result, your applications will grind to a halt while you make any database dumps.

Snapshots are available in most cloud environments and provide an important approach for maintaining database integrity without completely shutting down application processing— even with large data sets in databases such as MySQL.

You need to freeze the database only for an instant to create your snapshot. The process follows these steps:

1. Lock the database.
2. Sync the filesystem (this procedure is filesystem-dependent).
3. Take a snapshot.
4. Unlock the database.

The whole process should take about one second.

On Amazon EC2, you will store your snapshots directly onto block storage. Unfortunately, the snapshots are not portable, so you can't use them for off-site storage. You therefore will need to do database dumps, no matter how much you would rather avoid doing them. Because of this need, I run my backups against a database slave. The slave can afford to be locked for a period of time while a database dump completes. The fact that the slave may be a transaction or two (or ten) behind the master is unimportant for the purposes of making a backup. What matters is that you can create the dump without impacting your applications.

The steps for creating the database dump are:

1. Execute the database dump.
2. When complete, encrypt the dump and break it into small, manageable chunks.
3. Move the dump over to S3.

Amazon S3 limits your file size to 5 GB. As a result, you probably need to break your database into chunks, and you should definitely encrypt it and anything else you send into Amazon S3.

Now that you have a portable backup of your database server, you can copy that backup out of the Amazon cloud and be protected from the loss of your S3 backups.

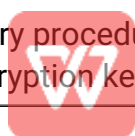
Backup security

If you are following the security practices I outlined earlier in the book, your filesystems are encrypted to protect the snapshots you are making for your backups from prying eyes. The harder part is securing your portable backups as you store them in S3 and move them off site.

I typically use PGP-compatible encryption for my portable backups. You need to worry about two issues:

1. Keeping your private decryption key out of the cloud.
2. Keeping your private decryption key some place that it will never, ever get lost.

Given the disaster recovery procedures I cover in this chapter, you really have no reason for ever giving out the private decryption key to an instance in the cloud unless you are automating the



failover between two different cloud infrastructures. The cloud needs only your public encryption key so it can encrypt the portable backups.

You can't store your decryption key with your backups. Doing so will defeat the purpose of encrypting the backups in the first place. Because you will store your decryption key somewhere else, you run the risk of losing your decryption key independent of your backups. On the other hand, keeping a bunch of copies of your decryption key will make it more likely it will fall into the wrong hands.

The best approach? Keep two copies:

10. One copy stored securely on a highly secure server in your internal network.
11. One copy printed out on a piece of paper and stored in a safety deposit box nowhere near the same building in which you house your highly secure server.

More than one person should know the locations of these copies. A true disaster can unfortunately result in the loss of personnel, so personnel redundancy is also important for a disaster recovery plan.

If you are automating the recovery from portable backups, you will also need to keep a copy of the private decryption key on the server that orchestrates your automated recovery efforts.

Geographic Redundancy

Everything I have discussed so far in this chapter focuses on the Recovery Point Objective. In the end, that's the easiest part of disaster recovery. The virtualization technologies behind the cloud simply make it a lot easier to automate those processes and have a relatively inexpensive mechanism for off-site backups.

Turning now to your Recovery Time Objective, the key is redundancy in infrastructure. If you can develop geographical redundancy, you can survive just about any physical disaster that might happen. With a physical infrastructure, geographical redundancy is expensive. In the cloud, however, it is relatively cheap.

You don't necessarily need to have your application running actively in all locations, but you need the ability to bring your application up from the redundant location in a state that meets your Recovery Point Objective within a timeframe that meets your Recovery Time Objective. If you have a 2-hour RTO with a 24-hour RPO, geographical redundancy means that your second location can be operational within two hours of the complete loss of your primary location using data that is no older than 24 hours.

Amazon provides built-in geographic redundancy in the form of regions and availability zones. If you have your instances running in a given availability zone, you can get them started back up in another availability zone in the same region without any effort. If you have specific requirements around what constitutes geographic redundancy,‡ Amazon's availability zones may not be enough—you may have to span regions.

‡ Some government agencies and other organizations mandate a separation of at least 50 miles.



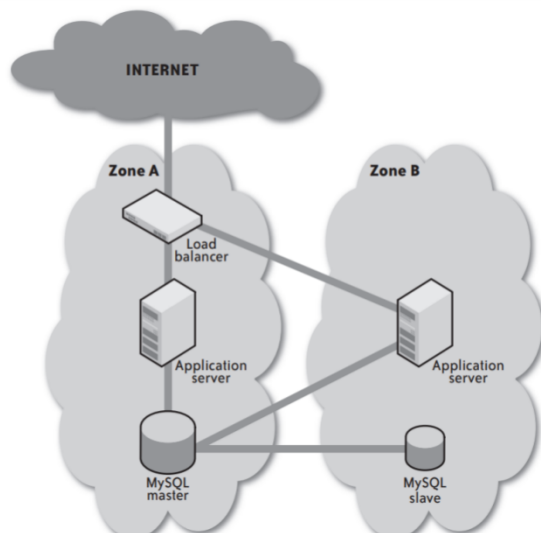


FIGURE 6-1. By spanning multiple availability zones, you can achieve geographic redundancy

Spanning availability zones

Just about everything in your Amazon infrastructure except block storage devices is available across all availability zones in a given region. Although there is a charge for network traffic that crosses availability zones, that charge is generally well worth the price for the leveraging ability to create redundancy across availability zones.

[Figure 6-1](#) illustrates an application environment that can easily tolerate the loss of an entire availability zone.

If you lose the entire availability zone B, nothing happens. The application continues to operate normally, although perhaps with degraded performance levels.

If you lose availability zone A, you will need to bring up a new load balancer in availability zone B and promote the slave in that availability zone to master. The system can return to operation in a few minutes with little or no data loss. If the database server were clustered and you had a spare load balancer running silently in the background, you could reassign the IP address from the old load balancer to the spare and see only a few seconds of downtime with no data loss.

The Amazon SLA provides for a 99.95% uptime of at least two availability zones in each region. If you span multiple availability zones, you can actually exceed the Amazon SLA in regions that have more than two availability zones. The U.S. East Coast, for example, has three availability zones. As a result, you have only a 33% chance of any given failure of two availability zones being exactly the two zones you are using.

Even in the event that you are unfortunate enough to be operating in exactly the two zones that fail, you can still exceed Amazon's SLA as long as the region you are operating in has more than two availability zones. The trick is to execute your disaster recovery procedures and bring your infrastructure back up in the remaining availability zone. As a result, you can be operational again while the other two availability zones are still down.

Operating across regions

At the time I write this chapter, Amazon supports two regions: us-east-1 (Eastern United States) and eu-west-1 (Western Europe). These regions share little or no meaningful infrastructure. The

advantage of this structure is that your application can basically survive a nuclear attack on the U.S. or EU (but not on both!) if you operate across regions. On the other hand, the lack of common infrastructure makes the task of replicating your environments across regions more difficult.

Each region has its own associated Amazon S3 region. Therefore, you cannot launch EC2 instances in the EU using AMIs from the U.S., and you cannot use IP addresses formerly associated with a load balancer in the EU with a replacement in the U.S.

How you manage operations across regions depends on the nature of your web application and your redundancy needs. It's entirely likely that just having the capability to rapidly launch in another region is good enough, without actually developing an infrastructure that simultaneously operates in both regions.

The issues you need to consider for simultaneous operation include:

§ The EU region has exactly two availability zones. As a result, the best uptime you can promise in the EU region alone is 99.95% availability.

DNS management

You can use round-robin DNS to work around the fact that IP addresses are not portable across regions, but you will end up sending European visitors to the U.S. and vice versa (very inefficient network traffic management) and lose half your traffic when one of the regions goes down. You can leverage a dynamic DNS system such as UltraDNS that will offer up the right DNS resolution based on source and availability.

Database management

Clustering across regions is likely not practical (but you can try it). You can also set up a master in one region with a slave in the other. Then you perform write operations against the master, but read against the slave for traffic from the region with the slave. Another option is to segment your database so that the European region has "European data" and the U.S. region has "American data." Each region also has a slave in the other region to act as a recovery point from the full loss of a region.

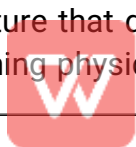
Regulatory issues

The EU does not allow the storage of certain data outside of the EU. As a result, legally you may not be allowed to operate across regions, no matter what clever technical solutions you devise. In reality, an Amazon+GoGrid or Amazon+Rackspace approach to redundancy may be more effective than trying to use Amazon's two cross-jurisdictional regions.

For most purposes, I recommend a process for regularly copying infrastructure elements (AMIs, backups, and configuration) over into the other region and then having the ability to rapidly start that infrastructure in the event of a total, prolonged failure of your core zone.

Organizational Redundancy

If you have an infrastructure that does everything I have recommended so far, you are pretty well protected against everything physical that can happen. You are still exposed at the business level,



however. Specifically, if Amazon or Rackspace or GoGrid or whoever you are using goes out of business or decides it is bored with cloud computing, you might find yourself in trouble.

Physical disasters are a relatively rare thing, but companies go out of business everywhere every day—even big companies like Amazon and Rackspace. Even if a company goes into bankruptcy restructuring, there's no telling what will happen to the hardware assets that run their cloud infrastructure. Your disaster recovery plan should therefore have contingencies that assume your cloud provider simply disappears from the face of the earth.

You probably won't run concurrent environments across multiple clouds unless it provides some level of geographic advantage. Even in that case, your environments are not likely to be redundant so much as segmented for the geographies they are serving. Instead, the best approach to organizational redundancy is to identify another cloud provider and establish a backup environment with that provider in the event your first provider fails.

The issues associated with organizational redundancy are similar to the issues I discussed earlier around operating across Amazon EC2 regions. In particular, you must consider all of the following concerns:

12. Storing your portable backups at your secondary cloud provider.
13. Creating machine images that can operate your applications in the secondary provider's virtualized environment.
14. Keeping the machine images up to date with respect to their counterparts with the primary provider.
15. Not all cloud providers and managed service providers support the same operation systems or filesystems. If your application is dependent on either, you need to make sure you select a cloud provider that can support your needs.

Disaster Management

You are performing your backups and have an infrastructure in place with all of the appropriate redundancies. To complete the disaster recovery scenario, you need to recognize when a disaster has happened and have the tools and processes in place to execute your recovery plan. One of the coolest things about the cloud is that all of this can be automated. You can recover from the loss of Amazon's U.S. data centers while you sleep.||

Monitoring

Monitoring your cloud infrastructure is extremely important. You cannot replace a failing server or execute your disaster recovery plan if you don't know that there has been a failure. The trick, however, is that your monitoring systems cannot live in either your primary or secondary cloud provider's infrastructure. They must be independent of your clouds. If you want to enable automated disaster recovery, they also need the ability to manage your EC2 infrastructure from the monitoring site.

|| Though it is probably not a good idea to automatically engage disaster recovery processes without some human intervention when data could be lost during the launch of those processes.

Your primary monitoring objective should be to figure out what is going to fail before it actually



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fails. The most common problem I have encountered in EC2 is servers that gradually decrease in local file I/O throughput until they become unusable. This problem is something you can easily watch for and fix before users even notice it. On the other hand, if you wait for your application to fail, chances are users have had to put up with poor performance for some period of time before it failed completely. It may also prove to be a precursor to a larger cloud failure event.

There are many other more mundane things that you should check on in a regular environment. In particular, you should be checking capacity issues such as disk usage, RAM, and CPU. I cover those things in more detail in Chapter 7.

In the end, however, you will need to monitor for failure at three levels:

1. Through the provisioning API (for Amazon, the EC2 web services API)
2. Through your own instance state monitoring tools
3. Through your application health monitoring tools

Your cloud provider's provisioning API will tell you about the health of your instances, any volumes they are mounting, and the data centers in which they are operating. When you detect a failure at this level, it likely means something has gone wrong with the cloud itself. Before engaging in any disaster recovery, you will need to determine whether the outage is limited to one server or affects indeterminate servers, impacting an entire availability zone or an entire region.

Monitoring is not simply about checking for disasters; mostly it is checking on the mundane. With enStratus, I put a Python service on each server that checks for a variety of server health indicators—mostly related to capacity management. The service will notify the monitoring system if there is a problem with the server or its configuration and allow the monitoring system to take appropriate action. It also checks for the health of the applications running on the instance.

Load Balancer Recovery

One of the reasons companies pay absurd amounts of money for physical load balancers is to greatly reduce the likelihood of load balancer failure. With cloud vendors such as GoGrid— and in the future, Amazon—you can realize the benefits of hardware load balancers without incurring the costs. Under the current AWS offering, you have to use less-reliable EC2 instances. Recovering a load balancer in the cloud, however, is lightning fast. As a result, the downside of a failure in your cloud-based load balancer is minor.

Recovering a load balancer is simply a matter of launching a new load balancer instance from the AMI and notifying it of the IP addresses of its application servers. You can further reduce any downtime by keeping a load balancer running in an alternative availability zone and then remapping your static IP address upon the failure of the main load balancer.

Application Server Recovery

If you are operating multiple application servers in multiple availability zones, your system as a whole will survive the failure of any one instance—or even an entire availability zone. You will still need to recover that server so that future failures don't affect your infrastructure.

The recovery of a failed application server is only slightly more complex than the recovery of a failed load balancer. Like the failed load balancer, you start up a new instance from the application server machine image. You then pass it configuration information, including where the database is. Once the server is operational, you must notify the load balancer of the existence of the new server (as well as deactivate its knowledge of the old one) so that the new server enters the load-balancing rotation.



Database Recovery

Database recovery is the hardest part of disaster recovery in the cloud. Your disaster recovery algorithm has to identify where an uncorrupted copy of the database exists. This process may involve promoting slaves into masters, rearranging your backup management, and reconfiguring application servers.

The best solution is a clustered database that can survive the loss of an individual database server without the need to execute a complex recovery procedure. Absent clustering, the best recovery plan is one that simply launches a new database instance and mounts the still- functional EC2 volume formerly in use by the failed instance. When an instance goes down, however, any number of related issues may also have an impact on that strategy:

16. The database could be irreparably corrupted by whatever caused the instance to crash.
17. The volume could have gone down with the instance.
18. The instance's availability zone (and thus the volume as well) could be unavailable.
19. You could find yourself unable to launch new instances in the volume's availability zone.

On the face of it, it might seem that the likelihood of both things going wrong is small, but it happens. As a result, you need a fallback plan for your recovery plan. The following process will typically cover all levels of database failure:

1. Launch a replacement instance in the old instance's availability zone and mount its old volume.
2. If the launch fails but the volume is still running, snapshot the volume and launch a new instance in any zone, and then create a volume in that zone based on the snapshot.
3. If the volume from step 1 or the snapshot from step 2 are corrupt, you need to fall back to the replication slave and promote it to database master.
4. If the database slave is not running or is somehow corrupted, the next step is to launch a replacement volume from the most recent database snapshot.
5. If the snapshot is corrupt, go further back in time until you find a backup that is not corrupt.

Step 4 typically represents your worst-case scenario. If you get to 5, there is something wrong with the way you are doing backups.

HOW DO I KNOW MY BACKUPS ARE WORKING?

You need to test your disaster recovery procedures to know for sure that your backups are solid and the processes you have in place will work. I recommend testing your processes and your backups once every quarter, and certainly no less than once each year.

The most significant area in which your disaster recovery process can go wrong is in the database backups. Specifically, if you are not backing up your databases properly, the backups can appear to succeed but instead result in corrupted database backups. Silent failures are a bad thing.



The simplest way to test your database backup tools is to set up some bots to place a very high write transaction load on your database. Execute the backup script while the bots are running, and then try to recover the database from the backup. If your backup process is questionable, this test should leave you with a corrupted database.

